

Project Data Page:





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Utility Air Reciever

DESIGN CALCULATION

In Accordance with ASME Section VIII Division 1

ASME Code Version : 2025

Analysis Performed by : SL User

Job File : ...\\Utility Air Reciever-7%.pvdb

Date of Analysis : Sep 16,2026 2:27pm

PV Elite 28, January 2026



Class From To : Basic Element Checks.

Note 20 40 There is a high jump in the Joint Eff.

Class From To: Check of Additional Element Data

There were no geometry errors or warnings.

PV Elite performs all calculations internally in Imperial Units to remain compliant with the ASME Code and any built in assumptions in the ASME Code formulas. The finalized results are reflected to show the set of selected units for this analysis.

Information Regarding Nozzle Loads

Please note that nozzle loadings, if included, are assumed to be local in nature and will not contribute to or create a net section bending moment. Therefore, the addition of nozzle loads will not affect the support load calculation. If you wish to create a load on the support(s) from nozzle loads, you can enable that feature in tools->configuration->Job Specific Settings.



One or more nozzles in this model was specified using the actual thickness basis and a standard flange is also specified. In many cases a matching nominal diameter cannot be determined and the weight of the flange and optional blind flange cannot be determined. When actual thickness basis is specified, you must calculate the weight of the assembly (by hand) and enter it into the 'Overriding Weight' field in the nozzle dialog when there is a standard flange specified. However, for FVC nozzles that have their nominal diameter specified in the FVC dialog, entering the overriding weight for that nozzle is not required.



The level of precision is set to a value of less than 3. At this level you may not obtain proper resolution on some results. This value can be changed in Tools->Set Configuration Parameters->Default Values->Level of Precision.



The ASME Code Section VIII Division 1 does not explicitly require stresses on elements to be computed under test conditions. Please see interpretation VIII-1-10-12 for more information. PV Elite also computes weight due to test fluid and uses the weight in various calculations such as saddle and leg loads. PV Elite also computes static pressure due to operating and test liquids. The static pressure is used in conjunction with the design and test pressures to determine a total pressure in the determination of required thickness, MAWP etc. This ensures compliance with UG-22 requirements.

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Units used in this Analysis (bars):

Name	System Unit	Constant	User Unit
Length	Feet	304.8000	mm.
Force	Pounds	0.4536	Kgf
Mass	Pounds	0.4536	kg.
Area	sq. inches	6.4516	cm ²
Moment	ft. lbs.	0.1383	Kg-m.
Stress	lbs./sq.in.	0.0069	N./mm ²
Temperature	Degrees F	0.5556	°C
Pressure	psig	0.0689	bars
Elast. Modulus	lbs./sq.in.	6.8946	KPa.
Pipe Density	lbs./cu.in.	0.0277	kg./cm ³
Ins. Density	lbs./cu.ft.	0.1600E-04	kg./cm ³
Fluid Density	lbs./cu.ft.	0.1600E-04	kg./cm ³
Wind Speed	miles/hr	1.6093	Km/hr
Tray Weight	lbs./sq.ft.	0.0005	Kgs/cm ²
Inertia	in.4	41.6231	cm4
G Load	G's	1.0000	g's
Wind Load	lbs./sq.ft.	4.8825	Kgs/m ²
Elevation	Feet	304.8000	mm.
Volume	in.3	16.3900	cm ³
Diameter	inches	25.4000	mm.
Thickness	inches	25.4000	mm.

PV Elite Vessel Analysis Program: Input Data

Utility Air Reciever

Design Internal Pressure (for Hydrotest) 12 bars
Design Internal Temperature 82.0 °C
Projection of Nozzle from Vessel Top 500 mm.
Projection of Nozzle from Vessel Bottom 500 mm.
Minimum Design Metal Temperature -1.0 °C
Type of Construction Welded
Special Service Air/Water/Steam
Degree of Radiography RT-2
Use Higher Longitudinal Stresses (Flag) Y
Select t for Internal Pressure (Flag) N
Select t for External Pressure (Flag) N
Select t for Axial Stress (Flag) N
Select Location for Stiff. Rings (Flag) N
Consider Vortex Shedding N

Shop Pressure Test:

Type of Pressure Test User Entered Pressure
Pressure Test Position Horizontal
Test Performed in Corroded Condition No

Field Pressure Test:

Type of Pressure Test UG-99(b) Note [35]
Pressure Test Position Vertical
Test Performed in Corroded Condition Yes

Load Case 1 NP+EW+1.1WI+FW+BW
Load Case 2 NP+EW+EE+FS+BS
Load Case 3 NP+OW+1.1WI+FW+BW
Load Case 4 NP+OW+EQ+FS+BS
Load Case 5 NP+HW+HI
Load Case 6 NP+HW+HE
Load Case 7 IP+OW+1.1WI+FW+BW
Load Case 8 IP+OW+EQ+FS+BS
Load Case 9 EP+OW+1.1WI+FW+BW
Load Case 10 EP+OW+EQ+FS+BS
Load Case 11 HP+HW+HI
Load Case 12 HP+HW+HE



Load Case 13	IP+WE+EW
Load Case 14	IP+WF+CW
Load Case 15	IP+VO+OW
Load Case 16	IP+VE+EW
Load Case 17	NP+VO+OW
Load Case 18	FS+BS+IP+OW
Load Case 19	FS+BS+EP+OW
Load Case 20	BL+IP+OW

Wind Design Code	ASCE/SEI 7-16
Wind Load Reduction Scale Factor	1.100
Basic Wind Speed [V]	97.2 Km/hr
Surface Roughness Category	C: Open Terrain
Type of Surface	Moderately Smooth
Base Elevation	300 mm.
Percent Wind for Hydrotest	50.0
Using User defined Wind Press. Vs Elev.	N
Height of Hill or Escarpment H or Hh	0 mm.
Distance Upwind of Crest Lh	0 mm.
Distance from Crest to the Vessel x	0 mm.
Type of Terrain (Hill, Escarpment)	Flat
Damping Factor (Beta) for Wind (Ope)	0.0100
Damping Factor (Beta) for Wind (Empty)	0.0000
Damping Factor (Beta) for Wind (Filled)	0.0000
Site Elevation above Sea Level	0 mm.

Seismic Design Code	ASCE/SEI 7-16
Seismic Load Reduction Scale Factor	1.000
Importance Factor	1.250
Table Value Fa	1.600
Table Value Fv	2.400
Max. Mapped Res. Acceleration [Ss]	0.147
Max. Eff. Ground Acceleration [S]	0.065
Force Modification Factor R	1.250
Site Class	D
Component Elevation Ratio z/h	0.020
Amplification Factor Ap	2.500
Force Factor	0.000
Consider Vertical Acceleration	Yes
Minimum Acceleration Multiplier	0.000
User Value of Sds (used if > 0)	0.000
User Value of Sd1 (used if > 0)	0.000
Moment Reduction Factor Tau	1.000

M.A.W.P + Static Head to Nozzle	Y
Consider MAP New and Cold in Noz. Design	N
Consider External Loads for Nozzle Des.	Y
Use ASME VIII-1 Appendix 1-9	N

Perform Blast Load Analysis	No
Material Database Year	Current w/Addenda or Code Year

Configuration Directives:

g0 is taken as min(g0, attached shell thk)	Yes
Do not use Nozzle MDMT Interpretation VIII-1 01-37	No
Use Table G instead of exact equation for "A"	Yes
Shell Head Joints are Tapered	Yes
Compute "K" in corroded condition	Yes
Use Code Case 2286	No
Use the MAWP to compute the MDMT	Yes
For thickness ratios <= 0.35, MDMT will be -155F (-104C)	Yes
For PWHT & P1 Materials the MDMT can be < -55F (-48C)	No
Using Metric Material Databases, ASME II D	Yes
Calculate B31.3 type stress for Nozzles with Loads	Yes
Reduce the MDMT due to lower membrane stress	Yes
Consider Longitudinal Stress in MDMT Calculations	Yes



Complete Listing of Vessel Elements and Details:

Element#	1/9
Element From Node	10
Element To Node	20
Element Type	Skirt Support
Description	Skirt2
Distance "FROM" to "TO"	1700 mm.
Skirt Outside Diameter	5756 mm.
Diameter of Skirt at Base	5756 mm.
Skirt Thickness	22 mm.
Internal Corrosion Allowance	1 mm.
Nominal Thickness	22 mm.
External Corrosion Allowance	0 mm.
Design Temperature Internal Pressure	82 °C
Design Temperature External Pressure	0 °C
Effective Diameter Multiplier	1.2
Material Name	SA-36
Allowable Stress, Ambient	114 N./mm ²
Allowable Stress, Operating	114 N./mm ²
Allowable Stress, Hydrotest	225 N./mm ²
Material Density	0.00775 kg./cm ³
P Number Thickness	32.004 mm.
Yield Stress, Operating	230.08 N./mm ²
UCS-66 Chart Curve Designation	A
External Pressure Chart Name	CS-2
UNS Number	K02600
Product Form	Plate; sheet
Efficiency, Longitudinal Seam	0.7
Efficiency, Head-to-Skirt or Circ. Seam	0.7
Weld is pre-Heated	No

Element#	2/9
Element From Node	20
Element To Node	30
Element Type	Skirt Support
Description	Skirt-1
Distance "FROM" to "TO"	1000 mm.
Skirt Outside Diameter	5752 mm.
Diameter of Skirt at Base	5752 mm.
Skirt Thickness	18 mm.
Internal Corrosion Allowance	1 mm.
Nominal Thickness	18 mm.
External Corrosion Allowance	0 mm.
Design Temperature Internal Pressure	82 °C
Design Temperature External Pressure	0 °C
Effective Diameter Multiplier	1.2
Material Name	SA-516 70
Allowable Stress, Ambient	138 N./mm ²
Allowable Stress, Operating	138 N./mm ²
Allowable Stress, Hydrotest	234 N./mm ²
Material Density	0.00775 kg./cm ³
P Number Thickness	32.004 mm.
Yield Stress, Operating	242.6 N./mm ²
UCS-66 Chart Curve Designation	B
External Pressure Chart Name	CS-2
UNS Number	K02700
Product Form	Plate
Efficiency, Longitudinal Seam	0.7
Efficiency, Head-to-Skirt or Circ. Seam	0.5
Weld is pre-Heated	No

Element#	3/9
Element From Node	30
Element To Node	40

Element Type	Elliptical
Description	Bottom Dish
Distance "FROM" to "TO"	50 mm.
Inside Diameter	5700 mm.
Element Thickness	27 mm.
Internal Corrosion Allowance	1.6 mm.
Nominal Thickness	32 mm.
External Corrosion Allowance	0 mm.
Design Internal Pressure	12 bars
Design Temperature Internal Pressure	82 °C
Design External Pressure	0 bars
Design Temperature External Pressure	0 °C
Effective Diameter Multiplier	1.2
Material Name	SA-516 70
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Elliptical Head Factor	2.0
Weld is pre-Heated	No
Element From Node	30
Detail Type	Platform
Detail ID	PLAT:[1 OF 1]
Dist. from "FROM" Node / Offset dist	0 mm.
Platform Start Angle (degrees)	360.0
Platform End Angle (degrees)	0.0
Platform Wind Area	30000 cm ²
Platform Weight	6061.5 Kg
Platform Railing Weight	0 Kg/mm.
Platform Grating Weight	0.017 Kgs/cm ²
Platform Width	1500 mm.
Platform Height	1000 mm.
Platform Clearance or End Offset	85 mm.
Platform Force Coefficient	1.0
Ladder Layout Angle	90.0
Ladder Start Elevation	0 mm.
Ladder End Elevation	3000 mm.
Unit Weight of Ladder	0.037 Kg/mm.
Platform Length (top head platform)	0 mm.
Element From Node	30
Detail Type	Liquid
Detail ID	Liquid: 10
Dist. from "FROM" Node / Offset dist	-1376.5 mm.
Height/Length of Liquid	1426.5 mm.
Liquid Density	0.12979E-05 kg./cm ³
Element From Node	30
Detail Type	Nozzle
Detail ID	N2-3"
Dist. from "FROM" Node / Offset dist	0 mm.
Nozzle Diameter	3 in.
Nozzle Schedule	160
Nozzle Class	300
Layout Angle	0.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	0 Kg
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-106 B
Element From Node	30
Detail Type	Weight
Detail ID	tailing lug bea
Dist. from "FROM" Node / Offset dist	0 mm.
Miscellaneous Weight	900 Kg
Offset from Element Centerline	0 mm.



Element From Node	40
Element To Node	50
Element Type	Cylinder
Description	Shell-1
Distance "FROM" to "TO"	3000 mm.
Inside Diameter	5700 mm.
Element Thickness	32 mm.
Internal Corrosion Allowance	1.6 mm.
Nominal Thickness	32 mm.
External Corrosion Allowance	0 mm.
Design Internal Pressure	12 bars
Design Temperature Internal Pressure	82 °C
Design External Pressure	0 bars
Design Temperature External Pressure	0 °C
Effective Diameter Multiplier	1.2
Material Name	SA-516 70
Efficiency, Longitudinal Seam	0.85
Efficiency, Circumferential Seam	0.85
Weld is pre-Heated	No

Element From Node	40
Detail Type	Liquid
Detail ID	Liquid: 20
Dist. from "FROM" Node / Offset dist	0 mm.
Height/Length of Liquid	3000 mm.
Liquid Density	0.12980E-05 kg./cm ³

Element From Node	40
Detail Type	Nozzle
Detail ID	K1B-2"
Dist. from "FROM" Node / Offset dist	600 mm.
Nozzle Diameter	2 in.
Nozzle Schedule	None
Nozzle Class	300
Layout Angle	120.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	0 Kg
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-105

Element From Node	40
Detail Type	Nozzle
Detail ID	N6-2"
Dist. from "FROM" Node / Offset dist	1450 mm.
Nozzle Diameter	2 in.
Nozzle Schedule	None
Nozzle Class	300
Layout Angle	270.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	0 Kg
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-105

Element From Node	40
Detail Type	Nozzle
Detail ID	K2B-2"
Dist. from "FROM" Node / Offset dist	600 mm.
Nozzle Diameter	2 in.
Nozzle Schedule	None
Nozzle Class	300
Layout Angle	90.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	0 Kg
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-105

Element From Node	40
Detail Type	Nozzle
Detail ID	K1A-2"



Dist. from "FROM" Node / Offset dist	1450	mm.
Nozzle Diameter	2	in.
Nozzle Schedule	None	
Nozzle Class	300	
Layout Angle	90.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	0	Kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-105	

Element From Node	40	
Detail Type	Nozzle	
Detail ID	K2A-2"	
Dist. from "FROM" Node / Offset dist	1450	mm.
Nozzle Diameter	2	in.
Nozzle Schedule	None	
Nozzle Class	300	
Layout Angle	120.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	0	Kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-105	

Element#	5/9	
Element From Node	50	
Element To Node	60	
Element Type	Cylinder	
Description	Shell2	
Distance "FROM" to "TO"	3000	mm.
Inside Diameter	5700	mm.
Element Thickness	32	mm.
Internal Corrosion Allowance	1.6	mm.
Nominal Thickness	32	mm.
External Corrosion Allowance	0	mm.
Design Internal Pressure	12	bars
Design Temperature Internal Pressure	82	°C
Design External Pressure	0	bars
Design Temperature External Pressure	0	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-516 70	
Efficiency, Longitudinal Seam	0.85	
Efficiency, Circumferential Seam	0.85	
Weld is pre-Heated	No	

Element From Node	50	
Detail Type	Platform	
Detail ID	PLAT:[1 OF 1]	
Dist. from "FROM" Node / Offset dist	1000	mm.
Platform Start Angle (degrees)	360.0	
Platform End Angle (degrees)	180.0	
Platform Wind Area	30000	cm ²
Platform Weight	3059.1	Kgf
Platform Railing Weight	0	Kgf/mm.
Platform Grating Weight	0.017	Kgs/cm ²
Platform Width	1500	mm.
Platform Height	1000	mm.
Platform Clearance or End Offset	0	mm.
Platform Force Coefficient	1.0	
Ladder Layout Angle	180.0	
Ladder Start Elevation	0	mm.
Ladder End Elevation	4000	mm.
Unit Weight of Ladder	0.037	Kgf/mm.
Platform Length (top head platform)	0	mm.

Element From Node	50	
Detail Type	Liquid	
Detail ID	Liquid: 30	



Dist. from "FROM" Node / Offset dist	0 mm.
Height/Length of Liquid	3000 mm.
Liquid Density	0.12979E-05 kg./cm ³

Element#	6/9
Element From Node	60
Element To Node	70
Element Type	Cylinder
Description	Shell3
Distance "FROM" to "TO"	3000 mm.
Inside Diameter	5700 mm.
Element Thickness	32 mm.
Internal Corrosion Allowance	1.6 mm.
Nominal Thickness	32 mm.
External Corrosion Allowance	0 mm.
Design Internal Pressure	12 bars
Design Temperature Internal Pressure	82 °C
Design External Pressure	0 bars
Design Temperature External Pressure	0 °C
Effective Diameter Multiplier	1.2
Material Name	SA-516 70
Efficiency, Longitudinal Seam	0.85
Efficiency, Circumferential Seam	0.85
Weld is pre-Heated	No

Element From Node	60
Detail Type	Platform
Detail ID	PLAT:[1 OF 1]
Dist. from "FROM" Node / Offset dist	2000 mm.
Platform Start Angle (degrees)	180.0
Platform End Angle (degrees)	0.0
Platform Wind Area	30000 cm ²
Platform Weight	3059.1 Kg
Platform Railing Weight	0 Kg/mm.
Platform Grating Weight	0.017 Kgs/cm ²
Platform Width	1500 mm.
Platform Height	1000 mm.
Platform Clearance or End Offset	0 mm.
Platform Force Coefficient	1.0
Ladder Layout Angle	180.0
Ladder Start Elevation	0 mm.
Ladder End Elevation	4000 mm.
Unit Weight of Ladder	0.037 Kg/mm.
Platform Length (top head platform)	0 mm.

Element From Node	60
Detail Type	Liquid
Detail ID	Liquid: 40
Dist. from "FROM" Node / Offset dist	0 mm.
Height/Length of Liquid	3000 mm.
Liquid Density	0.12979E-05 kg./cm ³

Element From Node	60
Detail Type	Nozzle
Detail ID	M1-24"
Dist. from "FROM" Node / Offset dist	2500 mm.
Nozzle Diameter	24 in.
Nozzle Schedule	None
Nozzle Class	150
Layout Angle	90.0
Blind Flange (Y/N)	Y
Weight of Nozzle (Used if > 0)	0 Kg
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-516 70

Element From Node	60
Detail Type	Nozzle



Detail ID	M2-24"
Dist. from "FROM" Node / Offset dist	1500 mm.
Nozzle Diameter	24 in.
Nozzle Schedule	None
Nozzle Class	150
Layout Angle	220.0
Blind Flange (Y/N)	Y
Weight of Nozzle (Used if > 0)	566.71 Kg
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-516 70

Element From Node	60
Detail Type	Weight
Detail ID	Davit-1
Dist. from "FROM" Node / Offset dist	1500 mm.
Miscellaneous Weight	50 Kg
Offset from Element Centerline	3500 mm.

Element From Node	60
Detail Type	Weight
Detail ID	Davit-2
Dist. from "FROM" Node / Offset dist	2500 mm.
Miscellaneous Weight	50 Kg
Offset from Element Centerline	3500 mm.

Element#	7/9
Element From Node	70
Element To Node	80
Element Type	Cylinder
Description	shell4
Distance "FROM" to "TO"	3000 mm.
Inside Diameter	5700 mm.
Element Thickness	32 mm.
Internal Corrosion Allowance	1.6 mm.
Nominal Thickness	32 mm.
External Corrosion Allowance	0 mm.
Design Internal Pressure	12 bars
Design Temperature Internal Pressure	82 °C
Design External Pressure	0 bars
Design Temperature External Pressure	0 °C
Effective Diameter Multiplier	1.2
Material Name	SA-516 70
Efficiency, Longitudinal Seam	0.85
Efficiency, Circumferential Seam	0.85
Weld is pre-Heated	No

Element From Node	70
Detail Type	Liquid
Detail ID	Liquid: 50
Dist. from "FROM" Node / Offset dist	0 mm.
Height/Length of Liquid	3000 mm.
Liquid Density	0.12980E-05 kg./cm ³

Element From Node	70
Detail Type	Nozzle
Detail ID	K5A-2"
Dist. from "FROM" Node / Offset dist	1950 mm.
Nozzle Diameter	2 in.
Nozzle Schedule	None
Nozzle Class	300
Layout Angle	120.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	0 Kg
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-105

Element From Node	70
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Detail Type	Nozzle
Detail ID	K5B-2"
Dist. from "FROM" Node / Offset dist	1950 mm.
Nozzle Diameter	2 in.
Nozzle Schedule	None
Nozzle Class	300
Layout Angle	90.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	0 Kgf
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-105

Element From Node	70
Detail Type	Nozzle
Detail ID	K3-2"
Dist. from "FROM" Node / Offset dist	1650 mm.
Nozzle Diameter	2 in.
Nozzle Schedule	None
Nozzle Class	300
Layout Angle	90.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	0 Kgf
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-105

Element#	8/9
Element From Node	80
Element To Node	90
Element Type	Cylinder
Description	shell15
Distance "FROM" to "TO"	2400 mm.
Inside Diameter	5700 mm.
Element Thickness	32 mm.
Internal Corrosion Allowance	1.6 mm.
Nominal Thickness	32 mm.
External Corrosion Allowance	0 mm.
Design Internal Pressure	12 bars
Design Temperature Internal Pressure	82 °C
Design External Pressure	0 bars
Design Temperature External Pressure	0 °C
Effective Diameter Multiplier	1.2
Material Name	SA-516 70
Efficiency, Longitudinal Seam	0.85
Efficiency, Circumferential Seam	0.85
Weld is pre-Heated	No

Element From Node	80
Detail Type	Platform
Detail ID	PLAT:[1 OF 1]
Dist. from "FROM" Node / Offset dist	0 mm.
Platform Start Angle (degrees)	360.0
Platform End Angle (degrees)	180.0
Platform Wind Area	30000 cm ²
Platform Weight	3096.1 Kgf
Platform Railing Weight	0 Kgf/mm.
Platform Grating Weight	0.017 Kgs/cm ²
Platform Width	1500 mm.
Platform Height	1000 mm.
Platform Clearance or End Offset	0 mm.
Platform Force Coefficient	1.0
Ladder Layout Angle	180.0
Ladder Start Elevation	0 mm.
Ladder End Elevation	5000 mm.
Unit Weight of Ladder	0.037 Kgf/mm.
Platform Length (top head platform)	0 mm.

Element From Node	80
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Detail Type	Liquid
Detail ID	Liquid: 60
Dist. from "FROM" Node / Offset dist	0 mm.
Height/Length of Liquid	2400 mm.
Liquid Density	0.12980E-05 kg./cm ³

Element From Node	80
Detail Type	Nozzle
Detail ID	N1-14"
Dist. from "FROM" Node / Offset dist	500 mm.
Nozzle Diameter	14 in.
Nozzle Schedule	None
Nozzle Class	150
Layout Angle	270.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	0 Kgf
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-516 70

Element From Node	80
Detail Type	Nozzle
Detail ID	K4-2"
Dist. from "FROM" Node / Offset dist	1850 mm.
Nozzle Diameter	2 in.
Nozzle Schedule	None
Nozzle Class	300
Layout Angle	270.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	0 Kgf
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-105

Element From Node	80
Detail Type	Weight
Detail ID	Desmister
Dist. from "FROM" Node / Offset dist	-0.001161 mm.
Miscellaneous Weight	1375 Kgf
Offset from Element Centerline	0 mm.

Element From Node	80
Detail Type	Weight
Detail ID	Feed line
Dist. from "FROM" Node / Offset dist	500 mm.
Miscellaneous Weight	4070 Kgf
Offset from Element Centerline	4056 mm.

Element From Node	80
Detail Type	Weight
Detail ID	Donut Ring
Dist. from "FROM" Node / Offset dist	0 mm.
Miscellaneous Weight	220 Kgf
Offset from Element Centerline	0 mm.

Element#	9/9
Element From Node	90
Element To Node	100
Element Type	Elliptical
Description	Top dish
Distance "FROM" to "TO"	50 mm.
Inside Diameter	5700 mm.
Element Thickness	27 mm.
Internal Corrosion Allowance	1.6 mm.
Nominal Thickness	32 mm.
External Corrosion Allowance	0 mm.
Design Internal Pressure	12 bars
Design Temperature Internal Pressure	82 °C
Design External Pressure	0 bars



Design Temperature External Pressure 0 °C
Effective Diameter Multiplier 1.2
Material Name SA-516 70
Efficiency, Longitudinal Seam 1.0
Efficiency, Circumferential Seam 1.0
Elliptical Head Factor 2.0
Weld is pre-Heated No

Element From Node 90
Detail Type Platform
Detail ID SQR PLT
Dist. from "FROM" Node / Offset dist 1500 mm.
Platform Start Angle (degrees) 0.0
Platform End Angle (degrees) 0.0
Platform Wind Area 60000 cm²
Platform Weight 6227.5 Kg
Platform Railing Weight 0 Kg/mm.
Platform Grating Weight 0.017 Kgs/cm²
Platform Width 6000 mm.
Platform Height 1000 mm.
Platform Clearance or End Offset 3000 mm.
Platform Force Coefficient 1.0
Ladder Layout Angle 0.0
Ladder Start Elevation 180 mm.
Ladder End Elevation 3000 mm.
Unit Weight of Ladder 0.037 Kg/mm.
Platform Length (top head platform) 6000 mm.

Element From Node 90
Detail Type Liquid
Detail ID Liquid: 70
Dist. from "FROM" Node / Offset dist 0 mm.
Height/Length of Liquid 1475 mm.
Liquid Density 0.12979E-05 kg./cm³

Element From Node 90
Detail Type Nozzle
Detail ID N3-14"
Dist. from "FROM" Node / Offset dist 0 mm.
Nozzle Diameter 14 in.
Nozzle Schedule None
Nozzle Class 150
Layout Angle 0.0
Blind Flange (Y/N) N
Weight of Nozzle (Used if > 0) 0 Kg
Grade of Attached Flange GR 1.1
Nozzle Matl SA-516 70

Element From Node 90
Detail Type Nozzle
Detail ID N4-6"
Dist. from "FROM" Node / Offset dist 1500 mm.
Nozzle Diameter 6 in.
Nozzle Schedule 120
Nozzle Class 150
Layout Angle 90.0
Blind Flange (Y/N) N
Weight of Nozzle (Used if > 0) 0 Kg
Grade of Attached Flange GR 1.1
Nozzle Matl SA-106 B

Element From Node 90
Detail Type Nozzle
Detail ID N5-4"
Dist. from "FROM" Node / Offset dist 1600 mm.
Nozzle Diameter 4 in.
Nozzle Schedule 160
Nozzle Class 150
Layout Angle 270.0



Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	0	Kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-106 B	
Element From Node	90	
Detail Type	Weight	
Detail ID	Clip wt	
Dist. from "FROM" Node / Offset dist	0	mm.
Miscellaneous Weight	1550	Kgf
Offset from Element Centerline	0	mm.
Element From Node	90	
Detail Type	Weight	
Detail ID	N3-Over head	
Dist. from "FROM" Node / Offset dist	0	mm.
Miscellaneous Weight	3631	Kgf
Offset from Element Centerline	4056	mm.
Element From Node	90	
Detail Type	Weight	
Detail ID	PSV-6"	
Dist. from "FROM" Node / Offset dist	1450	mm.
Miscellaneous Weight	2500	Kgf
Offset from Element Centerline	3000	mm.
Element From Node	90	
Detail Type	Weight	
Detail ID	PSV2-6"	
Dist. from "FROM" Node / Offset dist	1450	mm.
Miscellaneous Weight	2500	Kgf
Offset from Element Centerline	3000	mm.



XY Coordinate Calculations:

From	To	X (Horiz.) mm.	Y (Vert.) mm.	DX (Horiz.) mm.	DY (Vert.) mm.
Skirt2	...	1700	1700	...	1700
Skirt-1	...	2700	1000	...	1000
Bottom Dish	...	2750	50	...	50
Shell-1	...	5750	3000	...	3000
Shell2	...	8750	3000	...	3000
Shell3	...	11750	3000	...	3000
shell4	...	14750	3000	...	3000
shell5	...	17150	2400	...	2400
Top dish	...	17200	50	...	50

The calculated dimensions are based on the given element lengths. Due to variability in manufacturing (weld gaps etc.), the Tangent to Tangent length may not be exact. The length includes gasket thicknesses and raised face heights.

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Internal Pressure Results Summary:

Element Thickness, Pressure, Diameter and Allowable Stress :

From	To	Int. Press + Liq. Hd bars	Nominal Thickness mm.	Total Corr Allowance mm.	Element Diameter mm.	Allowable Stress(SE) N./mm ²	Element#
Skirt2	...	...	22	1	5756	...	1
Skirt-1	...	...	18	1	5752	...	2
Bottom Dish	12.002	12.002	32	1.6	5700	138	3
Shell-1	12.002	12.002	32	1.6	5700	117.3	4
Shell2	12.002	12.002	32	1.6	5700	117.3	5
Shell3	12.001	12.001	32	1.6	5700	117.3	6
shell4	12.001	12.001	32	1.6	5700	117.3	7
shell5	12	12	32	1.6	5700	117.3	8
Top dish	12	12	32	1.6	5700	138	9

Element Required Thickness and MAWP :

From	To	Design Pressure bars	M.A.W.P. Corroded bars	M.A.P. New & Cold bars	Minimum Thickness mm.	Required Thickness mm.	Element#
Skirt2	...	...	No Calc	No Calc	22	No Calc	1
Skirt-1	...	...	No Calc	No Calc	18	No Calc	2
Bottom Dish	12	12.2874	12.2874	13.0605	27	26.4056	3
Shell-1	12	12.4228	12.4228	13.0816	32	30.9592	4
Shell2	12	12.4231	12.4231	13.0816	32	30.9583	5
Shell3	12	12.4235	12.4235	13.0816	32	30.9574	6
shell4	12	12.4239	12.4239	13.0816	32	30.9564	7
shell5	12	12.4243	12.4243	13.0816	32	30.9555	8
Top dish	12	12.2894	12.2894	13.0605	27	26.4014	9

Minimum

12

13

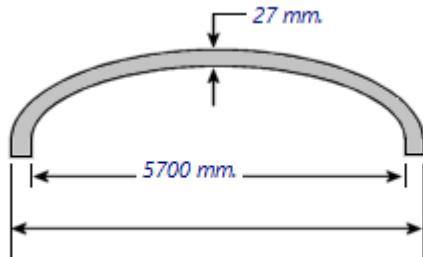
MAWP: 12 bars, limited by: Bottom Dish.

Elements Suitable for Design Internal Pressure.

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Internal Pressure Calculation Results:

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Internal Pressure Results for: Bottom Dish

Elliptical Head Design Information:					
Design Pressure	P	12.002 bars	Int. Design Temperature		82 °C
Material		SA-516 70	Ext. Design Temperature		0 °C
Length		50.000 mm.	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	1.6000 mm.	Allowable Stress (ope)	S	138.0 N./mm ²
Ext. Corr. All.	ce	0.0000 mm.	Allowable Stress (amb)	Sa	138.0 N./mm ²
Inside Diameter	Di	5700.000 mm.	Actual Stress		134.8 N./mm ²
Spec. Min. Thk t _e , or s _a		27.0000 mm.	Spec. Nominal Thk.		32.000 mm.
Surface Area		369194.1 cm ²	SG of Contents		0.001
Element Volume		25522158.0 cm ³	Weight of Contents		33.1 Kg _f
Empty Weight		9975.7 Kg _f	Operating Weight		10008.8 Kg _f
			Comp. All. Ope. Stress		105.9 N./mm ²
			Aspect Ratio		2.00

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 71.250 mm.
Head inner surface maximum deviation inside the specified shape, 0.625% of D: 35.625 mm.

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot D \cdot K_{cor}) / (2 \cdot S \cdot E - 0.2 \cdot P) \text{ Appendix 1-4(c)}$$

$$= (12.002 \cdot 5703.2002 \cdot 0.999) / (2 \cdot 138.0 \cdot 1.0 - 0.2 \cdot 12.002)$$

$$= 24.8056 + 1.6000 = 26.4056 \text{ mm.}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.002 bars

$$= (2 \cdot S \cdot E \cdot t) / (K_{cor} \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 138.0 \cdot 1.0 \cdot 25.4) / (0.999 \cdot 5703.2002 + 0.2 \cdot 25.4)$$

$$= 12 - 0 = 12 \text{ bars}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (2 \cdot S \cdot E \cdot t) / (K \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 138.0 \cdot 1.0 \cdot 27.0) / (1.0 \cdot 5700.0 + 0.2 \cdot 27.0)$$

$$= 13 \text{ bars}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (K_{cor} \cdot D + 0.2 \cdot t)) / (2 \cdot E \cdot t)$$

$$= (12.002 \cdot (0.999 \cdot 5703.2002 + 0.2 \cdot 25.4)) / (2 \cdot 1.0 \cdot 25.4)$$

$$= 134.773 \text{ N./mm}^2$$

Straight Flange Required Thickness:

$$\begin{aligned} &= (P \cdot R) / (S \cdot E - 0.6 \cdot P) + c \quad \text{per UG-27 (c)(1)} \\ &= (12.002 \cdot 2851.6001) / (138.0 \cdot 1.0 - 0.6 \cdot 12.002) + 1.6 \\ &= 26.533 \text{ mm.} \end{aligned}$$

Straight Flange Maximum Allowable Working Pressure:

Less Operating Hydrostatic Head Pressure of 0.002 bars

$$\begin{aligned} &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \quad \text{per UG-27 (c)(1)} \\ &= (138.0 \cdot 1.0 \cdot 30.4) / (2851.6001 + 0.6 \cdot 30.4) \\ &= 15 - 0 = 15 \text{ bars} \end{aligned}$$

Factor K, corroded condition [Kcor]:

$$\begin{aligned} &= (2 + (\text{Inside Diameter} / (2 \cdot \text{Inside Head Depth})))^2 / 6 \\ &= (2 + (5703.2 / (2 \cdot 1426.6)))^2 / 6 \\ &= 0.999252 \end{aligned}$$

Percent Elong. per UCS-79, VIII-1-01-57 $(75 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$ 2.437 %

MDMT Calculations in the Knuckle Portion:

Govrn. thk, $t_g = 27.0$, $t_r = 25.4$, $c = 1.6$ mm., $E^* = 1.0$
Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 1.$, Temp. Reduction = 0 °C

Min Metal Temp. w/o impact per UCS-66, Curve D -33 °C

MDMT Calculations in the Head Straight Flange:

Govrn. thk, $t_g = 32.0$, $t_r = 25.533$, $c = 1.6$ mm., $E^* = 1.0$
Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.84$, Temp. Reduction = 9 °C

Min Metal Temp. w/o impact per UCS-66, Curve D -29 °C

Min Metal Temp. at Required thickness (UCS 66.1) -38 °C

MDMT Design Information:			
MDMT Curve	D	Press. at MDMT	12.290 bars
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-1 °C
Governing Thk.	27.000 mm.	Basic MDMT	-33 °C
Stress/Thk. Ratio	1.000	Temperature Difference	0 °C
UCS-66(c) red.	No	Computed Min. Temp.	-33 °C
		Joint Efficiency E^*	100 %

Note: Temperature values displayed above are for the location with the highest basic MDMT

Tolerance for Formed Heads per UG-81(a):

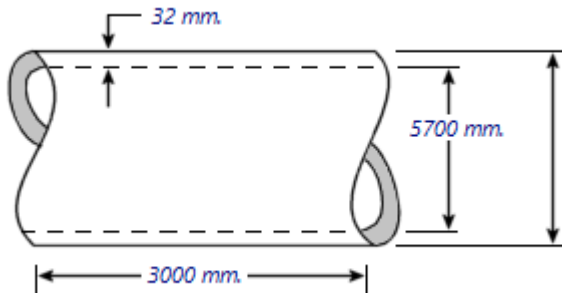
Head inner surface maximum deviation outside the specified shape, 1.25% of D: 71.250 mm.

Head inner surface maximum deviation inside the specified shape, 0.625% of D: 35.625 mm.

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Internal Pressure Calculation Results:

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Internal Pressure Results for: Shell-1

Cylindrical Shell Design Information:			
Design Pressure P	12.002 bars	Int. Design Temperature	82 °C
Material	SA-516 70	Ext. Design Temperature	0 °C
Length	3000.000 mm.	External Pressure Chart	CS-2
		UNS Number	K02700
Int. Corr. All. c	1.6000 mm.	Allowable Stress (ope) S	138.0 N./mm ²
Ext. Corr. All. ce	0.0000 mm.	Allowable Stress (amb) Sa	138.0 N./mm ²
Inside Diameter Di	5700.000 mm.	Actual Stress	133.3 N./mm ²
Spec. Min. Thk t _e , or sa	32.0000 mm.	Spec. Nominal Thk.	32.000 mm.
Surface Area	543244.2 cm ²	SG of Contents	0.001
Element Volume	76566472.0 cm ³	Weight of Contents	99.5 Kg
Empty Weight	13397.9 Kg	Operating Weight	13497.4 Kg
		Comp. All. Ope. Stress	112.0 N./mm ²

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	85 %	Long. Joint Eff. E or v	85 %

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (S \cdot E \cdot 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (12.002 \cdot 2851.6001) / (138.0 \cdot 0.85 \cdot 0.6 \cdot 12.002)$$

$$= 29.3592 + 1.6000 = 30.9592 \text{ mm.}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.002 bars

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 0.85 \cdot 30.4) / (2851.6001 + 0.6 \cdot 30.4)$$

$$= 12 - 0 = 12 \text{ bars}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 0.85 \cdot 32.0) / (2850.0 + 0.6 \cdot 32.0)$$

$$= 13 \text{ bars}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t)$$

$$= (12.002 \cdot (2851.6001 + 0.6 \cdot 30.4)) / (0.85 \cdot 30.4)$$

$$= 133.305 \text{ N./mm}^2$$

% Elongation per Table UG-79-1 (50 * t_{nom} / R_f * (1 - R_f / R_o)) 0.558 %

Minimum Design Metal Temperature Results:



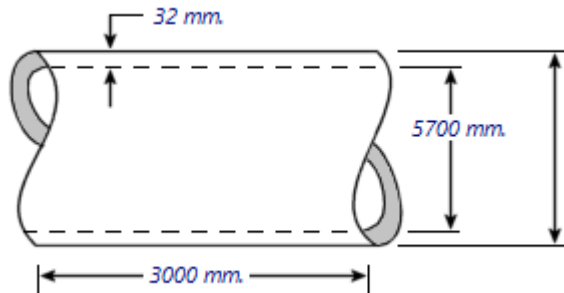
Govrn. thk, tg = 32.0, tr = 30.067, c = 1.6 mm., $E^* = 0.85$
 Thickness Ratio = $tr * E^* / (tg - c) = 0.841$, Temp. Reduction = 9 °C

Min Metal Temp. w/o impact per UCS-66, Curve D -29 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -38 °C

MDMT Design Information:			
MDMT Curve	D	Press. at MDMT	12.289 bars
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-1 °C
Governing Thk.	32.000 mm.	Basic MDMT	-29 °C
Stress/Thk. Ratio	0.841	Temperature Difference	9 °C
UCS-66(c) red.	No	Computed Min. Temp.	-38 °C
		Joint Efficiency E^*	85 %

Internal Pressure Calculation Results:

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Internal Pressure Results for: Shell2

Cylindrical Shell Design Information:			
Design Pressure P	12.002 bars	Int. Design Temperature	82 °C
Material	SA-516 70	Ext. Design Temperature	0 °C
Length	3000.000 mm.	External Pressure Chart	CS-2
		UNS Number	K02700
Int. Corr. All. c	1.6000 mm.	Allowable Stress (ope) S	138.0 N./mm ²
Ext. Corr. All. ce	0.0000 mm.	Allowable Stress (amb) Sa	138.0 N./mm ²
Inside Diameter Di	5700.000 mm.	Actual Stress	133.3 N./mm ²
Spec. Min. Thk t,e, or sa	32.0000 mm.	Spec. Nominal Thk.	32.000 mm.
Surface Area	543244.2 cm ²	SG of Contents	0.001
Element Volume	76566472.0 cm ³	Weight of Contents	99.5 Kg
Empty Weight	13397.9 Kg	Operating Weight	13497.4 Kg
		Comp. All. Ope. Stress	112.0 N./mm ²

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	85 %	Long. Joint Eff. E or v	85 %

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (12.002 \cdot 2851.6001) / (138.0 - 0.6 \cdot 12.002)$$

$$= 29.3583 + 1.6000 = 30.9583 \text{ mm.}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.002 bars

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 0.85 \cdot 30.4) / (2851.6001 + 0.6 \cdot 30.4)$$

$$= 12 - 0 = 12 \text{ bars}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 0.85 \cdot 32.0) / (2850.0 + 0.6 \cdot 32.0)$$

$$= 13 \text{ bars}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t)$$

$$= (12.002 \cdot (2851.6001 + 0.6 \cdot 30.4)) / (0.85 \cdot 30.4)$$

$$= 133.300 \text{ N./mm}^2$$

% Elongation per Table UG-79-1 (50 * tnom / Rf * (1 - Rf / Ro)) 0.558 %

Minimum Design Metal Temperature Results:



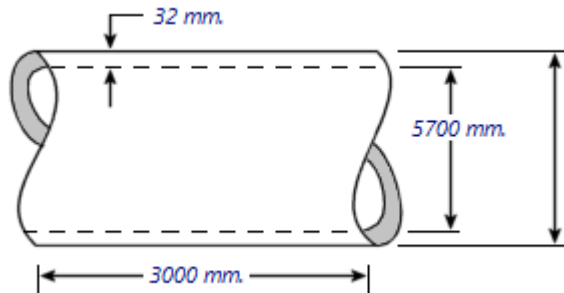
Govrn. thk, tg = 32.0, tr = 30.066, c = 1.6 mm., E* = 0.85
 Thickness Ratio = $tr * E^* / (tg - c) = 0.841$, Temp. Reduction = 9 °C

Min Metal Temp. w/o impact per UCS-66, Curve D -29 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -38 °C

MDMT Design Information:			
MDMT Curve	D	Press. at MDMT	12.289 bars
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-1 °C
Governing Thk.	32.000 mm.	Basic MDMT	-29 °C
Stress/Thk. Ratio	0.841	Temperature Difference	9 °C
UCS-66(c) red.	No	Computed Min. Temp.	-38 °C
		Joint Efficiency E*	85 %

Internal Pressure Calculation Results:

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Internal Pressure Results for: Shell3

Cylindrical Shell Design Information:					
Design Pressure	P	12.001 bars	Int. Design Temperature		82 °C
Material		SA-516 70	Ext. Design Temperature		0 °C
Length		3000.000 mm.	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	1.6000 mm.	Allowable Stress (ope)	S	138.0 N./mm ²
Ext. Corr. All.	ce	0.0000 mm.	Allowable Stress (amb)	Sa	138.0 N./mm ²
Inside Diameter	Di	5700.000 mm.	Actual Stress		133.3 N./mm ²
Spec. Min. Thk t,e, or sa		32.0000 mm.	Spec. Nominal Thk.		32.000 mm.
Surface Area		543244.2 cm ²	SG of Contents		0.001
Element Volume		76566472.0 cm ³	Weight of Contents		99.5 Kg _f
Empty Weight		13397.9 Kg _f	Operating Weight		13497.4 Kg _f
			Comp. All. Ope. Stress		112.0 N./mm ²

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	85 %	Long. Joint Eff.	E or v 85 %

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (12.001 \cdot 2851.6001) / (138.0 - 0.6 \cdot 12.001)$$

$$= 29.3574 + 1.6000 = 30.9574 \text{ mm.}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.001 bars

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 0.85 \cdot 30.4) / (2851.6001 + 0.6 \cdot 30.4)$$

$$= 12 - 0 = 12 \text{ bars}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 0.85 \cdot 32.0) / (2850.0 + 0.6 \cdot 32.0)$$

$$= 13 \text{ bars}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t)$$

$$= (12.001 \cdot (2851.6001 + 0.6 \cdot 30.4)) / (0.85 \cdot 30.4)$$

$$= 133.296 \text{ N./mm}^2$$

% Elongation per Table UG-79-1 (50 * tnom / Rf * (1 - Rf / Ro)) 0.558 %

Minimum Design Metal Temperature Results:



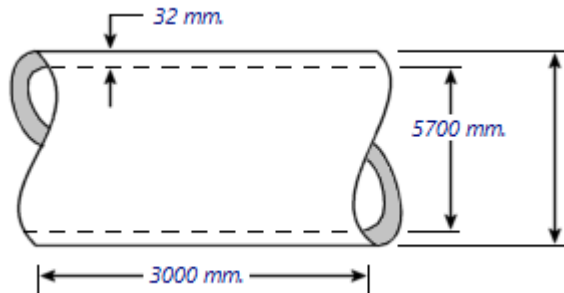
Govern. thk, $t_g = 32.0$, $t_r = 30.065$, $c = 1.6$ mm., $E^* = 0.85$
Thickness Ratio = $t_r * E^* / (t_g - c) = 0.841$, Temp. Reduction = $9\text{ }^{\circ}\text{C}$

Min Metal Temp. w/o impact per UCS-66, Curve D -29 $^{\circ}\text{C}$
Min Metal Temp. at Required thickness (UCS 66.1) -38 $^{\circ}\text{C}$

MDMT Design Information:			
MDMT Curve	D	Press. at MDMT	12.289 bars
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-1 $^{\circ}\text{C}$
Governing Thk.	32.000 mm.	Basic MDMT	-29 $^{\circ}\text{C}$
Stress/Thk. Ratio	0.841	Temperature Difference	9 $^{\circ}\text{C}$
UCS-66(c) red.	No	Computed Min. Temp.	-38 $^{\circ}\text{C}$
		Joint Efficiency E^*	85 %

Internal Pressure Calculation Results:

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Internal Pressure Results for: shell4

Cylindrical Shell Design Information:					
Design Pressure	P	12.001 bars	Int. Design Temperature		82 °C
Material		SA-516 70	Ext. Design Temperature		0 °C
Length		3000.000 mm.	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	1.6000 mm.	Allowable Stress (ope)	S	138.0 N./mm ²
Ext. Corr. All.	ce	0.0000 mm.	Allowable Stress (amb)	Sa	138.0 N./mm ²
Inside Diameter	Di	5700.000 mm.	Actual Stress		133.3 N./mm ²
Spec. Min. Thk t,e, or sa		32.0000 mm.	Spec. Nominal Thk.		32.000 mm.
Surface Area		543244.2 cm ²	SG of Contents		0.001
Element Volume		76566472.0 cm ³	Weight of Contents		99.5 Kg
Empty Weight		13397.9 Kg	Operating Weight		13497.4 Kg
			Comp. All. Ope. Stress		112.0 N./mm ²

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	85 %	Long. Joint Eff.	E or v 85 %

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (S \cdot E \cdot 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (12.001 \cdot 2851.6001) / (138.0 \cdot 0.85 \cdot 0.6 \cdot 12.001)$$

$$= 29.3564 + 1.6000 = 30.9564 \text{ mm.}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.001 bars

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 0.85 \cdot 30.4) / (2851.6001 + 0.6 \cdot 30.4)$$

$$= 12 - 0.001 = 12 \text{ bars}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 0.85 \cdot 32.0) / (2850.0 + 0.6 \cdot 32.0)$$

$$= 13 \text{ bars}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t)$$

$$= (12.001 \cdot (2851.6001 + 0.6 \cdot 30.4)) / (0.85 \cdot 30.4)$$

$$= 133.292 \text{ N./mm}^2$$

% Elongation per Table UG-79-1 (50 * tnom / Rf * (1 - Rf / Ro)) 0.558 %

Minimum Design Metal Temperature Results:



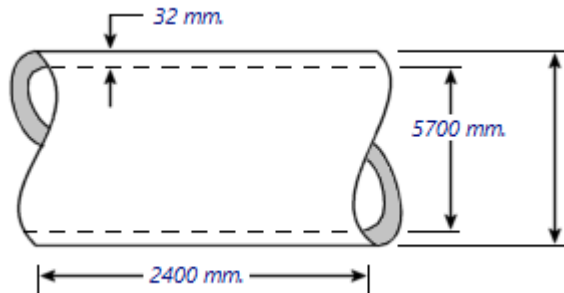
Govrn. thk, tg = 32.0, tr = 30.064, c = 1.6 mm., E* = 0.85
 Thickness Ratio = $tr * E^* / (tg - c) = 0.841$, Temp. Reduction = 9 °C

Min Metal Temp. w/o impact per UCS-66, Curve D -29 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -38 °C

MDMT Design Information:			
MDMT Curve	D	Press. at MDMT	12.288 bars
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-1 °C
Governing Thk.	32.000 mm.	Basic MDMT	-29 °C
Stress/Thk. Ratio	0.841	Temperature Difference	9 °C
UCS-66(c) red.	No	Computed Min. Temp.	-38 °C
		Joint Efficiency E*	85 %

Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: shell5

Cylindrical Shell Design Information:					
Design Pressure	P	12.000 bars	Int. Design Temperature		82 °C
Material		SA-516 70	Ext. Design Temperature		0 °C
Length		2400.000 mm.	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	1.6000 mm.	Allowable Stress (ope)	S	138.0 N./mm ²
Ext. Corr. All.	ce	0.0000 mm.	Allowable Stress (amb)	Sa	138.0 N./mm ²
Inside Diameter	Di	5700.000 mm.	Actual Stress		133.3 N./mm ²
Spec. Min. Thk t,e, or sa		32.0000 mm.	Spec. Nominal Thk.		32.000 mm.
Surface Area		434595.5 cm ²	SG of Contents		0.001
Element Volume		61253184.0 cm ³	Weight of Contents		79.6 Kg _f
Empty Weight		10718.3 Kg _f	Operating Weight		10797.9 Kg _f
			Comp. All. Ope. Stress		112.0 N./mm ²

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	85 %	Long. Joint Eff.	E or v 85 %

Required Thickness due to Internal Pressure [tr]:

$$\begin{aligned}
 &= (P \cdot R) / (S \cdot E \cdot 0.6 \cdot P) \text{ per UG-27 (c)(1)} \\
 &= (12.0 \cdot 2851.6001) / (138.0 \cdot 0.85 \cdot 0.6 \cdot 12.0) \\
 &= 29.3555 + 1.6000 = 30.9555 \text{ mm.}
 \end{aligned}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.000 bars

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (138.0 \cdot 0.85 \cdot 30.4) / (2851.6001 + 0.6 \cdot 30.4) \\
 &= 12 - 0 = 12 \text{ bars}
 \end{aligned}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (138.0 \cdot 0.85 \cdot 32.0) / (2850.0 + 0.6 \cdot 32.0) \\
 &= 13 \text{ bars}
 \end{aligned}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$\begin{aligned}
 &= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t) \\
 &= (12.0 \cdot (2851.6001 + 0.6 \cdot 30.4)) / (0.85 \cdot 30.4) \\
 &= 133.288 \text{ N./mm}^2
 \end{aligned}$$

% Elongation per Table UG-79-1 (50 * tnom / Rf * (1 - Rf / Ro)) 0.558 %

Minimum Design Metal Temperature Results:



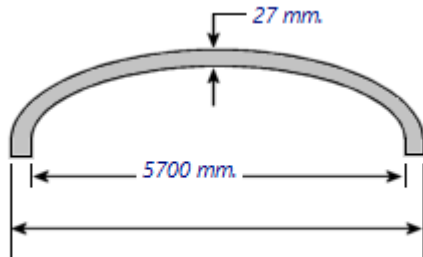
Govrn. thk, tg = 32.0, tr = 30.063, c = 1.6 mm., $E^* = 0.85$
 Thickness Ratio = $tr * E^* / (tg - c) = 0.841$, Temp. Reduction = 9 °C

Min Metal Temp. w/o impact per UCS-66, Curve D -29 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -38 °C

MDMT Design Information:			
MDMT Curve	D	Press. at MDMT	12.288 bars
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-1 °C
Governing Thk.	32.000 mm.	Basic MDMT	-29 °C
Stress/Thk. Ratio	0.841	Temperature Difference	9 °C
UCS-66(c) red.	No	Computed Min. Temp.	-38 °C
		Joint Efficiency E^*	85 %

Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: Top dish

Elliptical Head Design Information:					
Design Pressure	P	12.000 bars	Int. Design Temperature		82 °C
Material		SA-516 70	Ext. Design Temperature		0 °C
Length		50.000 mm.	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	1.6000 mm.	Allowable Stress (ope)	S	138.0 N./mm ²
Ext. Corr. All.	ce	0.0000 mm.	Allowable Stress (amb)	Sa	138.0 N./mm ²
Inside Diameter	Di	5700.000 mm.	Actual Stress		134.8 N./mm ²
Spec. Min. Thk t _e , or s _a		27.0000 mm.	Spec. Nominal Thk.		32.000 mm.
Surface Area		369194.1 cm ²	SG of Contents		0.001
Element Volume		25522158.0 cm ³	Weight of Contents		33.2 Kg _f
Empty Weight		9975.7 Kg _f	Operating Weight		10008.8 Kg _f
			Comp. All. Ope. Stress		105.9 N./mm ²
			Aspect Ratio		2.00

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 71.250 mm.
Head inner surface maximum deviation inside the specified shape, 0.625% of D: 35.625 mm.

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot D \cdot K_{cor}) / (2 \cdot S \cdot E - 0.2 \cdot P) \text{ Appendix 1-4(c)}$$

$$= (12.0 \cdot 5703.2002 \cdot 0.999) / (2 \cdot 138.0 \cdot 1.0 - 0.2 \cdot 12.0)$$

$$= 24.8014 + 1.6000 = 26.4014 \text{ mm.}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.000 bars

$$= (2 \cdot S \cdot E \cdot t) / (K_{cor} \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 138.0 \cdot 1.0 \cdot 25.4) / (0.999 \cdot 5703.2002 + 0.2 \cdot 25.4)$$

$$= 12 - 0 = 12 \text{ bars}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (2 \cdot S \cdot E \cdot t) / (K \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 138.0 \cdot 1.0 \cdot 27.0) / (1.0 \cdot 5700.0 + 0.2 \cdot 27.0)$$

$$= 13 \text{ bars}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (K_{cor} \cdot D + 0.2 \cdot t)) / (2 \cdot E \cdot t)$$

$$= (12.0 \cdot (0.999 \cdot 5703.2002 + 0.2 \cdot 25.4)) / (2 \cdot 1.0 \cdot 25.4)$$

$$= 134.750 \text{ N./mm}^2$$



Straight Flange Required Thickness:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) + c \text{ per UG-27 (c)(1)}$$

$$= (12.0 \cdot 2851.6001) / (138.0 \cdot 1.0 - 0.6 \cdot 12.0) + 1.6$$

$$= 26.528 \text{ mm.}$$

Straight Flange Maximum Allowable Working Pressure:

[Less Operating Hydrostatic Head Pressure of 0.000 bars](#)

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 30.4) / (2851.6001 + 0.6 \cdot 30.4)$$

$$= 15 - 0 = 15 \text{ bars}$$

Factor K, corroded condition [Kcor]:

$$= (2 + (\text{Inside Diameter} / (2 \cdot \text{Inside Head Depth})))^2 / 6$$

$$= (2 + (5703.2 / (2 \cdot 1426.6)))^2 / 6$$

$$= 0.999252$$

Percent Elong. per UCS-79, VIII-1-01-57 $(75 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$ 2.437 %

MDMT Calculations in the Knuckle Portion:

Govrn. thk, tg = 27.0, tr = 25.396, c = 1.6 mm., E* = 1.0
Thickness Ratio = tr * E* / (tg - c) = 1., Temp. Reduction = 0 °C

Min Metal Temp. w/o impact per UCS-66, Curve D -33 °C

MDMT Calculations in the Head Straight Flange:

Govrn. thk, tg = 32.0, tr = 25.529, c = 1.6 mm., E* = 1.0
Thickness Ratio = tr * E* / (tg - c) = 0.84, Temp. Reduction = 9 °C

Min Metal Temp. w/o impact per UCS-66, Curve D -29 °C

Min Metal Temp. at Required thickness (UCS 66.1) -38 °C

MDMT Design Information:			
MDMT Curve	D	Press. at MDMT	12.288 bars
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-1 °C
Governing Thk.	27.000 mm.	Basic MDMT	-33 °C
Stress/Thk. Ratio	1.000	Temperature Difference	0 °C
UCS-66(c) red.	No	Computed Min. Temp.	-33 °C
		Joint Efficiency E*	100 %

Note: Temperature values displayed above are for the location with the highest basic MDMT

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 71.250 mm.

Head inner surface maximum deviation inside the specified shape, 0.625% of D: 35.625 mm.

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Hydrostatic Test Pressure Results:

UG99(b) = 1.30 * M.A.W.P. * Sa/S 15.974 bars
UG99b(b)(35) = 1.30 * Design Pres * Sa/S 15.600 bars
UG99(c) = 1.30 * M.A.P. - Head(Hyd) 16.322 bars
UG100 = 1.10 * M.A.W.P. * Sa/S 13.516 bars
PED = max(1.43*DP, 1.25*DP*ratio) 17.160 bars
PED&UG99(b)(35) = max(1.43*DP, 1.3*DP*ratio) 17.160 bars
App 27-4 = M.A.W.P. 12.287 bars

User Defined Hydrostatic Test Pressure at High Point 12.750 bars

Horizontal Test performed per: User Hydro Pressure

Please note that Nozzle, Shell, Head, Flange, etc MAWPs are all considered when determining the hydrotest pressure for those test types that are based on the MAWP of the vessel.

Stresses on Elements due to Test Pressure (N./mm² & bars):

From To	Stress	Allowable	Ratio	Pressure
Bottom Dish	140.6	234.0	0.601	13.31
Shell-1	140.4	234.0	0.600	13.31
Shell2	140.4	234.0	0.600	13.31
Shell3	140.4	234.0	0.600	13.31
shell4	140.4	234.0	0.600	13.31
shell5	140.4	234.0	0.600	13.31
Top dish	140.6	234.0	0.601	13.31

Stress ratios for Nozzle and Pad Materials (N./mm²):

Description	Pad/Nozzle	Ambient	Operating	Ratio
N2-3"	Nozzle	118.00	118.00	1.000
N2-3"	Pad	138.00	138.00	1.000
K1B-2"	Nozzle	138.00	138.00	1.000
N6-2"	Nozzle	138.00	138.00	1.000
K2B-2"	Nozzle	138.00	138.00	1.000
K1A-2"	Nozzle	138.00	138.00	1.000
K2A-2"	Nozzle	138.00	138.00	1.000
M1-24"	Nozzle	138.00	138.00	1.000
M1-24"	Pad	138.00	138.00	1.000
M2-24"	Nozzle	138.00	138.00	1.000
M2-24"	Pad	138.00	138.00	1.000
K5A-2"	Nozzle	138.00	138.00	1.000
K5B-2"	Nozzle	138.00	138.00	1.000
K3-2"	Nozzle	138.00	138.00	1.000
N1-14"	Nozzle	138.00	138.00	1.000
N1-14"	Pad	138.00	138.00	1.000
K4-2"	Nozzle	138.00	138.00	1.000
N3-14"	Nozzle	138.00	138.00	1.000
N3-14"	Pad	138.00	138.00	1.000
N4-6"	Nozzle	118.00	118.00	1.000
N4-6"	Pad	138.00	138.00	1.000
N5-4"	Nozzle	118.00	118.00	1.000
N5-4"	Pad	138.00	138.00	1.000

Minimum 1.000

Stress ratios for Pressurized Vessel Elements (N./mm²):

Description	Ambient	Operating	Ratio
Bottom Dish	138.00	138.00	1.000
Shell-1	138.00	138.00	1.000
Shell2	138.00	138.00	1.000



Shell3	138.00	138.00	1.000
shell4	138.00	138.00	1.000
shell5	138.00	138.00	1.000
Top dish	138.00	138.00	1.000

Minimum 1.000

Hoop Stress in Nozzle Wall during Pressure Test (N./mm²):

Description	Ambient	Operating	Ratio
N2-3"	5.55	153.40	0.036
K1B-2"	2.81	179.40	0.016
N6-2"	2.81	179.40	0.016
K2B-2"	2.81	179.40	0.016
K1A-2"	2.81	179.40	0.016
K2A-2"	2.81	179.40	0.016
M1-24"	13.96	179.40	0.078
M2-24"	13.96	179.40	0.078
K5A-2"	2.81	179.40	0.016
K5B-2"	2.81	179.40	0.016
K3-2"	2.81	179.40	0.016
N1-14"	11.30	179.40	0.063
K4-2"	2.81	179.40	0.016
N3-14"	11.30	179.40	0.063
N4-6"	8.43	153.40	0.055
N5-4"	5.91	153.40	0.039

Elements Suitable for Test Pressure.

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Hydrostatic Test Pressure Results:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	15.974 bars
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	15.600 bars
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	16.785 bars
UG100	= 1.10 * M.A.W.P. * Sa/S	13.516 bars
PED	= max(1.43*DP, 1.25*DP*ratio)	17.160 bars
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	17.160 bars
App 27-4	= M.A.W.P.	12.287 bars

UG-99(b) Note 35, Test Pressure Calculation:
= Test Factor • Design Pressure • Stress Ratio
= 1.3 • 12.0 • 1.0
= 15.600 bars

Vertical Test performed per: UG-99b (Note 35)

Please note that Nozzle, Shell, Head, Flange, etc MAWPs are all considered when determining the hydrotest pressure for those test types that are based on the MAWP of the vessel.

Stresses on Elements due to Test Pressure (N./mm² & bars):

From To	Stress	Allowable	Ratio	Pressure
Bottom Dish	194.8	234.0	0.833	17.35
Shell-1	191.1	234.0	0.817	17.21
Shell2	187.8	234.0	0.803	16.91
Shell3	184.6	234.0	0.789	16.62
shell4	181.3	234.0	0.775	16.32
shell5	178.0	234.0	0.761	16.03
Top dish	177.3	234.0	0.758	15.79

Stress ratios for Nozzle and Pad Materials (N./mm²):

Description	Pad/Nozzle	Ambient	Operating	Ratio
N2-3"	Nozzle	118.00	118.00	1.000
N2-3"	Pad	138.00	138.00	1.000
K1B-2"	Nozzle	138.00	138.00	1.000
N6-2"	Nozzle	138.00	138.00	1.000
K2B-2"	Nozzle	138.00	138.00	1.000
K1A-2"	Nozzle	138.00	138.00	1.000
K2A-2"	Nozzle	138.00	138.00	1.000
M1-24"	Nozzle	138.00	138.00	1.000
M1-24"	Pad	138.00	138.00	1.000
M2-24"	Nozzle	138.00	138.00	1.000
M2-24"	Pad	138.00	138.00	1.000
K5A-2"	Nozzle	138.00	138.00	1.000
K5B-2"	Nozzle	138.00	138.00	1.000
K3-2"	Nozzle	138.00	138.00	1.000
N1-14"	Nozzle	138.00	138.00	1.000
N1-14"	Pad	138.00	138.00	1.000
K4-2"	Nozzle	138.00	138.00	1.000
N3-14"	Nozzle	138.00	138.00	1.000
N3-14"	Pad	138.00	138.00	1.000
N4-6"	Nozzle	118.00	118.00	1.000
N4-6"	Pad	138.00	138.00	1.000
N5-4"	Nozzle	118.00	118.00	1.000
N5-4"	Pad	138.00	138.00	1.000

Minimum 1.000

Stress ratios for Pressurized Vessel Elements (N./mm²):

Description	Ambient	Operating	Ratio
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Bottom Dish	138.00	138.00	1.000
Shell-1	138.00	138.00	1.000
Shell2	138.00	138.00	1.000
Shell3	138.00	138.00	1.000
shell4	138.00	138.00	1.000
shell5	138.00	138.00	1.000
Top dish	138.00	138.00	1.000

Minimum 1.000

Hoop Stress in Nozzle Wall during Pressure Test (N./mm²):

Description	Ambient	Operating	Ratio
N2-3"	7.23	153.40	0.047
K1B-2"	3.63	179.40	0.020
N6-2"	3.63	179.40	0.020
K2B-2"	3.63	179.40	0.020
K1A-2"	3.63	179.40	0.020
K2A-2"	3.63	179.40	0.020
M1-24"	17.43	179.40	0.097
M2-24"	17.43	179.40	0.097
K5A-2"	3.45	179.40	0.019
K5B-2"	3.45	179.40	0.019
K3-2"	3.45	179.40	0.019
N1-14"	13.61	179.40	0.076
K4-2"	3.39	179.40	0.019
N3-14"	13.41	179.40	0.075
N4-6"	10.01	153.40	0.065
N5-4"	7.02	153.40	0.046

Elements Suitable for Test Pressure.

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External Pressure Calculation Results Summary:

ASME Code, Section VIII Division 1, 2025

3

Elliptical Head From 30 to 40 Ext. Chart: CS-2 at 0 °C

Elastic Modulus from Chart: CS-2 at 0 °C: 199943392 KPa.

Results for Maximum Allowable Ext. Pressure				MAEP
Tca	Outer Dia	Do/t	Factor A	Factor B
25.400	5754.00	226.54	0.0006131	61.30

$$MAEP = B / (K0 * Do / t) = 61.2962 / (0.9 * 226.5354) = 3.0063 \text{ bars}$$

4

Cylindrical Shell From 40 to 50 Ext. Chart: CS-2 at 0 °C

Elastic Modulus from Chart: CS-2 at 0 °C: 199943392 KPa.

Results for Maximum Allowable Ext. Pressure				MAEP		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
30.400	5764.00	15450.00	189.61	2.6804	0.0001833	18.33

$$MAEP = (4 * B) / (3 * (Do / t)) = (4 * 18.3267) / (3 * 189.6053) = 1.2887 \text{ bars}$$

Results for Maximum Stiffened Length				Slen		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
30.400	5764.00	15450.00	189.61	2.6804	0.0001833	18.33

$$MAEP = (4 * B) / (3 * (Do / t)) = (4 * 18.3267) / (3 * 189.6053) = 1.2887 \text{ bars}$$

5

Cylindrical Shell From 50 to 60 Ext. Chart: CS-2 at 0 °C

Elastic Modulus from Chart: CS-2 at 0 °C: 199943392 KPa.

Results for Maximum Allowable Ext. Pressure				MAEP		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
30.400	5764.00	15450.00	189.61	2.6804	0.0001833	18.33

$$MAEP = (4 * B) / (3 * (Do / t)) = (4 * 18.3267) / (3 * 189.6053) = 1.2887 \text{ bars}$$

Results for Maximum Stiffened Length				Slen		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
30.400	5764.00	15450.00	189.61	2.6804	0.0001833	18.33

$$MAEP = (4 * B) / (3 * (Do / t)) = (4 * 18.3267) / (3 * 189.6053) = 1.2887 \text{ bars}$$

6

Cylindrical Shell From 60 to 70 Ext. Chart: CS-2 at 0 °C

Elastic Modulus from Chart: CS-2 at 0 °C: 199943392 KPa.

Results for Maximum Allowable Ext. Pressure					MAEP	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
30.400	5764.00	15450.00	189.61	2.6804	0.0001833	18.33

$$MAEP = (4*B)/(3*(Do/t)) = (4*18.3267)/(3*189.6053) = 1.2887 \text{ bars}$$

Results for Maximum Stiffened Length					Slen	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
30.400	5764.00	15450.00	189.61	2.6804	0.0001833	18.33

$$MAEP = (4*B)/(3*(Do/t)) = (4*18.3267)/(3*189.6053) = 1.2887 \text{ bars}$$

7

Cylindrical Shell From 70 to 80 Ext. Chart: CS-2 at 0 °C

Elastic Modulus from Chart: CS-2 at 0 °C: 199943392 KPa.

Results for Maximum Allowable Ext. Pressure					MAEP	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
30.400	5764.00	15450.00	189.61	2.6804	0.0001833	18.33

$$MAEP = (4*B)/(3*(Do/t)) = (4*18.3267)/(3*189.6053) = 1.2887 \text{ bars}$$

Results for Maximum Stiffened Length					Slen	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
30.400	5764.00	15450.00	189.61	2.6804	0.0001833	18.33

$$MAEP = (4*B)/(3*(Do/t)) = (4*18.3267)/(3*189.6053) = 1.2887 \text{ bars}$$

8

Cylindrical Shell From 80 to 90 Ext. Chart: CS-2 at 0 °C

Elastic Modulus from Chart: CS-2 at 0 °C: 199943392 KPa.

Results for Maximum Allowable Ext. Pressure					MAEP	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
30.400	5764.00	15450.00	189.61	2.6804	0.0001833	18.33

$$MAEP = (4*B)/(3*(Do/t)) = (4*18.3267)/(3*189.6053) = 1.2887 \text{ bars}$$

Results for Maximum Stiffened Length					Slen	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
30.400	5764.00	15450.00	189.61	2.6804	0.0001833	18.33

$$MAEP = (4*B)/(3*(Do/t)) = (4*18.3267)/(3*189.6053) = 1.2887 \text{ bars}$$



9

Elliptical Head From 90 to 100 Ext. Chart: CS-2 at 0 °C

Elastic Modulus from Chart: CS-2 at 0 °C: 199943392 KPa.

Results for Maximum Allowable Ext. Pressure				MAEP
Tca	Outer Dia	Do/t	Factor A	Factor B
25.400	5754.00	226.54	0.0006131	61.30

$$MAEP = B / (K0 * Do / t) = 61.2962 / (0.9 * 226.5354) = 3.0063 \text{ bars}$$

External Pressure Calculations:

From	To	Section Length mm.	Outside Diameter mm.	Corroded Thickness mm.	Factor A	Factor B N./mm ²
10	20	2700	...	...	No Calc	No Calc
20	30	2700	...	...	No Calc	No Calc
30	40	No Calc	5754	25.4	0.0006131	61.2962
40	50	15450	5764	30.4	0.00018331	18.3267
50	60	15450	5764	30.4	0.00018331	18.3267
60	70	15450	5764	30.4	0.00018331	18.3267
70	80	15450	5764	30.4	0.00018331	18.3267
80	90	15450	5764	30.4	0.00018331	18.3267
90	100	No Calc	5754	25.4	0.0006131	61.2962

External Pressure Calculations:

From	To	External Actual T. mm.	External Required T. mm.	External Design Pressure bars	External M.A.W.P. bars
10	20	...	No Calc	...	No Calc
20	30	...	No Calc	...	No Calc
30	40	27	No Calc	...	3.00628
40	50	32	No Calc	...	1.28868
50	60	32	No Calc	...	1.28868
60	70	32	No Calc	...	1.28868
70	80	32	No Calc	...	1.28868
80	90	32	No Calc	...	1.28868
90	100	27	No Calc	...	3.00628

Minimum

1

External Pressure Calculations:

From	To	Actual Length Bet. Stiffeners mm.	Allowable Length Bet. Stiffeners mm.	Ring Inertia Required cm ⁴	Ring Inertia Available cm ⁴
10	20	2700	No Calc	No Calc	No Calc
20	30	2700	No Calc	No Calc	No Calc
30	40	No Calc	No Calc	No Calc	No Calc
40	50	15450	No Calc	No Calc	No Calc
50	60	15450	No Calc	No Calc	No Calc
60	70	15450	No Calc	No Calc	No Calc
70	80	15450	No Calc	No Calc	No Calc
80	90	15450	No Calc	No Calc	No Calc
90	100	No Calc	No Calc	No Calc	No Calc



Elements Suitable for External Pressure.

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Element and Detail Weights:

From	To	Element Metal Wgt. kg.	Element ID Volume cm³	Corroded Metal Wgt. kg.	Corroded ID Volume cm³	Extra due Misc % kg.
10	20	8952.57	...	8716.09	...	626.68
20	30	2513.11	...	2373.91	...	175.918
30	40	9975.68	25522158	9476.89	25564454	698.297
40	50	13397.9	76566472	12731.4	76652480	937.852
50	60	13397.9	76566472	12731.4	76652480	937.852
60	70	13397.9	76566472	12731.4	76652480	937.852
70	80	13397.9	76566472	12731.4	76652480	937.852
80	90	10718.3	61253184	10185.2	61321988	750.28
90	100	9975.68	25522158	9476.89	25564454	698.297
Total		95726	0.419E+09	91154	0.419E+09	6700

Weight of Details:

From	Type	Weight of Detail kg.	X Offset, Dtl. Cent. mm.	Y Offset, Dtl. Cent. mm.	Z Offset Dtl. Cent. mm.	Description
30	Plat	6485.77	...	...	0.33080E-04	PLAT:[1 OF 1]
30	Liqd	33.1045	...	-688.25	...	Liquid: 10
30	Nozl	18.4493	...	-1561.5	...	N2-3"
30	Wght	900	...	...	...	tailing lug bea
40	Liqd	99.4818	...	1500	...	Liquid: 20
40	Nozl	11.6308	2557.54	600	1476.59	K1B-2"
40	Nozl	11.6308	-2953.19	1450	...	N6-2"
40	Nozl	11.6308	2953.19	600	...	K2B-2"
40	Nozl	11.6308	2953.19	1450	...	K1A-2"
40	Nozl	11.6308	2557.54	1450	1476.59	K2A-2"
50	Plat	3273.25	-2324.99	1000	...	PLAT:[1 OF 1]
50	Liqd	99.4757	...	1500	...	Liquid: 30
60	Plat	3273.25	2324.99	2000	...	PLAT:[1 OF 1]
60	Liqd	99.4757	...	1500	...	Liquid: 40
60	Nozl	577.844	2975.01	2500	...	M1-24"
60	Nozl	606.376	-1912.3	1500	2278.99	M2-24"
60	Wght	50	-2249.76	1500	2681.16	Davit-1
60	Wght	50	3500	2500	...	Davit-2
70	Liqd	99.4818	...	1500	...	Liquid: 50
70	Nozl	11.6308	2557.54	1950	1476.59	K5A-2"
70	Nozl	11.6308	2953.19	1950	...	K5B-2"
70	Nozl	11.6308	2953.19	1650	...	K3-2"
80	Plat	3312.84	-2324.99	...	...	PLAT:[1 OF 1]
80	Liqd	79.5855	...	1200	...	Liquid: 60
80	Nozl	130.543	-2950	500	...	N1-14"
80	Nozl	11.6308	-2953.19	1850	...	K4-2"
80	Wght	1375	...	-0.0011613	...	Desmister
80	Wght	4070	-4056	500	...	Feed line
80	Wght	220	...	...	...	Donut Ring
90	Plat	6663.42	...	1500	...	SQR PLT
90	Liqd	33.1585	...	762.5	...	Liquid: 70
90	Nozl	159.919	...	1648	...	N3-14"
90	Nozl	43.1675	1500	1453.71	...	N4-6"
90	Nozl	27.9739	-1600	1427.65	...	N5-4"
90	Wght	1550	...	...	...	Clip wt
90	Wght	3631	4056	...	...	N3-Over head
90	Wght	2500	2121.32	1450	-2121.32	PSV-6"
90	Wght	2500	2121.32	1450	2121.32	PSV2-6"
80	Trun	598.326	...	1800	...	Trunnions

Total Weight of Each Detail Type:

Platforms

23008.5



Liquid 543.8
Nozzles 1669.0
Weights 16846.0
Trunnions 598.3

Sum of the Detail Weights 42665.6 kg.

Weight Summation Results: (kg.)

	Fabricated	Shop Test	Shipping	Erected	Empty	Operating
Main Elements	102427.8	102427.8	102427.8	102427.8	102427.8	102427.8
Nozzles	1669.0	1669.0	1669.0	1669.0	1669.0	1669.0
Wld Weights	2770.0	2770.0	2770.0	2770.0	2770.0	2770.0
Trunnions	598.3	598.3	598.3	598.3	598.3	598.3
Platforms	...	...	...	23008.5	23008.5	23008.5
Empty Weights	...	...	...	14076.0	14076.0	...
Test Weights	...	12701.0	...	...	...	...
Ope Weights	...	...	...	...	...	14076.0
Ope. Liquid	...	...	...	...	...	543.8
Test Liquid	...	418307.9	...	...	...	...
Totals	107465.0	538474.0	107465.0	144549.6	144549.6	145093.3

Field Installation Options:

Platforms installed before lifting.

Miscellaneous Weight Percent: 7.0 %

Note that the above value for the miscellaneous weight percent has been applied to the shells/heads/flange/tubesheets/tubes etc. in the weight calculations for metallic components.

Weight Summary:

Fabricated Wt.	- Bare weight without removable internals	107465.0 kg.
Shop Test Wt.	- Fabricated weight + water (full)	538474.0 kg.
Shipping Wt.	- Fab. weight + removable intls.+ shipping app.	107465.0 kg.
Erected Wt.	- Fab. wt + or - loose items (trays,platforms etc.)	144549.6 kg.
Ope. Wt. no Liq	- Fab. weight + internals. + details + weights	144549.6 kg.
Operating Wt.	- Empty weight + operating liq. uncorroded	145093.3 kg.
Field Test Wt.	- Empty weight + test medium (full)	561482.5 kg.

Weights used in Vortex Shedding Calculations:

Mass of the Upper 1/3 of the Vertical Vessel (operating)	62399.6 kg.
Mass of the Upper 1/3 of the Vertical Vessel (filled w/water)	231805.5 kg.
Mass of the Upper 1/3 of the Vertical Vessel (empty)	64062.6 kg.

Surface Areas of Elements:

From	To	Outside Surface Area cm ²	Inside Surface Area cm ²
10	20	307411	305061
20	30	180704	179573
30	40	369194	361140
40	50	543244	537212
50	60	543244	537212
60	70	543244	537212
70	80	543244	537212
80	90	434595	429770
90	100	369194	361140

Total	3834076.2	3785535.2 cm ²
Total	4127.0	4074.7 ft ²

Element and Detail Weights:



From	To	Total Ele. Empty Wgt. kg.	Total. Ele. Oper. Wgt. kg.	Total. Ele. Hydro. Wgt. kg.	Total Dtl. Offset Mom. Kg-m.	Oper. Wgt. No Liquid kg.
10	20	9579.25	9579.25	9579.25	...	9579.25
20	30	2689.02	2689.02	2689.02	...	2689.02
30	40	11592.4	18111.3	43584.8	0.00021456	18078.2
40	50	14393.9	14493.4	90913.6	34.3487	14393.9
50	60	14335.7	17708.5	94128.7	...	17609
60	70	15620	18992.7	95413	1516.01	18893.2
70	80	14370.6	14470.1	90890.4	17.1744	14370.6
80	90	17874.1	21266.5	81027.7	...	21186.9
90	100	21086	27782.6	53256	...	27749.5

Cumulative Vessel Weight

From	To	Cumulative Ope Wgt. No Liquid kg.	Cumulative Oper. Wgt. kg.	Cumulative Hydro. Wgt. kg.
10	20	144550	145093	561482
20	30	134970	135514	551903
30	40	132281	132825	549214
40	50	114203	114714	505629
50	60	99809.2	100220	414716
60	70	82200.2	82511.9	320587
70	80	63307	63519.2	225174
80	90	48936.4	49049.1	134284
90	100	27749.5	27782.6	53256

The cumulative operating weights no liquid in the column above are the cumulative operating weights minus the operating liquid weight minus any weights absent in the empty condition.

Cumulative Vessel Moment

From	To	Cumulative Empty Mom. Kg-m.	Cumulative Oper. Mom. Kg-m.	Cumulative Hydro. Mom. Kg-m.
10	20	1567.54	1567.54	1567.54
20	30	1567.54	1567.54	1567.54
30	40	1567.54	1567.54	1567.54
40	50	1567.53	1567.53	1567.53
50	60	1533.19	1533.19	1533.19
60	70	1533.19	1533.19	1533.19
70	80	17.1744	17.1744	17.1744
80	90	...	...	...
90	100	...	...	...



Nozzle Flange MAWP Results: (bars & °C)

Nozzle Description	Flange Rating Ope.	Ambient	Des. Temp	Class	Grade/Group	Equiv. Press	- - - UG-44	Max Pressure 50%	DNV
N2-3"	47.82	51.00	82	300	GR 1.1	...	...	...	...
K1B-2"	47.82	51.00	82	300	GR 1.1	...	...	...	...
N6-2"	47.82	51.00	82	300	GR 1.1	...	...	...	...
K2B-2"	47.82	51.00	82	300	GR 1.1	...	...	...	...
K1A-2"	47.82	51.00	82	300	GR 1.1	...	...	...	...
K2A-2"	47.82	51.00	82	300	GR 1.1	...	...	...	...
M1-24"	18.20	19.60	82	150	GR 1.1	...	...	...	...
M2-24"	18.20	19.60	82	150	GR 1.1	...	...	...	...
K5A-2"	47.82	51.00	82	300	GR 1.1	...	...	...	...
K5B-2"	47.82	51.00	82	300	GR 1.1	...	...	...	...
K3-2"	47.82	51.00	82	300	GR 1.1	...	...	...	...
N1-14"	18.20	19.60	82	150	GR 1.1	...	...	...	...
K4-2"	47.82	51.00	82	300	GR 1.1	...	...	...	...
N3-14"	18.20	19.60	82	150	GR 1.1	...	...	...	...
N4-6"	18.20	19.60	82	150	GR 1.1	...	...	...	...
N5-4"	18.20	19.60	82	150	GR 1.1	...	...	...	...
Min Rating	18.204	19.600, [for Core Elements]					...	...	...

Flange Pressure Ratings are per ASME B16.5 2025 Metric Edition

Warning:

There are nozzles in this model, but no flange MAWPs were de-rated. Be sure this is what you intended. There is a check box in the nozzle dialog that instructs PV Elite to perform the de-rating for each nozzle flange. See ASME VIII-1, UG-44(b) for more information.

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The Natural Frequencies for the vessel have been computed iteratively by solving a system of matrices. These matrices describe the mass and the stiffness of the vessel. This is the generalized eigenvalue/eigenvector problem and is referenced in some mathematical texts.

The Natural Frequency for the Vessel (Empty.) is 8.72142 Hz.

The Natural Frequency for the Vessel (Ope...) is 8.70755 Hz.

The Natural Frequency for the Vessel (Filled) is 4.72813 Hz.

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Wind Load Calculation Results:

Input Values:

Wind Design Code	ASCE/SEI 7-16
Wind Load Reduction Scale Factor	1.100
Basic Wind Speed	97.2 Km/hr
Surface Roughness Category	C: Open Terrain
Type of Surface	Moderately Smooth
Base Elevation	300 mm.
Percent Wind for Hydrotest	50.0
Using User defined Wind Press. Vs Elev.	N
Height of Hill or Escarpment H or Hh	0 mm.
Distance Upwind of Crest Lh	0 mm.
Distance from Crest to the Vessel x	0 mm.
Type of Terrain (Hill, Escarpment)	Flat
Damping Factor (Beta) for Wind (Ope)	0.0100
Damping Factor (Beta) for Wind (Empty)	0.0000
Damping Factor (Beta) for Wind (Filled)	0.0000
Site Elevation above Sea Level	0 mm.

Wind Analysis Results

The importance factor is not used in the calculation of wind pressure vs. elevation for the selected wind code.

Static Gust-Effect Factor, Operating Case [G]:
 $= \min(0.85, 0.925((1 + 1.7 \cdot gQ \cdot I_{zbar} \cdot Q) / (1 + 1.7 \cdot gV \cdot I_{zbar})))$
 $= \min(0.85, 0.925((1 + 1.7 \cdot 3.4 \cdot 0.199 \cdot 0.916) / (1 + 1.7 \cdot 3.4 \cdot 0.199)))$
 $= \min(0.85, 0.884)$
 $= 0.850$

Natural Frequency of Vessel (Operating)	8.708 Hz
Natural Frequency of Vessel (Empty)	8.721 Hz
Natural Frequency of Vessel (Test)	4.728 Hz

User Entered Importance Factor is	1.000
Force Coefficient	[Cf] 0.538
Structure Height to Diameter ratio	3.301
Height to top of Structure	18652.000 mm.

This is classified as a rigid structure. Static analysis performed.

Sample Calculation for the First Element:

The ASCE code performs calculations in Imperial Units. The wind pressure is therefore computed in these units.

Value of $[\alpha]$ and $[Z_g]$:
 Exposure Category: C from Table 26.11-1
 $\alpha = 9.5$; $Z_g = 274320.0$ mm.

Effective Height $[z]$:
 $= \text{Centroid Height} + \text{Vessel Base Elevation}$
 $= 850.0 + 300.0 = 1150.0$ mm.
 $= 3.773$ ft. Imperial Units

Velocity Pressure coefficient evaluated at height z $[K_z]$:
 Because z (3.773 ft.) < 15 ft.
 $= 2.01(15 / Z_g)^{2/\alpha}$
 $= 2.01(15 / 274320.0)^{2/9.5}$
 $= 0.849$

Type of Hill: No Hill

Wind Directionality Factor $[K_d]$:
 $= 1$, per Table 26.6-1

As there is No Hill Present: [Kzt]:
 $K1 = 0, K2 = 0, K3 = 0$

Topographical Factor [Kzt]:
 $= (1 + K1 \cdot K2 \cdot K3)^2$
 $= (1 + 0.0 \cdot 0.0 \cdot 0.0)^2$
 $= 1.0$

Velocity Pressure evaluated at height z, Imperial Units [qz]:
 $= \max(16, 0.00256 \cdot Kz \cdot Kzt \cdot Kd \cdot Ke \cdot V(\text{mph})^2)$
 $= \max(16, 0.00256 \cdot 0.849 \cdot 1.0 \cdot 1.0 \cdot 1.0 \cdot 60.399^2)$
 $= 16.0 \text{ psf } [78.12] \text{ Kgs/m}^2$

Force on the first element [F]:
 $= qz \cdot G \cdot Cf \cdot \text{WindArea}$
 $= 16.0 \cdot 0.85 \cdot 0.538 \cdot 126.392$
 $= 925.4 \text{ lbs. } [419.8] \text{ Kgf}$

Element	Hgt (z) mm.	K1	K2	K3	Kz	Kzt	qz Kgs/m ²
Skirt2	1150.0	0.000	0.000	0.000	0.849	1.000	78.120
Skirt-1	2500.0	0.000	0.000	0.000	0.849	1.000	78.120
Bottom Dish	3025.0	0.000	0.000	0.000	0.849	1.000	78.120
Shell-1	4550.0	0.000	0.000	0.000	0.849	1.000	78.120
Shell2	7550.0	0.000	0.000	0.000	0.943	1.000	78.120
Shell3	10550.0	0.000	0.000	0.000	1.012	1.000	78.120
shell4	13550.0	0.000	0.000	0.000	1.067	1.000	78.120
shell5	16250.0	0.000	0.000	0.000	1.109	1.000	78.120
Top dish	18089.3	0.000	0.000	0.000	1.134	1.000	78.120

Wind Vibration Calculations

This evaluation is based on work by Kanti Mahajan and Ed Zorilla

Nomenclature

Cf - Correction factor for natural frequency
 D - Average outer insulated diameter of vessel, mm.
 Df - Damping Factor < 0.75 Unstable, > 0.95 Stable
 Dr - Average outer insulated of top half of vessel, mm.
 f - Natural frequency of vibration (Hertz)
 f1 - Natural frequency of bare vessel based on a unit value of $(D/L^2)(10^4)$
 L - Total height of structure mm.
 Lc - Total length of conical section(s) of vessel, mm.
 tb - Uncorroded plate thickness at bottom of vessel, mm.
 V30 - Design Wind Speed provided by user, Km/hr
 Vc - Critical wind velocity, Km/hr
 Vw - Maximum wind speed at top of structure, Km/hr
 W - Total corroded weight of structure, Kgf
 Ws - Cor. vessel weight excl. weight of parts which do not effect stiff. Kgf
 Z - Maximum amplitude of vibration at top of vessel, mm.
 D1 - Logarithmic decrement (taken as 0.03 for Welded Structures)
 Vp - Vib. Chance, $\leq 0.320\text{E-}06$ (High); $0.320\text{E-}06 < 0.400\text{E-}06$ (Probable)
 P30 - wind pressure 30 feet above the base

Check other Conditions and Basic Assumptions:

:: Total Cone Length / Total Length < 0.5
 :: $0.0/17200.0 = 0.0$

:: $[D / L^2] \cdot 10^4 < 8$ (English Units, ft.)
 :: $[18.91/56.43^2] \cdot 10^4 = 59.37$ [Geometry Violation]

Compute the vibration possibility. If $Vp > 0.400\text{E-}06$ no chance. [Vp]:
 $= W / (L \cdot Dr^2)$
 $= 140201 / (17200.0 \cdot 5763.941^2)$
 $= 0.24535\text{E-}06$

Compute the damping factor Df which is a measure of instability [Df]:

$$= W \cdot DI / (L \cdot Dr^2)$$

$$= 140201 \cdot 0.03 / (17200.0 \cdot 5763.941^2)$$

$$= 0.73605E-08$$

Compute the critical wind velocity [Vc]:

$$= 3.4 \cdot f \cdot Dr$$

$$= 3.4 \cdot 8.708 \cdot 18.911$$

$$= 559.860 \text{ mile/hr } [900.983 \text{ Km/hr}]$$

Compute the velocity at the top of the tower [Vw]:

$$= V30 \cdot (L / (30 + \text{BaseHeight}))^{0.143}$$

$$= 60.4 \cdot (56.43 / (30 + 1.0))^{0.143}$$

$$= 65.805 \text{ mile/hr } [105.900 \text{ Km/hr}]$$

Compute the maximum gust velocity using the gust response factor Gh [Vg]:

$$= Vw \cdot Gh$$

$$= 105.9 \cdot 0.85$$

$$= 90.015 \text{ Km/hr}$$

Since Vc is greater than Vg the dynamic deflection Z, does not need to be computed.

Platform Load Calculations

ID	Wind Area cm ²	Elevation mm.	Pressure Kgs/m ²	Force Kgf	Cf
PLAT:[1 OF 1]	30000.00	3000.00	78.12	234.36	1.00
PLAT:[1 OF 1]	30000.00	7050.00	78.12	234.36	1.00
PLAT:[1 OF 1]	30000.00	11050.00	78.12	234.36	1.00
PLAT:[1 OF 1]	30000.00	15050.00	78.12	234.36	1.00
SQR PLT	60000.00	18950.00	78.12	468.72	1.00

Wind Loads on Masses/Equipment/Piping

ID	Wind Area cm ²	Elevation mm.	Pressure Kgs/m ²	Force Kgf
tailing lug b	0.00	3000.00	78.12	0.00
Davit-1	0.00	10550.00	78.12	0.00
Davit-2	0.00	11550.00	78.12	0.00
Desmister	0.00	15050.00	78.12	0.00
Feed line	0.00	15550.00	78.12	0.00
Donut Ring	0.00	15050.00	78.12	0.00
Clip wt	0.00	17450.00	78.12	0.00
N3-Over head	0.00	17450.00	78.12	0.00
PSV-6"	0.00	18900.00	78.12	0.00
PSV2-6"	0.00	18900.00	78.12	0.00

The Natural Frequency for the Vessel (Ope...) is 8.70755 Hz.

Wind Load Calculation:

From	To	Wind Height mm.	Wind Diameter mm.	Wind Area cm ²	Wind Pressure Kgs/m ²	Element Wind Load Kgf
10	20	1150	6907.2	117422	78.12	461.734
20	30	2500	6902.4	69024	78.12	271.419
30	40	3025	6904.8	3452.4	78.12	269.242
40	50	4550	6916.8	207504	78.12	815.957
50	60	7550	6916.8	207504	78.12	1056.68
60	70	10550	6916.8	207504	78.12	1085.97
70	80	13550	6916.8	207504	78.12	815.957
80	90	16250	6916.8	166003	78.12	891.525
90	100	18089.3	6904.8	82194.6	78.12	863.334



The Wind Loads calculated and printed in the Wind Load calculation report have been factored by the input scalar/load reduction factor of: 1.100.

Be sure the wind speed is in accordance with the specified wind design code.

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Earthquake Load Calculation Results:

Earthquake Load Calculation:

Input Values:

Seismic Design Code	ASCE/SEI 7-16
Seismic Load Reduction Scale Factor	1.000
Importance Factor	1.250
Table Value Fa	1.600
Table Value Fv	2.400
Max. Mapped Res. Acceleration [Ss]	0.147
Max. Eff. Ground Acceleration [S]	0.065
Force Modification Factor R	1.250
Site Class	D
Component Elevation Ratio z/h	0.020
Amplification Factor Ap	2.500
Force Factor	0.000
Consider Vertical Acceleration	Yes
Minimum Acceleration Multiplier	0.000
User Value of Sds (used if > 0)	0.000
User Value of Sd1 (used if > 0)	0.000
Moment Reduction Factor Tau	1.000

Seismic Analysis Results:

$$\begin{aligned} SMS &= Fa * Ss = 1.6 * 0.147 = 0.235 \\ SM1 &= Fv * S1 = 2.4 * 0.065 = 0.156 \\ SDS &= 2/3 * Sms = 2/3 * 0.235 = 0.157 \\ SD1 &= 2/3 * Sm1 = 2/3 * 0.156 = 0.104 \end{aligned}$$

Check Approximate Fundamental Period from 12.8-7 [Ta]:
 $= Ct \cdot hn^x$ where $Ct = 0.020$, $x = 0.75$ and hn = Structural Height (ft.)
 $= 0.020(61.10560.75)$
 $= 0.437$ seconds

The Coefficient Cu from Table 12.8-1 is : 1.700

Fundamental Period (1/Frequency) [T]:
 $= (1/\text{Natural Frequency}) = (1/8.708)$
 $= 0.115$

Check the Value of T which is the smaller of $Cu \cdot Ta$ and T:
 $= \text{Minimum Value of } (1.7 \cdot 0.437, 0.115)$ per 12.8.2
 $= 0.115$

Loads on Building/Structure Supported Equipment:

$$\begin{aligned} &\text{Seismic Force per equation 13.3-1, [Fp]:} \\ &= 0.4 \cdot Ap \cdot Sds \cdot W(1 + 2 \cdot (z/h)) / (R / Ie) \\ &= 0.4 \cdot 2.5 \cdot 0.157 \cdot 145093(1 + 2 \cdot 0.02) / (1.25/1.25) \\ &= 23660.664 \text{ Kgf} \end{aligned}$$

Maximum value of Fp per equation 13.3-2:
 $= 1.6 \cdot Sds \cdot I \cdot W$
 $= 1.6 \cdot 0.157 \cdot 1.25 \cdot 145093 = 45501.28 \text{ Kgf}$

Minimum value of Fp per equation 13.3-3:
 $= 0.3 \cdot 0.16 \cdot 1.25 \cdot 145093$
 $= 8531.490 \text{ Kgf}$

Total Base Shear $V = Fp$, [V]:
 $= 23660.664 \text{ Kgf}$

Vertical load per 12.4-4, [Ev]:
 $= 0.2 \cdot Sds \cdot W$
 $= 0.2 \cdot 0.157 \cdot 145093$
 $= 4550.1 \text{ Kgf}$



Distribute the Base shear force to each element according to the equations
 $F_x = C_{vx} * V$ (eqn. 12.8-11) and the vertical distribution factor
 $C_{vx} = W_x * h_{xk} / (\text{Sum of } W_i * h_{ik})$ and k is an exponent which is related
to the period of Vibration.

In this case, the value of k was 1.0

The Natural Frequency for the Vessel (Ope...) is 8.70755 Hz.

Earthquake Load Calculation:

From	To	Earthquake Height mm.	Earthquake Weight Kgf	Element Ope Load Kgf	Element Emp Load Kgf	Element Vertical Load Kgf
10	20	850	9579.25	132.302	133.537	4550.13
20	30	2200	2689.02	96.1244	97.0214	4249.72
30	40	2725	18111.3	801.922	518.073	4165.39
40	50	4250	14493.4	1000.86	1003.27	3597.42
50	60	7250	17708.5	2086.1	1704.54	3142.91
60	70	10250	18992.7	3163.2	2625.75	2587.57
70	80	13250	14470.1	3115.33	3122.78	1991.96
80	90	15950	21266.5	5511.54	4675.57	1538.18
90	100	17175	27782.6	7753.29	5939.39	871.263



The following table is for the Operating Case.

Wind/Earthquake Shear, Bending:

From	To	Distance to Support mm.	Cumulative Wind Shear Kgf	Earthquake Shear Kgf	Wind Bending Kg-m.	Earthquake Bending Kg-m.
10	20	850	6531.82	23660.7	64373.9	321429
20	30	2200	6070.09	23528.4	53662.1	281318
30	40	2725	5798.67	23432.2	47727.6	257837
40	50	4250	5529.42	22630.3	47444.4	256685
50	60	7250	4713.47	21629.5	32079.7	190294
60	70	10250	3656.79	19543.4	19524.1	128534
70	80	13250	2570.82	16380.2	10182.5	74647.5
80	90	15950	1754.86	13264.8	3693.85	30179.1
90	100	17175	863.334	7753.29	551.953	4956.89

Note:
The Wind Shears/Moments and the Earthquake Shears/Moments calculated and printed in the Wind/Earthquake Shear and Bending report have been factored by the input Scalar/Load reductions factors of;
Wind: 1.100; Earthquake: 1.000.

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Wind Deflection Calculations:

The following table is for Wind Deflection; Operating Case.

Wind Load Deflection:

From	To	Cumulative Shear Kgf	Centroid Deflection mm.	Elem. End Deflection mm.	Elem. Ang. Rotation
10	20	6531.82	0.00069846	0.0027134	0.30984E-05
20	30	6070.09	0.0045126	0.0067929	0.50230E-05
30	40	5798.67	0.0069188	0.0070456	0.50855E-05
40	50	5529.42	0.015775	0.026448	0.76815E-05
50	60	4713.47	0.038703	0.052212	0.93574E-05
60	70	3656.79	0.066686	0.081873	0.10314E-04
70	80	2570.82	0.097565	0.11359	0.10756E-04
80	90	1754.86	0.12654	0.13956	0.10858E-04
90	100	863.334	0.13983	0.1401	0.10858E-04

Critical Wind Velocity for Tower Vibration:

From	To	1st Crit. Wind Speed Km/hr	2nd Crit. Wind Speed Km/hr
10	20	1079.69	6748.07
20	30	1078.94	6743.38
30	40	1079.32	6745.72
40	50	1081.19	6757.44
50	60	1081.19	6757.44
60	70	1081.19	6757.44
70	80	1081.19	6757.44
80	90	1081.19	6757.44
90	100	1079.32	6745.72

Allowable deflection at the Tower Top (Ope)(6.000"/100ft. Criteria)
Allowable deflection : 86.000 Actual deflection : 0.140 mm.

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Longitudinal Stress Constants:

From	To	Metal Area New cm ²	Metal Area Corroded cm ²	Section Modulus New mm. ³	Section Modulus Corroded mm. ³
10	20	3963.07	3783.6	565942784	540502528
20	30	3242.51	3062.93	463362784	437852192
30	40	4857.8	4571.17	692268224	651784000
40	50	5762.43	5475.81	821198592	780784576
50	60	5762.43	5475.81	821198592	780784576
60	70	5762.43	5475.81	821198592	780784576
70	80	5762.43	5475.81	821198592	780784576
80	90	5762.43	5475.81	821198592	780784576
90	100	4857.8	4571.17	692268224	651784000



Longitudinal Allowable Stresses:

From	To	Tensile N./mm ²	Hydrotest Tensile N./mm ²	Compressive N./mm ²	Hydrotest Compressive N./mm ²
10	20	95.7583	188.997	-98.9047	-98.9047
20	30	82.7985	140.397	-88.6449	-88.6449
30	40	165.6	280.8	-105.927	-105.927
40	50	140.76	238.68	-112.017	-112.017
50	60	140.76	238.68	-112.017	-112.017
60	70	140.76	238.68	-112.017	-112.017
70	80	140.76	238.68	-112.017	-112.017
80	90	140.76	238.68	-112.017	-112.017
90	100	165.6	280.8	-105.927	-105.927

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Longitudinal Stress Report



Longitudinal Operating and Empty Stresses are computed in the corroded condition. Stresses due to loads in the hydrostatic test cases have also been computed in the corroded condition.

Longitudinal Pressure Stresses due to:

From	To	Longitudinal Stress Internal Pressure N./mm ²	Longitudinal Stress External Pressure N./mm ²	Longitudinal Stress Hydrotest Pressure N./mm ²
10	20	...	...	...
20	30	...	...	...
30	40	67.1245	...	87.2619
40	50	56.0448	...	72.8583
50	60	56.0448	...	72.8583
60	70	56.0448	...	72.8583
70	80	56.0448	...	72.8583
80	90	56.0448	...	72.8583
90	100	67.1245	...	87.2619

Longitudinal Stresses due to Weight Loads for these Conditions:

From	To	Wght. Str. Empty N./mm ²	Wght. Str. Operating N./mm ²	Wght. Str. Hydrotest N./mm ²	Wght. Str. Emp. Mom. N./mm ²	Wght. Str. Opr. Mom. N./mm ²
10	20	-3.74662	-3.76072	-14.5532	0.028441	0.028441
20	30	-4.32146	-4.33887	-17.6707	0.035108	0.035108
30	40	-2.83791	-2.83791	-2.83791	0.023585	0.023585
40	50	-2.04531	-2.04531	-2.04531	0.019688	0.019688
50	60	-1.78752	-1.78752	-1.78752	0.019257	0.019257
60	70	-1.47216	-1.47216	-1.47216	0.019257	0.019257
70	80	-1.13379	-1.13379	-1.13379	0.00021571	0.00021571
80	90	-0.87642	-0.87642	-0.87642	...	...
90	100	-0.59533	-0.59533	-0.59533	...	...

Longitudinal Stresses due to Weight Loads and Bending for these Conditions:

From	To	Wght. Str. Hyd. Mom. N./mm ²	Bend. Str. Oper. Wind N./mm ²	Bend. Str. Oper. Equ. N./mm ²	Bend. Str. Hyd. Wind N./mm ²	Bend. Str. Hyd. Equ. N./mm ²
10	20	0.028441	1.16797	5.83186	0.58399	...
20	30	0.035108	1.20188	6.30071	0.60094	...
30	40	0.023585	0.7181	3.87937	0.35905	...
40	50	0.019688	0.5959	3.22396	0.29795	...
50	60	0.019257	0.40292	2.39009	0.20146	...
60	70	0.019257	0.24522	1.61438	0.12261	...
70	80	0.00021571	0.12789	0.93757	0.063946	...
80	90	...	0.046395	0.37905	0.023197	...
90	100	...	0.0083046	0.074581	0.0041523	...

Longitudinal Stresses due to these Conditions:

From	To	Vortex Shedding Operating Case N./mm ²	Vortex Shedding Empty Case N./mm ²	Vortex Shedding Test Case N./mm ²	Earthquake Empty Case N./mm ²
10	20	...	...	...	4.84233
20	30	...	...	...	5.22544
30	40	...	...	...	3.21485



40	50	...	...	...	2.67155
50	60	...	...	...	1.97184
60	70	...	...	...	1.32313
70	80	...	...	...	0.75602
80	90	...	...	...	0.2972
90	100	...	...	...	0.057132

Longitudinal Stresses due to Applied Axial Forces:

From	To	Longitudinal Stress Y Forces Wind N./mm ²	Longitudinal Stress Y Forces Seismic N./mm ²
10	20	...	0.11794
20	30	...	0.13607
30	40	...	0.089363
40	50	...	0.064428
50	60	...	0.056288
60	70	...	0.046342
70	80	...	0.035675
80	90	...	0.027548
90	100	...	0.018692

Longitudinal Stresses due to User Forces and Moments:

From	To	Wind For/Mom Corroded N./mm ²	Earthquake For/Mom Corroded N./mm ²	Wind For/Mom No Corrosion N./mm ²	Earthquake For/Mom No Corrosion N./mm ²
10	20	...	...	...	...
20	30	...	...	...	...
30	40	...	...	...	...
40	50	...	...	...	...
50	60	...	...	...	...
60	70	...	...	...	...
70	80	...	...	...	...
80	90	...	...	...	...
90	100	...	...	...	...

Bending Stresses due to Blast Load:

From	To	Bending Stress N./mm ²
10	20	...
20	30	...
30	40	...
40	50	...
50	60	...
60	70	...
70	80	...
80	90	...
90	100	...

Stress Combination Load Cases for Vertical Vessels:

Load Case Definition Key

IP = Longitudinal Stress due to Internal Pressure
EP = Longitudinal Stress due to External Pressure
HP = Longitudinal Stress due to Hydrotest Pressure
NP = No Pressure
EW = Longitudinal Stress due to Weight (No Liquid)
OW = Longitudinal Stress due to Weight (Operating)
HW = Longitudinal Stress due to Weight (Hydrotest)
WI = Bending Stress due to Wind Moment (Operating)
EQ = Bending Stress due to Earthquake Moment (Operating)
EE = Bending Stress due to Earthquake Moment (Empty)
HI = Bending Stress due to Wind Moment (Hydrotest)
HE = Bending Stress due to Earthquake Moment (Hydrotest)
WE = Bending Stress due to Wind Moment (Empty) (no CA)
WF = Bending Stress due to Wind Moment (Filled) (no CA)
CW = Longitudinal Stress due to Weight (Empty) (no CA)
VO = Bending Stress due to Vortex Shedding Loads (Ope)
VE = Bending Stress due to Vortex Shedding Loads (Emp)
VF = Bending Stress due to Vortex Shedding Loads (Test No CA.)
FW = Axial Stress due to Vertical Forces for the Wind Case
FS = Axial Stress due to Vertical Forces for the Seismic Case
BW = Bending Stress due to Lat. Forces for the Wind Case, Corroded
BS = Bending Stress due to Lat. Forces for the Seismic Case, Corroded
BN = Bending Stress due to Lat. Forces for the Wind Case, UnCorroded
BU = Bending Stress due to Lat. Forces for the Seismic Case, UnCorroded
BL = Bending Stress due to Blast Loads, Corroded

General Notes:

Case types HI and HE are in the Corroded condition.

Case types WE, WF, and CW are in the Un-Corroded condition.

A blank stress and stress ratio indicates that the corresponding stress comprising those components that did not contribute to that type of stress.

An asterisk (*) in the final column denotes overstress.

Analysis of Load Case 1 : NP+EW+WI+FW+BW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		95.76	-4.94	98.90		0.0500
20		82.80	-5.56	88.64		0.0627
30		165.60	-3.58	105.93		0.0338
40		140.76	-2.66	112.02		0.0238
50		140.76	-2.21	112.02		0.0197
60		140.76	-1.74	112.02		0.0155
70		140.76	-1.26	112.02		0.0113
80		140.76	-0.92	112.02		0.0082
90		165.60	-0.60	105.93		0.0057

Analysis of Load Case 2 : NP+EW+EE+FS+BS

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	1.24	95.76	-8.50	98.90	0.0130	0.0859
20	1.08	82.80	-9.45	88.64	0.0130	0.1066
30	0.49	165.60	-5.99	105.93	0.0030	0.0565
40	0.71	140.76	-4.67	112.02	0.0050	0.0417
50	0.26	140.76	-3.72	112.02	0.0018	0.0332
60		140.76	-2.77	112.02		0.0247
70		140.76	-1.85	112.02		0.0166
80		140.76	-1.15	112.02		0.0102
90		165.60	-0.63	105.93		0.0060



Analysis of Load Case 3 : NP+OW+WI+FW+BW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		95.76	-4.96	98.90		0.0501
20		82.80	-5.58	88.64		0.0629
30		165.60	-3.58	105.93		0.0338
40		140.76	-2.66	112.02		0.0238
50		140.76	-2.21	112.02		0.0197
60		140.76	-1.74	112.02		0.0155
70		140.76	-1.26	112.02		0.0113
80		140.76	-0.92	112.02		0.0082
90		165.60	-0.60	105.93		0.0057

Analysis of Load Case 4 : NP+OW+EQ+FS+BS

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	2.22	95.76	-9.50	98.90	0.0232	0.0961
20	2.13	82.80	-10.54	88.64	0.0258	0.1189
30	1.15	165.60	-6.65	105.93	0.0070	0.0628
40	1.26	140.76	-5.22	112.02	0.0090	0.0466
50	0.68	140.76	-4.14	112.02	0.0048	0.0370
60	0.21	140.76	-3.06	112.02	0.0015	0.0273
70		140.76	-2.04	112.02		0.0182
80		140.76	-1.23	112.02		0.0110
90		165.60	-0.65	105.93		0.0061

Analysis of Load Case 5 : NP+HW+HI

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		189.00	-15.17	98.90		0.1533
20		140.40	-18.31	88.64		0.2065
30		280.80	-3.22	105.93		0.0304
40		238.68	-2.36	112.02		0.0211
50		238.68	-2.01	112.02		0.0179
60		238.68	-1.61	112.02		0.0144
70		238.68	-1.20	112.02		0.0107
80		238.68	-0.90	112.02		0.0080
90		280.80	-0.60	105.93		0.0057

Analysis of Load Case 6 : NP+HW+HE

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		189.00	-14.58	98.90		0.1474
20		140.40	-17.71	88.64		0.1997
30		280.80	-2.86	105.93		0.0270
40		238.68	-2.06	112.02		0.0184
50		238.68	-1.81	112.02		0.0161
60		238.68	-1.49	112.02		0.0133
70		238.68	-1.13	112.02		0.0101
80		238.68	-0.88	112.02		0.0078
90		280.80	-0.60	105.93		0.0056

Analysis of Load Case 7 : IP+OW+WI+FW+BW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		95.76	-4.96	98.90		0.0501
20		82.80	-5.58	88.64		0.0629
30	65.03	165.60		105.93	0.3927	
40	54.62	140.76		112.02	0.3880	
50	54.68	140.76		112.02	0.3885	
60	54.84	140.76		112.02	0.3896	
70	55.04	140.76		112.02	0.3910	
80	55.21	140.76		112.02	0.3923	



90 66.54 165.60 105.93 0.4018

Analysis of Load Case 8 : IP+OW+EQ+FS+BS

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	2.22	95.76	-9.50	98.90	0.0232	0.0961
20	2.13	82.80	-10.54	88.64	0.0258	0.1189
30	68.28	165.60		105.93	0.4123	
40	57.31	140.76		112.02	0.4071	
50	56.72	140.76		112.02	0.4030	
60	56.25	140.76		112.02	0.3996	
70	55.88	140.76		112.02	0.3970	
80	55.58	140.76		112.02	0.3948	
90	66.62	165.60		105.93	0.4023	

Analysis of Load Case 9 : EP+OW+WI+FW+BW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		95.76	-4.96	98.90		0.0501
20		82.80	-5.58	88.64		0.0629
30		165.60	-3.58	105.93		0.0338
40		140.76	-2.66	112.02		0.0238
50		140.76	-2.21	112.02		0.0197
60		140.76	-1.74	112.02		0.0155
70		140.76	-1.26	112.02		0.0113
80		140.76	-0.92	112.02		0.0082
90		165.60	-0.60	105.93		0.0057

Analysis of Load Case 10 : EP+OW+EQ+FS+BS

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	2.22	95.76	-9.50	98.90	0.0232	0.0961
20	2.13	82.80	-10.54	88.64	0.0258	0.1189
30	1.15	165.60	-6.65	105.93	0.0070	0.0628
40	1.26	140.76	-5.22	112.02	0.0090	0.0466
50	0.68	140.76	-4.14	112.02	0.0048	0.0370
60	0.21	140.76	-3.06	112.02	0.0015	0.0273
70		140.76	-2.04	112.02		0.0182
80		140.76	-1.23	112.02		0.0110
90		165.60	-0.65	105.93		0.0061

Analysis of Load Case 11 : HP+HW+HI

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		189.00	-15.17	98.90		0.1533
20		140.40	-18.31	88.64		0.2065
30	84.81	280.80		105.93	0.3020	
40	71.13	238.68		112.02	0.2980	
50	71.29	238.68		112.02	0.2987	
60	71.53	238.68		112.02	0.2997	
70	71.79	238.68		112.02	0.3008	
80	72.01	238.68		112.02	0.3017	
90	86.67	280.80		105.93	0.3087	

Analysis of Load Case 12 : HP+HW+HE

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		189.00	-14.58	98.90		0.1474
20		140.40	-17.71	88.64		0.1997
30	84.45	280.80		105.93	0.3007	
40	70.83	238.68		112.02	0.2968	
50	71.09	238.68		112.02	0.2978	
60	71.41	238.68		112.02	0.2992	
70	71.72	238.68		112.02	0.3005	



80	71.98	238.68	112.02	0.3016
90	86.67	280.80	105.93	0.3086

Analysis of Load Case 13 : IP+WE+EW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		95.76	-3.78	98.90		0.0382
20		82.80	-4.36	88.64		0.0491
30	64.31	165.60		105.93	0.3883	
40	54.02	140.76		112.02	0.3838	
50	54.28	140.76		112.02	0.3856	
60	54.59	140.76		112.02	0.3878	
70	54.91	140.76		112.02	0.3901	
80	55.17	140.76		112.02	0.3919	
90	66.53	165.60		105.93	0.4017	

Analysis of Load Case 14 : IP+WF+CW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		95.76	-3.59	98.90		0.0363
20		82.80	-4.10	88.64		0.0462
30	64.45	165.60		105.93	0.3892	
40	54.10	140.76		112.02	0.3844	
50	54.35	140.76		112.02	0.3861	
60	54.65	140.76		112.02	0.3882	
70	54.97	140.76		112.02	0.3905	
80	55.21	140.76		112.02	0.3922	
90	66.56	165.60		105.93	0.4020	

Analysis of Load Case 15 : IP+VO+OW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		95.76	-3.79	98.90		0.0383
20		82.80	-4.37	88.64		0.0493
30	64.31	165.60		105.93	0.3883	
40	54.02	140.76		112.02	0.3838	
50	54.28	140.76		112.02	0.3856	
60	54.59	140.76		112.02	0.3878	
70	54.91	140.76		112.02	0.3901	
80	55.17	140.76		112.02	0.3919	
90	66.53	165.60		105.93	0.4017	

Analysis of Load Case 16 : IP+VE+EW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		95.76	-3.78	98.90		0.0382
20		82.80	-4.36	88.64		0.0491
30	64.31	165.60		105.93	0.3883	
40	54.02	140.76		112.02	0.3838	
50	54.28	140.76		112.02	0.3856	
60	54.59	140.76		112.02	0.3878	
70	54.91	140.76		112.02	0.3901	
80	55.17	140.76		112.02	0.3919	
90	66.53	165.60		105.93	0.4017	

Analysis of Load Case 17 : NP+VO+OW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		95.76	-3.79	98.90		0.0383
20		82.80	-4.37	88.64		0.0493
30		165.60	-2.86	105.93		0.0270
40		140.76	-2.06	112.02		0.0184
50		140.76	-1.81	112.02		0.0161
60		140.76	-1.49	112.02		0.0133



70	140.76	-1.13	112.02	0.0101
80	140.76	-0.88	112.02	0.0078
90	165.60	-0.60	105.93	0.0056

Analysis of Load Case 18 : FS+BS+IP+OW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		79.80	-3.67	82.42		0.0445
20		69.00	-4.24	73.87		0.0574
30	64.40	138.00		88.27	0.4667	
40	54.08	117.30		93.35	0.4611	
50	54.33	117.30		93.35	0.4632	
60	54.64	117.30		93.35	0.4658	
70	54.95	117.30		93.35	0.4684	
80	55.20	117.30		93.35	0.4706	
90	66.55	138.00		88.27	0.4822	

Analysis of Load Case 19 : FS+BS+EP+OW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		79.80	-3.67	82.42		0.0445
20		69.00	-4.24	73.87		0.0574
30		138.00	-2.77	88.27		0.0314
40		117.30	-2.00	93.35		0.0214
50		117.30	-1.75	93.35		0.0188
60		117.30	-1.45	93.35		0.0155
70		117.30	-1.10	93.35		0.0118
80		117.30	-0.85	93.35		0.0091
90		138.00	-0.58	88.27		0.0065

Analysis of Load Case 20 : BL+IP+OW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		95.76	-3.79	98.90		0.0383
20		82.80	-4.37	88.64		0.0493
30	64.31	165.60		105.93	0.3883	
40	54.02	140.76		112.02	0.3838	
50	54.28	140.76		112.02	0.3856	
60	54.59	140.76		112.02	0.3878	
70	54.91	140.76		112.02	0.3901	
80	55.17	140.76		112.02	0.3919	
90	66.53	165.60		105.93	0.4017	

Absolute Maximum of the all of the Stress Ratio's 0.4822

Governing Element: Top dish

Governing Load Case 18 : FS+BS+IP+OW

Element Description	Long Pressure Stress	Weight + Axial Load Stress	Eccentric Moment Stress	Wind or Earthquake Stress	Combined Stress	Which Side
Skirt2	...	-3.6	...	...	...	+ Tens
Skirt2	...	-3.6	...	...	-3.7	- Comp
Skirt-1	...	-4.2	...	...	...	+ Tens
Skirt-1	...	-4.2	...	...	-4.2	- Comp
Bottom Dish	67.1	-2.7	...	...	64.4	+ Tens
Bottom Dish	67.1	-2.7	...	...	...	- Comp
Shell-1	56.0	-2.0	...	...	54.1	+ Tens
Shell-1	56.0	-2.0	...	...	...	- Comp
Shell12	56.0	-1.7	...	...	54.3	+ Tens
Shell12	56.0	-1.7	...	...	...	- Comp



Shell3	56.0	-1.4	...	...	54.6	+ Tens
Shell3	56.0	-1.4	...	...	...	- Comp
shell14	56.0	-1.1	...	...	54.9	+ Tens
shell14	56.0	-1.1	...	...	...	- Comp
shell15	56.0	-0.8	...	...	55.2	+ Tens
shell15	56.0	-0.8	...	...	...	- Comp
Top dish	67.1	-0.6	...	...	66.5	+ Tens
Top dish	67.1	-0.6	...	...	...	- Comp



Skirt Support Analysis Per EN 13445-3 Section 16.12:

The analysis of this skirt has been conducted referencing the EN 13445-3 design code. It is important to note that the selected design code does not explicitly cover the analysis of this component. Therefore, the results provided herein should be regarded as informational only.

Item: Skirt-1

Total vessel (empty) weight	Fg	133083.891 Kg
Head Weight below section 2-2	DeltaFg	18078.197 Kg
Weight of vessel contents (liquid)	Ff	543.763 Kg
Vessel Gross Weight	Fg + Ff	133627.672 Kg
Skirt Mean Diameter	Dz	5735.000 mm.
Skirt Analysis Thickness (Corroded)	ez	17.000 mm.
Skirt Height	H	1000.000 mm.
Additional Axial Force at section 1-1	F1	0.000 Kg
Moment at section 1-1	M1	257836.734 Kg-m.
Applied Moment - Bottom of Skirt	Mb	281317.500 Kg-m.
Weight at Bottom of Skirt	W	135514.094 Kg

Structural Type B Skirt per EN 13445 Figure 16.12-2:

Type of Head:		Elliptical
Head Mean Diameter at Skirt Top	Db	5728.600 mm.
Head Corroded Thickness	eb	25.400 mm.

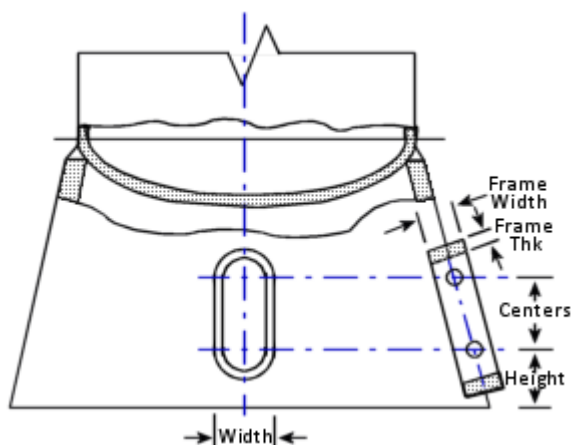
Per EN 13445 Section 7.5.4 [R, r]:

$$h_i = 1426.6 \text{ mm.}$$

$$K = D_i / (2 * h_i) = 5703.2 / (2 * 1426.6) = 1.9989$$

$$R = D_i (0.44 * K + 0.02) = 5703.2 * (0.44 * 2.0 + 0.02) = 5130.0659 \text{ mm.}$$

$$r = D_i ((0.5 / K) - 0.08) = 5703.2 * ((0.5 / 2.0) - 0.08) = 970.3441 \text{ mm.}$$



There are no Access Openings in the Skirt - Plain Skirt Considered

Moment of Inertia	Ixx =	125924040.000 cm ⁴
Moment of Inertia	Iyy =	125924040.000 cm ⁴
Section Modulus	W =	439142656.000 mm. ³
Section Metal Area	A =	3062.896 cm ²

Analysis of the Section at weakest section Modulus [Z min]:

Radius of Gyration of Skirt at 4-4 [rg]:

$$= \sqrt{(\min(I_{xx}, I_{yy}) / \text{Area})}$$

$$= \sqrt{(\min(0.12592E+09, 0.12592E+09) / 3062.896)}$$

$$= 2027.629 \text{ mm.}$$

Skirt Slenderness Ratio [rg/L]:

$$= r_g / L$$

$$= 2027.629/1000.0$$

$$= 2.028$$

Correct the Moment [M4]:

top of skirt is the critical end
 Controlling Moment [M4] = 257836.734 Kg-m.

Axial Stress at Residual (Weakest) Surface Area [Sigma]:

Downwind Side:

$$= -M / Z - W / A$$

$$= -257836.734/.17566E+10 - 135514.094/3062.896$$

$$= -5.778 \text{ N./mm}^2$$

Upwind Side:

$$= +(M) / Z - W / A$$

$$= +257836.734/.17566E+10 - 135514.094/3062.896$$

$$= -2.899 \text{ N./mm}^2$$

16.12.3.4 Checking at connection areas:

Force at the joint, 16.12-1 [Fzp]:

$$= F_1 - F_g - F_f + 4 \cdot M_1 / D_z$$

$$= 0.0 - 133083.891 - 543.763 + 4 \cdot 257836.734/5735.0$$

$$= 46202.6 \text{ Kg}$$

Force at the joint, 16.12-2 [Fzq]:

$$= F_1 - F_g - F_f - 4 \cdot M_1 / D_z$$

$$= 0.0 - 133083.891 - 543.763 - 4 \cdot 257836.734/5735.0$$

$$= -313457.9 \text{ Kg}$$

16.12.3.4.1 Membrane stresses:

Membrane stress at 1-1, 16.12-3 & 4:

Stress [σ_{1pm}]:

$$= (F_{zp} + \Delta F_g) / (\pi \cdot D_b \cdot e_b) + P \cdot D_b / (4 \cdot e_b)$$

$$= (46202.602 + 18078.197 + 543.763) / (\pi \cdot 5728.6 \cdot 25.4) + 12.0 \cdot 5728.6 / (4 \cdot 25.4)$$

$$= 69.06 \text{ N./mm}^2$$

Stress [σ_{1qm}]:

$$= (F_{zq} + \Delta F_g + F_f) / (\pi \cdot D_b \cdot e_b) + P \cdot D_b / (4 \cdot e_b)$$

$$= (-0.3E+06 + 18078.197 + 543.763) / (\pi \cdot 5728.6 \cdot 25.4) + 12.0 \cdot 5728.6 / (4 \cdot 25.4)$$

$$= 61.34 \text{ N./mm}^2$$

Membrane stress at 2-2, 16.12-9:

Stress [σ_{2pm} & σ_{2qm}]:

$$= (F_f + \Delta F_g) / (\pi \cdot D_b \cdot e_b) + P \cdot D_b / (4 \cdot e_b)$$

$$= (543.763 + 18078.197) / (\pi \cdot 5728.6 \cdot 25.4) + 12.0 \cdot 5728.6 / (4 \cdot 25.4)$$

$$= 68.06 \text{ N./mm}^2$$

Membrane stress at 3-3, 16.12 12 & 13:

Stress [σ_{3pm}]:

$$= F_{zp} / (\pi \cdot D_z \cdot e_z)$$

$$= 46202.602 / (\pi \cdot 5735.0 \cdot 17.0)$$

$$= 1.48 \text{ N./mm}^2$$

Stress [σ_{3qm}]:

$$= F_{zq} / (\pi \cdot D_z \cdot e_z)$$

$$= -313457.938 / (\pi \cdot 5735.0 \cdot 17.0)$$

$$= -10.04 \text{ N./mm}^2$$

16.12.3.4.2 Bending stresses:

Cosine of angle, 16.12-25 [cos(γ)]:

$$= 1 - 0.5(D_b + e_b - D_z + e_z) / (r + e_b)$$



$$= 1 - 0.5(5728.6 + 25.4 - 5735.0 + 17.0) / (970.34 + 25.4)$$

$$= 0.9819$$

$$\gamma = \arccos(\cos(\gamma)) = \arccos(0.982) = 10.911 \text{ deg}$$

Dimension [a]:

$$= 0.5 \cdot \sqrt{(eb^2 + ez^2 + 2 \cdot eb \cdot ez \cdot \cos(\gamma))}$$

$$= 0.5 \cdot \sqrt{(25.4^2 + 17.0^2 + 2 \cdot 25.4 \cdot 17.0 \cdot 0.982)}$$

$$= 21.108 \text{ mm.}$$

Bending moment at verification points p and q [Mp & Mq]:

$$M_p = a \cdot F_{zp} = 21.10 \cdot 46202.6 = 975.25 \text{ Kg-m.}$$

$$M_q = a \cdot F_{zq} = 21.10 \cdot -313457.9 = -6616.52 \text{ Kg-m.}$$

Correction factor [C]:

$$= 0.63 - 0.057(eb/ez)^2$$

$$= 0.63 - 0.057(25.4/17.0)^2$$

$$= 0.503$$

Bending stresses in sections 1-1 to 3-3 at the outer surface (o), 16.12-26 to 29:

Stress [$\sigma_{1pb(o)}$ & $\sigma_{2pb(o)}$]:

$$= C \cdot 6 \cdot M_p / (\pi \cdot D_b \cdot eb^2)$$

$$= 0.503 \cdot 6 \cdot 975.252 / (\pi \cdot 5728.6 \cdot 25.4^2)$$

$$= 2.48 \text{ N./mm}^2$$

Stress [$\sigma_{1qb(o)}$ & $\sigma_{2qb(o)}$]:

$$= C \cdot 6 \cdot M_q / (\pi \cdot D_b \cdot eb^2)$$

$$= 0.503 \cdot 6 \cdot -6616.523 / (\pi \cdot 5728.6 \cdot 25.4^2)$$

$$= -16.86 \text{ N./mm}^2$$

Stress [$\sigma_{3pb(o)}$]:

$$= C \cdot 6 \cdot M_p / (\pi \cdot D_z \cdot ez^2)$$

$$= 0.503 \cdot 6 \cdot 975.252 / (\pi \cdot 5735.0 \cdot 17.0^2)$$

$$= 5.54 \text{ N./mm}^2$$

Stress [$\sigma_{3qb(o)}$]:

$$= C \cdot 6 \cdot M_q / (\pi \cdot D_z \cdot ez^2)$$

$$= 0.503 \cdot 6 \cdot -6616.523 / (\pi \cdot 5735.0 \cdot 17.0^2)$$

$$= -37.59 \text{ N./mm}^2$$

Geometry variables, 16.12-32 to 36:

$$eb/D_b = 1. / 225.535 = 0.0044$$

$$y = 125 \cdot eb/D_b = 125 \cdot 25.4/5728.6 = 0.5542$$

As this an elliptical head, it will be treated a a Korbbogen type head

Geometric value [α]:

$$= 1.51861 \cdot 2.71828 - 4.2335 \cdot RYY + 3.994$$

$$= 1.51861 \cdot 2.71828 - 4.2335 \cdot 0.554 + 3.994 = 4.1394$$

Bending stress in sections 1-1 and 2-2 Equation 16.12-31 [$\sigma_{1b(p)}$ & $\sigma_{2b(p)}$]:

$$= (P + Ph) \cdot D_b ((\gamma / \gamma_A) \cdot \alpha - 1) / (4 \cdot eb)$$

$$= (12.0 + 0.0) \cdot 5728.6 ((10.91/40.0) \cdot 4.139 - 1) / (4 \cdot 25.4)$$

$$= 8.73 \text{ N./mm}^2$$

16.12.3.4.3 Total stresses and strength conditions:

Structural shape B and C:

Stress [σ_{1pitot}]:

$$= \sigma_{1pm} - \sigma_{1pb(o)} + \sigma_{1b(p)}$$

$$= 69.055 - 2.485 + 8.735$$

$$= 75.31 \text{ N./mm}^2$$

Stress [σ_{1potot}]:

$$= \sigma_{1pm} + \sigma_{1pb(o)} - \sigma_{1b(p)}$$

$$= 69.055 + 2.485 - 8.735$$

$$= 62.81 \text{ N./mm}^2$$



Stress [σ_{1qitot}]:
 $= \sigma_{1qm} - \sigma_{1qb(o)} + \sigma_{1b(p)}$
 $= 61.339 - -16.857 + 8.735$
 $= 86.93 \text{ N./mm}^2$

Stress [σ_{1qotot}]:
 $= \sigma_{1qm} + \sigma_{1qb(o)} - \sigma_{1b(p)}$
 $= 61.339 + -16.857 - 8.735$
 $= 35.75 \text{ N./mm}^2$

Stress [σ_{2pitot}]:
 $= \sigma_{2pm} + \sigma_{2pb(o)} + \sigma_{2b(p)}$
 $= 68.064 + 2.485 + 8.735$
 $= 79.28 \text{ N./mm}^2$

Stress [σ_{2potot}]:
 $= \sigma_{2pm} - \sigma_{2pb(o)} - \sigma_{2b(p)}$
 $= 68.064 - 2.485 - 8.735$
 $= 56.84 \text{ N./mm}^2$

Stress [σ_{2qitot}]:
 $= \sigma_{2qm} + \sigma_{2qb(o)} + \sigma_{2b(p)}$
 $= 68.064 + -16.857 + 8.735$
 $= 59.94 \text{ N./mm}^2$

Stress [σ_{2qotot}]:
 $= \sigma_{2pm} - \sigma_{2pb(o)} - \sigma_{2b(p)}$
 $= 68.064 - -16.857 - 8.735$
 $= 76.19 \text{ N./mm}^2$

Stress [σ_{3pitot}]:
 $= \sigma_{3pm} - \sigma_{3pb}$
 $= 1.479 - 5.541$
 $= -4.06 \text{ N./mm}^2$

Stress [σ_{3potot}]:
 $= \sigma_{3pm} + \sigma_{3pb}$
 $= 1.479 + 5.541$
 $= 7.02 \text{ N./mm}^2$

Stress [σ_{3qitot}]:
 $= \sigma_{3qm} - \sigma_{3qb}$
 $= -10.036 - -37.59$
 $= 27.55 \text{ N./mm}^2$

Stress [σ_{3qotot}]:
 $= \sigma_{3qm} + \sigma_{3qb}$
 $= -10.036 + -37.59$
 $= -47.63 \text{ N./mm}^2$

Allowable stresses:

For σ_{1pitot} and σ_{1potot} , the stress must not exceed:
 $= fs(3 - (1 / 1.5)) \cdot (\sigma_{1pm} / f)^2$
 $= 138.0(3 - (1/1.5)) \cdot (69.055/137.997)^2$
 $= 390.96 \text{ N./mm}^2$

For σ_{1qitot} and σ_{1qotot} , the stress must not exceed:
 $= fs(3 - (1 / 1.5)) \cdot (\sigma_{1qm} / f)^2$
 $= 138.0(3 - (1/1.5)) \cdot (61.339/137.997)^2$
 $= 395.82 \text{ N./mm}^2$

For σ_{2pitot} and σ_{2potot} , the stress must not exceed:
 $= fs(3 - (1 / 1.5)) \cdot (\sigma_{2pm} / f)^2$
 $= 138.0(3 - (1/1.5)) \cdot (68.064/137.997)^2$
 $= 391.62 \text{ N./mm}^2$

For σ_{2qitot} and σ_{2qotot} , the stress must not exceed:



$$= fs(3 - (1 / 1.5) \cdot (\sigma_{2qm} / f)^2)$$

$$= 138.0(3 - (1/1.5) \cdot (68.064/137.997)^2)$$

$$= 391.62 \text{ N./mm}^2$$

For σ_{3pitot} & σ_{3potot} , the stress must not exceed:

$$= fs(3 - (1 / 1.5) \cdot (\sigma_{3pm} / f)^2)$$

$$= 138.0(3 - (1/1.5) \cdot (1.479/137.997)^2)$$

$$= 413.99 \text{ N./mm}^2$$

For σ_{3qitot} & σ_{3qotot} , the stress must not exceed:

$$= fs(3 - (1 / 1.5) \cdot (\sigma_{3qm} / f)^2)$$

$$= 138.0(3 - (1/1.5) \cdot (-10.036/137.997)^2)$$

$$= 413.51 \text{ N./mm}^2$$

Please note: None of the following stresses in the table below are in the region where there is an access opening.

Summary of Stresses: (N./mm²)

Stress	Actual	Allowed	Result	Equation
Stot1pi	75.31	390.96	Pass	46
Stot1po	62.81	390.96	Pass	47
Stot1qi	86.93	395.82	Pass	48
Stot1qo	35.75	395.82	Pass	49
Stot2pi	79.28	391.62	Pass	50
Stot2po	56.84	391.62	Pass	51
Stot2qi	59.94	391.62	Pass	52
Stot2qo	76.19	391.62	Pass	53
Stot3pi	-4.06	413.99	Pass	54
Stot3po	7.02	413.99	Pass	55
Stot3qi	27.55	413.51	Pass	56
Stot3qo	-47.63	413.51	Pass	57

The stresses are satisfactory in the region without openings.

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Shop/Field Installation Options :

Platform(s) installed in the Field before being lifted.

Note : The CG is computed from the first Element From Node

Center of Gravity of the Platforms	10775.592 mm.
Center of Gravity of the Liquid	9952.295 mm.
Center of Gravity of the Nozzles	11952.957 mm.
Center of Gravity of the Added Weights (Operating)	16084.108 mm.
Center of Gravity of the Added Weights (Empty)	16084.108 mm.
Center of Gravity of the Trunnions	16549.998 mm.
Center of Gravity of Bare Shell New and Cold	8895.488 mm.
Center of Gravity of Bare Shell Corroded	8878.041 mm.
Vessel CG in the Operating Condition	10128.812 mm.
Vessel CG in the Fabricated (Shop/Empty) Condition	9971.522 mm.
Vessel CG in the Test Condition	9954.845 mm.

Rigging Analysis Results:

- Lifting weight based on the user selected calculated erected weight

Total Effective Length of Vessel for this analysis	17200.00 mm.
Total vessel weight (No Liquid)	Twt 144549.59 Kgf
Impact weight multiplication factor	Imp 1.30
Design lifting weight, DWT = Imp * Twt	187914.45 Kgf
Elevation of the Tailing Lug (bottom)	150.00 mm.
Elevation of the Lifting Lug (top)	16550.00 mm.
Design Reaction force at the tailing lug	75377.50 Kgf
Design Reaction force at the Trunnions	112536.95 Kgf

CG distance during field lifting operation	9971.52 mm.
CG distance from tailing lug	9821.52 mm.
CG distance from the nearer lifting lug/trunnion	6578.48 mm.

Compressive Allowable Increase Factor	1.00
---------------------------------------	------

Critical Values: (N./mm², mm.)

	Max Stress	Elevation	Allowables
Bending	3.00	2500.00	77.39 (UG-23)
Shear	1.50	16190.00	96.60 (0.7*S)

The compressive bending allowable stress is the minimum of the tensile allowable and the compressive allowable times the specified compressive allowable increase factor.

Forces and Moments at selected elevations (not all analysis points shown):

Distance mm.	Bending Moment Kg-m.	Bending Stress N./mm ²	Shear Force Kgf	Shear Stress N./mm ²
0.00	0.0	0.0	42751.2	1.1
2300.00	130933.2	2.8	23651.2	0.7
3350.00	164914.2	2.0	14779.9	0.3
8150.00	246787.3	2.9	20755.2	0.4
12950.00	152973.7	1.8	58225.5	1.0
17150.00	21946.0	0.3	5015.6	0.1

Unity Check (Actual Stress / Allowable Stress):

Maximum Unity Check is 0.0388 at elevation 2500.0 mm. - Must be <=1

Note: The rigging analysis is performed using a uniformly distributed load.



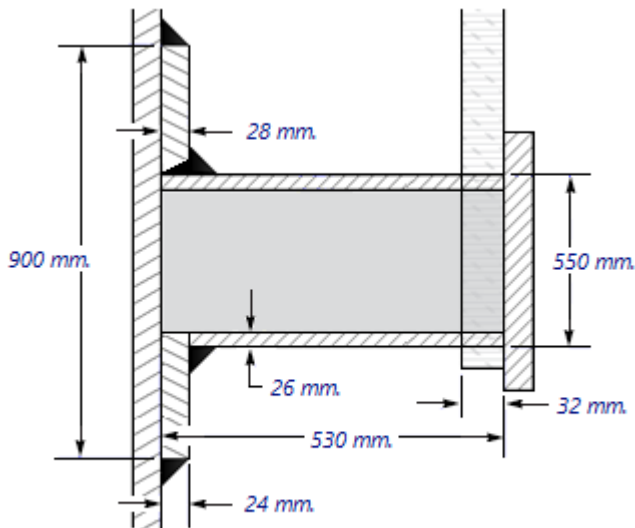
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Trunnion Analysis:

Input Data:

Lifting Lug Material	SA-516 70
Lifting Lug Yield Stress	Yield 262.00 N./mm ²
Trunnion Outside Diameter	Tod 550.000 mm.
Trunnion Thickness	Tthk 26.0000 mm.
Trunnion Length	Tl 530.0000 mm.
Reinforcing Pad Used	Yes
Pad Outside Diameter	Dp 900.000 mm.
Pad Thickness	Thkp 28.0000 mm.
Pad Edge Fillet Weld Thk	Tfp 24.0000 mm.
Bail/Sling Width	Tt 32.0000 mm.
Sling Angle from Horizontal	0.0 degrees
Occasional Load Factor	Occfac 1.0
Impact Factor	Impfac 1.30
Number of Lugs in Group	2



Trunnion may be inserted or abutting

Trunnion Force and Stress Results for different Lift Angles (Kgf&N./mm²):

Bending Stress Allowable:	172.9 N./mm ²
Shear Stress Allowable:	104.8 N./mm ²
Bearing Stress Allowable:	196.5 N./mm ²

Angle	Trunnion Lifting Load	Bending Stress	Shear Stress	Bearing Stress	Unity Check	Pass Fail
0	56268.5	53.0	12.9	31.4	0.429	Pass
5	56888.6	53.5	13.0	31.7	0.434	Pass
10	57497.8	54.1	13.2	32.0	0.439	Pass
15	58105.4	54.7	13.3	32.4	0.443	Pass
20	58720.9	55.3	13.5	32.7	0.448	Pass
25	59354.0	55.9	13.6	33.1	0.453	Pass
30	60015.7	56.5	13.8	33.4	0.458	Pass
35	60718.9	57.2	13.9	33.8	0.463	Pass
40	61479.6	57.9	14.1	34.3	0.469	Pass
45	62318.4	58.7	14.3	34.7	0.475	Pass
50	63263.2	59.5	14.5	35.3	0.483	Pass
55	64353.1	60.6	14.7	35.9	0.491	Pass
60	65645.4	61.8	15.0	36.6	0.501	Pass
65	67229.0	63.3	15.4	37.5	0.513	Pass
70	69249.3	65.2	15.9	38.6	0.528	Pass
75	71963.9	67.7	16.5	40.1	0.549	Pass



80	75876.4	71.4	17.4	42.3	0.579	Pass
85	82126.4	77.3	18.8	45.8	0.627	Pass
90	93957.2	88.4	21.5	52.4	0.717	Pass

Trunnion Calculation Results:

Trunnion Lifting Load [Tlf]:
= Resultant Load • Impact Factor
= 72274.8 • 1.3
= 93957.2 Kgf

Trunnion Section Modulus [Z]:
= $\pi (T_{od}^4 - T_{id}^4) / (32 \cdot T_{od})$
= $\pi (550.0^4 - 498.0^4) / (32 \cdot 550.0)$
= 5355033.0 mm.³

Trunnion Bending Moment [Tmom]:
= $(T_l - 0.5 \cdot T_t) / 9.807 \cdot T_{lf}$
= $(530.0 - 0.5 \cdot 32.0) / 9.807 \cdot 93957.23$
= 48295.0 Kg-m.

Trunnion Calculated Bending Stress [Sbc]:
= $T_{mom} \cdot 9.807 / Z$
= $48294.97 \cdot 9.807 / 5355033$
= 88.44 N./mm² [Passed] $S_{bc} < \min(0.66 \cdot SY \cdot Occfac, 0.75 \cdot Sy) (172.92)$

Trunnion Cross Section Area [Txsa]:
= $\pi/4 (T_{od}^2 - T_{id}^2)$
= $\pi/4 (550.0^2 - 498.0^2)$
= 428.0 cm²

Trunnion Calculated Shear Stress [Ssc]:
= T_{lf} / T_{xsa}
= $93957.23 / 428.01$
= 21.53 N./mm² [Passed] $S_{sc} < 0.4 \cdot Sy \cdot Occfac (104.80)$

Trunnion Unity Check [Uchk]:
= $S_{bc} / S_{ba} + S_{sc} / S_{sa}$
= $88.44 / 172.92 + 21.53 / 104.8$
= 0.717 [Passed]

Weld Section Modulus at Edge of Trunnion [Sw]:
= $\pi (T_{od}/2)^2$
= $\pi (550.0/2)^2$
= 237582.7 mm.²

Shear Stress in Weld due to Bending Moment [fs]:
= T_{mom} / S_w
= $48295.0 / 237582.7$
= 203.272 Kgf/mm.

Required Fillet Weld Leg Size at Edge of Trunnion [Wsr]:
= $\max(10\text{mm}, fs / (0.4 \cdot Sy))$
= $\max(10\text{mm}, 203.272 / (0.4 \cdot 262.0))$
= 19.022 mm.

Trunnion Bearing Contact Area [Tbca]:
= $T_{od} \cdot T_t$
= $550.0 \cdot 32.0$
= 176.0 cm²

Trunnion Bearing Stress [Sbec]:
= T_{lf} / T_{bca}
= $93957.23 / 176.0$
= 52.35 N./mm² [Passed] $S_{bec} < \min(0.75 \cdot SY \cdot Occfac, 0.9 \cdot Sy) (196.50)$

Summary of Results:



Trunnion Unity Check Value: 0.7169 Must be less than or equal to 1

Comparison of Stress Values versus Allowables: (N./mm²)

Type of Stress	Actual	Allowable	Pass/Fail
Bending Stress	88.44	172.92	Pass
Shear Stress	21.53	104.80	Pass
Bearing Stress	52.35	196.50	Pass

Results of local shell stress analysis due to loads on the trunnion:

This following Stress Analysis assumes that the Trunnion passes through the pad and is welded to the shell via a groove weld. Additionally the re-pad is welded to the shell at the outside edge via a fillet weld.

In the local stress results below, the 'Nozzle' is the 'Trunnion'.

WRC 107/537 Stress Analysis for the Lifting Lug to Shell Junction in the new and Cold Condition (no corrosion applied).

Loads on shell and WRC 107 results at different Lift Angles (Kgf & Kg-m.):

Angle	PT	PL	MC	ML	Rat1	Rat2	P/F
0	56268.5	0.0	28922.6	0.0	0.488	0.758	Pass
5	56672.1	4958.2	29130.1	2548.5	0.492	0.763	Pass
7	56707.2	6962.8	29148.1	3578.9	0.492	0.764	Pass
10	56624.3	9984.4	29105.4	5132.1	0.491	0.763	Pass
15	56125.5	15038.8	28849.1	7730.1	0.487	0.756	Pass
20	55179.6	20083.7	28362.9	10323.2	0.479	0.743	Pass
25	53793.0	25084.1	27650.1	12893.5	0.467	0.725	Pass
30	51975.1	30007.8	26715.7	15424.3	0.452	0.701	Pass
35	49738.0	34826.9	25565.8	17901.4	0.433	0.671	Pass
40	47096.1	39518.3	24207.9	20312.8	0.410	0.636	Pass
45	44065.8	44065.8	22650.3	22650.3	0.385	0.595	Pass
50	40664.8	48462.4	20902.1	24910.2	0.356	0.550	Pass
55	36911.4	52714.9	18972.8	27096.0	0.367	0.500	Pass
60	32822.7	56850.6	16871.2	29221.8	0.395	0.446	Pass
65	28412.2	60930.1	14604.2	31318.7	0.422	0.415	Pass
70	23684.7	65073.1	12174.2	33448.2	0.450	0.441	Pass
75	18625.6	69511.7	9573.8	35729.7	0.480	0.470	Pass
80	13175.8	74723.7	6772.5	38408.7	0.516	0.504	Pass
85	7157.8	81813.9	3679.2	42053.2	0.565	0.551	Pass
90	0.0	93957.2	0.0	48295.0	0.648	0.633	Pass

- PT - Transverse load per lug
- PL - Longitudinal load per lug
- MC - Circumferential moment in shell
- ML - Longitudinal moment in shell
- Rat1 - Computed WRC 107 stress ratio at trunnion edge
- Rat2 - Computed WRC 107 stress ratio at pad edge

Worst case combination loads were at: 7.0 degrees from horizontal.

Input Echo, WRC107/537 Item 1, Description: Trunnion

Diameter Basis for Vessel	Vbasis	ID
Cylindrical or Spherical Vessel	Cylsph	Cylindrical
Internal Corrosion Allowance	Cas	0.0000 mm.
Vessel Diameter	Dv	5700.000 mm.
Vessel Thickness	Tv	32.000 mm.
Design Temperature	T1	37.8 C
Attachment Type	Type	Round
Diameter Basis for Nozzle	Nbasis	OD

Corrosion Allowance for Nozzle	Can	0.0000	mm.
Nozzle Diameter	Dn	550.000	mm.
Nozzle Thickness	Tn	26.000	mm.
Nozzle Cold S.I. Allowable	SNmc	138.00	N./mm ²
Nozzle Hot S.I. Allowable	SNmh	138.00	N./mm ²
Thickness of Reinforcing Pad	Tpad	28.000	mm.
Diameter of Reinforcing Pad	Dpad	948.000	mm.
Design Internal Pressure	Dp	0.000	bars
Include Pressure Thrust	No		

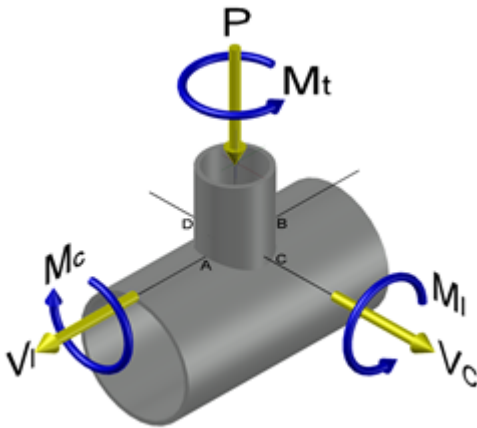
External Forces and Moments in WRC 107/537 Convention:

Radial Load (SUS)	P	0.0	Kgf
Longitudinal Shear (SUS)	Vl	6962.8	Kgf
Circumferential Shear (SUS)	Vc	56707.2	Kgf
Circumferential Moment (SUS)	Mc	29148.1	Kg-m.
Longitudinal Moment (SUS)	Ml	3578.9	Kg-m.
Torsional Moment (SUS)	Mt	0.0	Kg-m.

Use Interactive Control No
WRC107 Version Version March 1979

Include Pressure Stress Indices per Div. 2 No
Compute Pressure Stress per WRC-368 No
Local Loads applied at end of Nozzle/Attachment No

WRC Bulletin 537 provides equations for the dimensionless curves found in bulletin 107. As noted in the foreword to bulletin 537, "537 is equivalent to WRC 107". Where 107 is printed in the results below, "537" can be interchanged with "107".



Worst case combination loads were at: 7.0 degrees from horizontal.

WRC 107/537 has dimensional limitations that can be found in the bulletin's Foreword and which can also be inferred by review of the non-dimensional curve limits.

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
= NozzleOD + 2 • 1.65 • $\sqrt{R_{mean}(t - ca)}$
= 550.0 + 2 • 1.65 • $\sqrt{2866.0 (32.0 - 0)}$
= 1549.372 mm.

WRC 107 Stress Calculation for SUStained loads:

Radial Load	P	0.0	Kgf
Circumferential Shear	VC	56707.2	Kgf
Longitudinal Shear	VL	6962.8	Kgf
Circumferential Moment	MC	29148.1	Kg-m.
Longitudinal Moment	ML	3578.9	Kg-m.
Torsional Moment	MT	0.0	Kg-m.

Dimensionless Parameters used : Gamma = 48.00



Dimensionless Loads for Cylindrical Shells at Attachment Junction:

Curves read for 1979	Beta	Figure	Value	Location
N(PHI) / (P/Rm)	0.084	4C	8.521	(A,B)
N(PHI) / (P/Rm)	0.084	3C	7.696	(C,D)
M(PHI) / (P)	0.084	2C1	0.095	(A,B)
M(PHI) / (P)	0.084	1C	0.130	(C,D)
N(PHI) / (MC/(Rm**2 * Beta))	0.084	3A	1.357	(A,B,C,D)
M(PHI) / (MC/(Rm * Beta))	0.084	1A	0.098	(A,B,C,D)
N(PHI) / (ML/(Rm**2 * Beta))	0.084	3B	4.810	(A,B,C,D)
M(PHI) / (ML/(Rm * Beta))	0.084	1B	0.048	(A,B,C,D)
N(x) / (P/Rm)	0.084	3C	7.696	(A,B)
N(x) / (P/Rm)	0.084	4C	8.521	(C,D)
M(x) / (P)	0.084	1C1	0.134	(A,B)
M(x) / (P)	0.084	2C	0.095	(C,D)
N(x) / (MC/(Rm**2 * Beta))	0.084	4A	1.834	(A,B,C,D)
M(x) / (MC/(Rm * Beta))	0.084	2A	0.054	(A,B,C,D)
N(x) / (ML/(Rm**2 * Beta))	0.084	4B	1.281	(A,B,C,D)
M(x) / (ML/(Rm * Beta))	0.084	2B	0.077	(A,B,C,D)

Stress Concentration Factors: Kn = 1.00, Kb = 1.00

Stresses in the Vessel at the Attachment Junction (N./mm²)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Memb. P		-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0
Circ. Bend. P		-0.0	0.0	-0.0	0.0	-0.0	0.0	-0.0	0.0
Circ. Memb. MC		0.0	0.0	0.0	0.0	-9.3	-9.3	9.3	9.3
Circ. Bend. MC		0.0	0.0	0.0	0.0	-194.3	194.3	194.3	-194.3
Circ. Memb. ML		-4.1	-4.1	4.1	4.1	0.0	0.0	0.0	0.0
Circ. Bend. ML		-11.7	11.7	11.7	-11.7	0.0	0.0	0.0	0.0
Tot. Circ. Str.		-15.8	7.6	15.8	-7.6	-203.6	185.0	203.6	-185.0
Long. Memb. P		-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0
Long. Bend. P		-0.0	0.0	-0.0	0.0	-0.0	0.0	-0.0	0.0
Long. Memb. MC		0.0	0.0	0.0	0.0	-12.6	-12.6	12.6	12.6
Long. Bend. MC		0.0	0.0	0.0	0.0	-107.2	107.2	107.2	-107.2
Long. Memb. ML		-1.1	-1.1	1.1	1.1	0.0	0.0	0.0	0.0
Long. Bend. ML		-18.8	18.8	18.8	-18.8	0.0	0.0	0.0	0.0
Tot. Long. Str.		-19.9	17.7	19.9	-17.7	-119.8	94.6	119.8	-94.6
Shear VC		10.7	10.7	-10.7	-10.7	0.0	0.0	0.0	0.0
Shear VL		0.0	0.0	0.0	0.0	-1.3	-1.3	1.3	1.3
Shear MT		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Tot. Shear		10.7	10.7	-10.7	-10.7	-1.3	-1.3	1.3	1.3
Str. Int.		28.8	24.5	28.8	24.5	203.6	185.0	203.6	185.0

Dimensionless Parameters used : Gamma = 89.56

Dimensionless Loads for Cylindrical Shells at Pad Edge:

Curves read for 1979	Beta	Figure	Value	Location
N(PHI) / (P/Rm)	0.145	4C	12.217	(A,B)
N(PHI) / (P/Rm)	0.145	3C	7.602	(C,D)
M(PHI) / (P)	0.145	2C1	0.034	(A,B)
M(PHI) / (P)	0.145	1C	0.068	(C,D)
N(PHI) / (MC/(Rm**2 * Beta))	0.145	3A	3.547	(A,B,C,D)



M(PHI) / (MC/(Rm * Beta)) 0.145 1A 0.072 (A,B,C,D)
N(PHI) / (ML/(Rm**2 * Beta)) 0.145 3B 8.852 (A,B,C,D)
M(PHI) / (ML/(Rm * Beta)) 0.145 1B 0.023 (A,B,C,D)

N(x) / (P/Rm) 0.145 3C 7.602 (A,B)
N(x) / (P/Rm) 0.145 4C 12.217 (C,D)
M(x) / (P) 0.145 1C1 0.066 (A,B)
M(x) / (P) 0.145 2C ! 0.037 (C,D)
N(x) / (MC/(Rm**2 * Beta)) 0.145 4A 7.039 (A,B,C,D)
M(x) / (MC/(Rm * Beta)) 0.145 2A 0.034 (A,B,C,D)
N(x) / (ML/(Rm**2 * Beta)) 0.145 4B 3.535 (A,B,C,D)
M(x) / (ML/(Rm * Beta)) 0.145 2B 0.029 (A,B,C,D)

Note - The ! mark next to the figure name denotes curve value exceeded.

Stress Concentration Factors: Kn = 1.00, Kb = 1.00

Stresses in the Vessel at the Edge of Reinforcing Pad (N./mm²)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Memb. P		-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0
Circ. Bend. P		-0.0	0.0	-0.0	0.0	-0.0	0.0	-0.0	0.0
Circ. Memb. MC		0.0	0.0	0.0	0.0	-26.7	-26.7	26.7	26.7
Circ. Bend. MC		0.0	0.0	0.0	0.0	-289.5	289.5	289.5	-289.5
Circ. Memb. ML		-8.2	-8.2	8.2	8.2	0.0	0.0	0.0	0.0
Circ. Bend. ML		-11.3	11.3	11.3	-11.3	0.0	0.0	0.0	0.0
Tot. Circ. Str.		-19.4	3.1	19.4	-3.1	-316.1	262.8	316.1	-262.8
Long. Memb. P		-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0
Long. Bend. P		-0.0	0.0	-0.0	0.0	-0.0	0.0	-0.0	0.0
Long. Memb. MC		0.0	0.0	0.0	0.0	-52.9	-52.9	52.9	52.9
Long. Bend. MC		0.0	0.0	0.0	0.0	-137.7	137.7	137.7	-137.7
Long. Memb. ML		-3.3	-3.3	3.3	3.3	0.0	0.0	0.0	0.0
Long. Bend. ML		-14.3	14.3	14.3	-14.3	0.0	0.0	0.0	0.0
Tot. Long. Str.		-17.6	11.0	17.6	-11.0	-190.6	84.8	190.6	-84.8
Shear VC		11.7	11.7	-11.7	-11.7	0.0	0.0	0.0	0.0
Shear VL		0.0	0.0	0.0	0.0	-1.4	-1.4	1.4	1.4
Shear MT		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Tot. Shear		11.7	11.7	-11.7	-11.7	-1.4	-1.4	1.4	1.4
Str. Int.		30.2	24.7	30.2	24.7	316.2	262.8	316.2	262.8

WRC 107/537 Stress Summations:

Vessel Stress Summation at Attachment Junction (N./mm²)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Circ. Pl (SUS)		-4.1	-4.1	4.1	4.1	-9.3	-9.3	9.3	9.3
Circ. Q (SUS)		-11.7	11.7	11.7	-11.7	-194.3	194.3	194.3	-194.3
Long. Pm (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Long. Pl (SUS)		-1.1	-1.1	1.1	1.1	-12.6	-12.6	12.6	12.6
Long. Q (SUS)		-18.8	18.8	18.8	-18.8	-107.2	107.2	107.2	-107.2



Shear Pm (SUS)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear Pl (SUS)	10.7	10.7	-10.7	-10.7	-1.3	-1.3	1.3	1.3
Shear Q (SUS)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Pm (SUS)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Pm+Pl (SUS)	21.7	21.7	21.7	21.7	13.1	13.1	13.1	13.1
Pm+Pl+Q (Total)	28.8	24.5	28.8	24.5	203.6	185.0	203.6	185.0

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	0.00	138.00	Passed
Pm+Pl (SUS)	21.66	207.00	Passed
Pm+Pl+Q (TOTAL)	203.65	414.00	Passed

The Pm+Pl+Q allowable above was based on the rules of ASME VIII-2 Part 4, 4.1.6.3 (a)&(b) or per ASME VIII-2, 5.5.6.1(1)&(2) Part 5 as applicable.

The Pm+Pl+Q allowable was based on a temperature range cycling from ambient to design temperature.

WRC 107/537 Stress Summations:

Vessel Stress Summation at Reinforcing Pad Edge (N./mm²)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Circ. Pl (SUS)		-8.2	-8.2	8.2	8.2	-26.7	-26.7	26.7	26.7
Circ. Q (SUS)		-11.3	11.3	11.3	-11.3	-289.5	289.5	289.5	-289.5
Long. Pm (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Long. Pl (SUS)		-3.3	-3.3	3.3	3.3	-52.9	-52.9	52.9	52.9
Long. Q (SUS)		-14.3	14.3	14.3	-14.3	-137.7	137.7	137.7	-137.7
Shear Pm (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear Pl (SUS)		11.7	11.7	-11.7	-11.7	-1.4	-1.4	1.4	1.4
Shear Q (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Pm (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Pm+Pl (SUS)		23.9	23.9	23.9	23.9	53.0	53.0	53.0	53.0
Pm+Pl+Q (Total)		30.2	24.7	30.2	24.7	316.2	262.8	316.2	262.8

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	0.00	138.00	Passed
Pm+Pl (SUS)	52.98	207.00	Passed
Pm+Pl+Q (TOTAL)	316.15	414.00	Passed

The Pm+Pl+Q allowable above was based on the rules of ASME VIII-2 Part 4, 4.1.6.3 (a)&(b) or per ASME VIII-2, 5.5.6.1(1)&(2) Part 5 as applicable.

The Pm+Pl+Q allowable was based on a temperature range cycling from ambient to design temperature.



Skirt Data :

Skirt Outside Diameter at Base	SOD, D6	5756.0000	mm.
Skirt Thickness	STHK	22.0000	mm.
Skirt Internal Corrosion Allowance	SCA	1.0000	mm.
Skirt External Corrosion Allowance		0.0000	mm.
Skirt Material	SA-36		

Basing Input: Type of Geometry: Continuous Top Ring W/Gussets

Thickness of Basing	TBA, ea4	38.0000	mm.
Design Temperature of the Basing		38.00	°C
Basing Material		SA-36	
Basing Ope. All. Stress	BASOPE, f4	114.00	N./mm ²
Basing Yield Stress		248.00	N./mm ²
Inside Diameter of Basing	DI, D4	5588.0000	mm.
Outside Diameter of Basing	DOU, D5	6188.0000	mm.
Nominal Diameter of Bolts	BND, d(b0)	64.0000	mm.
Bolt Corrosion Allowance	BCA	0.0000	mm.
Root Area of a Single Bolt	Area	21.5100	cm ²
Bolt Material		SA-307 A	
Bolt Material Class/Thickness Range		6 <= t <= 100	
Bolt Operating Allowable Stress	SA, fB	138.00	N./mm ²
Number of Bolts	RN, nb	24	
Diameter of Bolt Circle	DC, DBC	5972.0000	mm.
Bolt Allowable Shear Stress		82.800	N./mm ²
Ultimate Comp. Strength of Concrete	FPC	41.0	N./mm ²
Allowable Comp. Strength of Concrete	FC	22.5	N./mm ²
Modular ratio Steel/Concrete		9.833	
Thickness of Gusset Plates	TGA, e7	22.0000	mm.
Width of Gussets at Top Plate	TWDT	206.0000	mm.
Width of Gussets at Base Plate	BWDT	206.0000	mm.
Gusset Plate Elastic Modulus	E	201734000.0	KPa.
Gusset Plate Yield Stress	SY	248.0	N./mm ²
Height of Gussets	HG, h1	312.0000	mm.
Distance between Gussets	RG, b6	120.0000	mm.
Dist. from Bolt Center to Gusset (Rg/2)	CG	60.0000	mm.
Number of Gussets per bolt	NG	2	
Thickness of Top Plate or Ring	TTA, e5	50.0000	mm.
Radial Width of the Top Plate	TOPWTH, b5	216.0000	mm.
Anchor Bolt Hole Dia. in Top Plate	dia,H	90.0000	mm.
External Corrosion Allowance	CA	1.0000	mm.
Dead Weight of Vessel	DW	144549.6	Kgf
Operating Weight of Vessel	ROW	145093.4	Kgf
Earthquake Moment on Basing	EQMOM	330317.7	Kg-m.
Wind Moment on Basing	WIMOM	73262.6	Kg-m.
Percent Bolt Preload	ppl	100.0	
Use AISC A5.2 Increase in Fc and Bolt Stress		No	
Use Allowable Weld Stress per AISC J2.5		No	
Factor for Increase of Allowables	Fact	1.0000	

Results for Brownell and Young Basing Analysis : Analyze Option

Note: This analysis is based on Neutral Axis shift method for Steel on Concrete (or a material with significantly different Young's modulus).

Calculation of Bolt Loads for Multiple Load Conditions:

Bolt Load, Earthquake Operating Case:

$$= ((4 \cdot M/DC) - W) / nb \text{ per Jawad \& Farr, Eq. 12.3}$$

$$= ((4 \cdot 330318/5972.0) - 140543) / 24$$

= 3362.4 Kgf

Bolt Load, Windload Empty Case:

$$= ((4 \cdot M/DC) - W) / nb \quad \text{per Jawad \& Farr, Eq. 12.3}$$

$$= ((4 \cdot 73263/5972.0) - 144550) / 24$$

$$= -3978.3 \text{ Kgf}$$

Bolt Load, Windload Vortex Shedding Case:

$$= ((4 \cdot M/DC) - W) / nb \quad \text{per Jawad \& Farr, Eq. 12.3}$$

$$= ((4 \cdot 0/5972.0) - 145093) / 24$$

$$= -6045.6 \text{ Kgf}$$

Governing Bolt Load Condition, Earthquake + Operating Condition:

Area Available in one Bolt Abs : 21.5100 cm²
Area Available in all the Bolts Abs * RN : 516.2400 cm²

Trial#	k	knew	Cc	Ct	z	j	Ft	Fc	Fs
2	0.800	0.701	2.662	1.217	0.316	0.775	13994.6	154537.9	903.1
4	0.651	0.626	2.335	1.639	0.356	0.783	6709.3	147252.5	321.5
6	0.613	0.620	2.253	1.732	0.366	0.784	5025.3	145568.5	227.8
8	0.623	0.624	2.273	1.709	0.363	0.784	5442.3	145985.5	250.0
10	0.623	0.624	2.275	1.707	0.363	0.784	5477.2	146020.4	251.9
12	0.624	0.624	2.276	1.706	0.363	0.784	5503.3	146046.5	253.3

The Actual Stress in a Single Bolt [Sbolt]:

$$= 2 \cdot Ft / (T1 \cdot Dc \cdot Ct)$$

$$= 2 \cdot 5503.3 / (2.752 \cdot 5972.0 \cdot 1.706)$$

$$= 3.851 \text{ N./mm}^2, \text{ Should be } \leq 138.0 \text{ N./mm}^2$$

Thickness of the Band of Bolting Steel [T1]

$$= RN \cdot \text{Bolt Area} / (\pi \cdot Dc)$$

$$= 24 \cdot 21.51 / (\pi \cdot 5972.0)$$

$$= 2.752 \text{ mm.}$$

Check the Bearing Stress in the Concrete [fc(max)]

$$= fc / [(2kd + t3) / (2kd)]$$

$$= 0.65 [(2 \cdot 0.624 \cdot 5972.0 + 300.0) / (2 \cdot 0.624 \cdot 5972.0)]$$

$$= 0.676 \text{ N./mm}^2, \text{ Should be } \leq 22.5 \text{ N./mm}^2$$

Values for table 10.3, l = 216.0, b = 390.867, l/b = 0.552618

Maximum Moment per unit width [Mmax]:

$$= \max(Mx, My) = \max(396.606, 932.354) = 932.354 \text{ Kgf}$$

Reqd Thickness of Basing, Brownell & Young Method [T]:

$$= \sqrt{(6 \cdot Mmax / fallow)} + Ca$$

$$= \sqrt{(6 \cdot 932.354 / 163.7)} + 1.0$$

$$= 19.308 \text{ mm.}$$

Nomenclature:

a = (Dc - Ds) / 2	Skirt Distance to Bolt Circle
P = Sa * Abs	Maximum Load on one Bolt
l	Radial Top Plate Width
lavg	Average Gusset Plate Width
g1 = Gamma 1	Constant Term f(b/l)
g2 = Gamma 2	Constant Term f(b/l)
g = Flat distance / 2	Nut 1/2 Dimension (from Tema)
Fb	Allowable Bending Stress

Values for table 10.6, l = 216.0, b = 120.0, b/l = 0.555556

As b/l (0.556) is less than 1, inverting b/l = 1.8.

Moment Term, based on geometry [Mo]:

$$= P / (4 \cdot \pi) [1.3 (\ln((2l \sin(\pi \cdot a/l)) / (\pi \cdot g))) + 1] - [(0.7 - g2)P / (4\pi)]$$

$$= 30268.49 / (4 \cdot \pi) [1.3 (\ln((2 \cdot 216.0 \cdot \sin(\pi \cdot 108.0/216.0)) / (\pi \cdot 53.975))) + 1] - [(0.7 - 0.04) \cdot 30268.49 / (4 \cdot \pi)]$$

$$= 3746.5251 \text{ Kgf}$$



Required Thickness of Continuous Top Ring [Tc]:

$$= \sqrt{\frac{6 \cdot \text{Abs}(\text{Mo})}{F_b}} + C_a$$

$$= \sqrt{\frac{6 \cdot \text{Abs}(3746.53)}{165.33}} + 1.0$$

$$= 37.5156 \text{ mm.}$$

Required Gusset Plate Thickness [tg]:

$$= P / (\text{Stress Term} \cdot l_{avg}) + C_a$$

$$= 30268.49 / (124.11 \cdot 206.0) + 1.0$$

$$= 12.61 \text{ (not less than } 9.525 + 1.0 \text{) mm.}$$

Bolt spacing [m]:

$$= \pi \cdot \text{Bolt Circle Diameter} / \text{number of Bolts}$$

$$= \pi \cdot 5972.0 / 24 = 781.733 \text{ mm.}$$

Req. Skirt Thk. to withstand Local Bending, (Brownell and Young) [t]:

$$= 1.76 \left(\frac{P \cdot a}{m \cdot (h + t_{ba})} \cdot 1.5 \cdot \text{Sklope} \right)^{2/3} \cdot r^{1/3} + C_a$$

$$= 1.76 (30268 \cdot 108.0 / (781.73 \cdot 350.0 \cdot 171))^{2/3} \cdot 2878.01^{1/3} + C_a$$

$$= 19.458 + 1.0 = 20.458 \text{ mm.}$$

Shear Stress in a Single Bolt [tb]:

$$= \text{Shear Force} / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 23660.7 / (21.51 \cdot 24)$$

$$= 4.5 \text{ N./mm}^2. \text{ Must be less than } 82.8 \text{ N./mm}^2.$$

Summary of Basing Thickness Calculations

Required Basing Thickness (tension)	19.3078	mm.
Actual Basing Thickness as entered by user	38.0000	mm.
Required Thickness of Chair Cap	37.5156	mm.
Actual Top Ring Thickness as entered by user	50.0000	mm.
Required Gusset thickness, + CA	12.6104	mm.
Actual Gusset Thickness as entered by user	22.0000	mm.
Required Thickness of Skirt for Local Stress	20.4579	mm.
Given Thickness of Skirt	22.0000	mm.
Required Gusset Height to meet local stress	274.1634	mm.

Weld Size Calculations per Steel Plate Engineering Data - Vol. 2

Compute the Weld load at the Skirt/Base Junction [W]:

$$= \text{SkirtStress} \cdot (\text{SkirtThickness} - C_A)$$

$$= 15.166 \cdot (22.0 - 1.0)$$

$$= 32.48 \text{ Kg/mm.}$$

Results for Computed Minimum Basing Weld Size [BWeld]:

$$= W / [(0.4 \cdot \text{Yield}) \cdot 2 \cdot 0.707]$$

$$= 32 / [(0.4 \cdot 230) \cdot 2 \cdot 0.707]$$

$$= 2.447 \text{ mm.}$$

Results for Computed Minimum Gusset and Top Plate to Skirt Weld Size

Vertical Plate Load [Wv]:

$$= \text{Bolt Load} / (\text{Cmwth} + 2 \cdot (\text{Hg} + \text{Tta}))$$

$$= 30268.5 / (170.0 + 2 \cdot (312.0 + 50.0))$$

$$= 33.857 \text{ Kg/mm.}$$

Horizontal Plate Load [Wh]:

$$= \text{Bolt Load} \cdot e / (\text{Cmwth} \cdot (\text{Hg} + \text{Tta}) + 0.6667 \cdot (\text{Hg} + \text{Tta})^2)$$

$$= 30268.5 \cdot 108.0 / (170.0 \cdot (362.0) + 0.6667 \cdot (362.0)^2)$$

$$= 21.954 \text{ Kg/mm.}$$

Resultant Weld Load [Wr]:

$$= \sqrt{W_v^2 + W_h^2}$$

$$= \sqrt{33.86^2 + 21.95^2}$$

$$= 40.352 \text{ Kg/mm.}$$

Results for Computed Min Gusset and Top Plate to Skirt Weld Size [GsWeld]:



$$= W_r / [(0.4 \cdot \text{Yield}) \cdot 2 \cdot 0.707]$$

$$= 40.35 / [(0.4 \cdot 230) \cdot 2 \cdot 0.707]$$

$$= 3.041 \text{ mm.}$$

Results for Computed Minimum Gusset to Top Plate Weld Size

Weld Load [Wv]:

$$= \text{Bolt Load} / (2 \cdot \text{TopWth})$$

$$= 30268.5 / (2 \cdot 206.0)$$

$$= 73.467 \text{ Kg/mm.}$$

Weld Load [Wh]:

$$= \text{Bolt Load} \cdot e / (2 \cdot \text{Hgt} \cdot \text{TopWth})$$

$$= 30268.5 \cdot 108.0 / (2 \cdot 362.0 \cdot 206.0)$$

$$= 21.918 \text{ Kg/mm.}$$

Resultant Weld Load [Wr]:

$$= \sqrt{W_v^2 + W_h^2}$$

$$= \sqrt{73.47^2 + 21.92^2}$$

$$= 76.667 \text{ Kg/mm.}$$

Results for Computed Min Gusset to Top Plate Weld Size [GtpWeld]:

$$= W_r / [(0.4 \cdot \text{Yield}) \cdot 2 \cdot 0.707]$$

$$= 76.67 / [(0.4 \cdot 230) \cdot 2 \cdot 0.707]$$

$$= 5.778 \text{ mm.}$$

Note: The calculated weld sizes need not exceed the component thickness framing into the weld. At the same time, the weld must meet a minimum size specification which is 3/16 in. (4.76 mm) or 1/4 in. (6.35 mm), depending on the component thickness.

Summary of Required Weld Sizes:

Required Basinger to Skirt Double Fillet Weld Size	6.3500	mm.
Required Gusset to Skirt Double Fillet Weld Size	6.3500	mm.
Required Top Plate to Skirt Weld Size	6.3500	mm.
Required Gusset to Top Plate Double Fillet Weld Size	5.7776	mm.

Tailing Lug Stress Calculation Results :

Tail Lug Type:	Dual			
Tail lug yield stress	Syt	248.00	N./mm ²	
Tail lug allowable stress	Sat	114.00	N./mm ²	
Tail lug thickness	TTL	80.0000	mm.	
Pin hole diameter	DPH	60.0000	mm.	
Fillet weld size	WS	30.0000	mm.	
Tail lug load	TLOAD	37688	Kgf	

Decay Length Along the Skirt (e):

$$= 0.5 \cdot \sqrt{(DS/2 \cdot T_s)}$$

$$= 0.5 \cdot \sqrt{(5756.0/2 \cdot 22.0)}$$

$$= 125.81 \text{ mm.}$$

Skirt Effective Length :

$$= \text{HG} + \text{TTA} + e$$

$$= 312.0 + 50.0 + 125.81$$

$$= 487.81 \text{ mm.}$$

Section	CG (Y) mm.	Width (H) mm.	Area (A) cm ²	Moment Area mm. ³
Top plate, t	192.00	216.00	108.00	2073600.75
Skirt, s	73.00	487.81	107.32	783429.31
Bottom plate, b	150.00	300.00	114.00	1710001.12
TOTAL			329.32	4567031.00

Calculated Overall CG (Y):

$$= M_{\text{total}} / A_{\text{total}}$$

$$= 4567031/329.32$$



$$= 138.68 \text{ mm.}$$

Calculated Moment of Inertia (I):

$$= (A_t \cdot Y_t^2 + Topw_{th}^3 \cdot TTA/12 + A_s \cdot Y_s^2 + Ts^3 \cdot Hs/12 + Ab \cdot Y_b^2 + Hb^3 \cdot Tb/12) - M_{total}^2/A_{total}$$

$$= (108.0 \cdot 192.0^2 + 216.0^3 \cdot 50.0/12 + 107.32 \cdot 73.0^2 + 22.0^3 \cdot 487.81/12 + 114.0 \cdot 150.0^2 + 300.0^3 \cdot 38.0/12) - 4567031^2/329.32$$

$$= 20638.44 \text{ cm}^4$$

$$\alpha = \arcsin(OS / (DS / 2))$$

$$= \arcsin(1000.0/(5756.0/2))$$

$$= 0.355 \text{ rad. (20.332 deg.)}$$

$$K = \sin(\alpha) \cdot (\sin(\alpha) - \pi + \alpha) + k_2 \cdot (1 + \cos(\alpha))$$

$$= \sin(0.35) \cdot (\sin(0.35) - 3.14 + 0.35) + 1.0 \cdot (1 + \cos(0.35))$$

$$= 1.09$$

Required Section Modulus (Zr) - per Roark's Table 17 ref 4

$$= K \cdot TLOAD \cdot (DI/2 + Y) / (0.66 \cdot S_y \cdot \pi)$$

$$= 1.09 \cdot 37688.75 \cdot (5588.0/2 + 138.68) / (0.66 \cdot 248.0 \cdot 3.142)$$

$$= 2297995.25 \text{ mm.}^3$$

Calculated Section Modulus (Zc):

$$= I / Y$$

$$= 20638.44/138.68$$

$$= 1488195.75 \text{ mm.}^3 \text{ ...[Insufficient section modulus!]}$$

Allowable Shear at the Pin Hole (Ssa):

$$= 0.4 \cdot S_{yt}$$

$$= 0.4 \cdot 248.0$$

$$= 99.20 \text{ N./mm}^2$$

Tail Lug Radius (R):

$$= (TBA + HG + TTA) / 2$$

$$= (38.0 + 312.0 + 50.0) / 2$$

$$= 200.00 \text{ mm.}$$

Calculated Shear at the Pin Hole (Ssc):

$$= TLOAD / (2 \cdot TTL \cdot (R - DPH / 2))$$

$$= 37688.8 / (2 \cdot 80.0 \cdot (200.0 - 60.0/2))$$

$$= 13.59 \text{ N./mm}^2$$

Allowable Bearing Stress at the Pin Hole (Sba):

$$= 0.75 \cdot S_{yt}$$

$$= 0.75 \cdot 248.0$$

$$= 186.00 \text{ N./mm}^2$$

Calculated Bearing Stress at the Pin Hole (Sbc):

$$= TLOAD / (DPH \cdot TTL)$$

$$= 37688.8 / (60.0 \cdot 80.0)$$

$$= 77.00 \text{ N./mm}^2$$

Weld Length (LW):

$$= 2 \cdot (TOPWTH + TBA + TTA + HG + (DOU - DS) / 2)$$

$$= 2 \cdot (216.0 + 38.0 + 50.0 + 312.0 + (6188.0 - 5756.0)/2)$$

$$= 1651.00 \text{ mm.}$$

Allowable Weld Stress (Swa):

$$= 0.49 \cdot S_{at}$$

$$= 0.49 \cdot 114.0$$

$$= 55.86 \text{ N./mm}^2$$

Calculated Weld Stress (Swc):

$$= TLOAD / (0.707 \cdot WS \cdot LW)$$

$$= 37688.8 / (0.707 \cdot 30.0 \cdot 1651.0)$$

$$= 10.55 \text{ N./mm}^2$$



Tailing Lug Results Summary (mm.³ & N./mm²):

	Actual	Allowable or Required	Pass/Fail
<i>Section Modulus</i>	<i>1488195.75</i>	<i>2297995.25</i>	<i>Failed</i>
Shear Stress	13.59	99.20	Passed
Bearing Stress	77.00	186.00	Passed
Weld Stress	10.55	55.86	Passed

Review notes about nozzle loadings and supports in the Warnings report.

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Input, Nozzle Desc: N2-3"

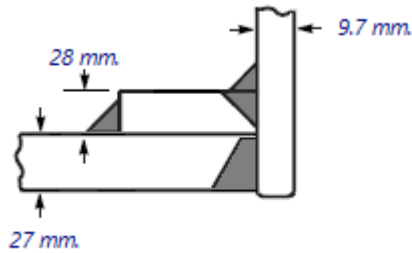
From: 30

Pressure for Reinforcement Calculations	P	12.290	bars
Temperature for Internal Pressure	Temp	82	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Elliptical Head	D	5700.00	mm.
Aspect Ratio of Elliptical Head	Ar	2.00	
Head Finished (Minimum) Thickness	t	27.0000	mm.
Head Internal Corrosion Allowance	c	1.6000	mm.
Head External Corrosion Allowance	co	0.0000	mm.
Distance from Head Centerline	L1	0.0000	mm.
User Entered Minimum Design Metal Temperature		-1.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-106 B	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	118.00	N./mm ²
Allowable Stress At Ambient	Sna	118.00	N./mm ²
Diameter Basis (for tr calc only)		Outside	
Layout Angle		0.00	deg
Diameter		3.0000	in.
Size and Thickness Basis		Minimum	
Nominal Thickness		160	
Flange Material		SA-105	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	1.6000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	200.0000	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	26.5000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Pad Material		SA-516 70	
Pad Allowable Stress at Temperature	Sp	138.00	N./mm ²
Pad Allowable Stress At Ambient	Spa	138.00	N./mm ²
Diameter of Pad along vessel surface	Dp	190.0000	mm.
Thickness of Pad	te	28.0000	mm.
Weld leg size between Pad and Shell	Wp	20.0000	mm.
Groove weld depth between Pad and Nozzle	Wgpn	28.0000	mm.
Reinforcing Pad Width		50.5500	mm.
Flange Class		300	
Flange Grade		GR 1.1	
Piping Exit Angle [North Clockwise reference]		0.0	deg
Bend Radius Multiplier		1.50	
Horizontal Run Length		3000.0	mm.
Centerline Distance to Tangent		1800.0	mm.

The Pressure Design option was MAWP + static head (to the nozzle).



Reinforcement CALCULATION, Description: N2-3"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	3.500 in.
Actual Thickness Used in Calculation	0.383 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Elliptical Head, tr [Int. Press]
 $= P \cdot D \cdot K1 / (2 \cdot Sv \cdot E - 0.2 \cdot P)$ per Appendix 1-4(c)
 $= 12.29 \cdot 5703.2 \cdot 0.8995 / (2 \cdot 138.0 \cdot 1.0 - 0.2 \cdot 12.29)$
 $= 22.8648 \text{ mm.}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot Ro / (Sn \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 12.29 \cdot 44.45 / (118 \cdot 1.0 + 0.4 \cdot 12.29)$
 $= 0.4610 \text{ mm.}$

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	145.2618 mm.
Parallel to Vessel Wall, opening length	d	72.6309 mm.
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		48.3364 mm.

The Pad diameter is greater than the Diameter Limit. The excess will not be considered.

Weld Strength Reduction Factor [fr1]:
 $= \min(1, Sn/Sv)$
 $= \min(1, 118.0/138.0)$
 $= 0.855$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, Sn/Sv)$
 $= \min(1, 118.0/138.0)$
 $= 0.855$

Weld Strength Reduction Factor [fr4]:
 $= \min(1, Sp/Sv)$
 $= \min(1, 138.0/138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(0.855, 1.0)$
 $= 0.855$

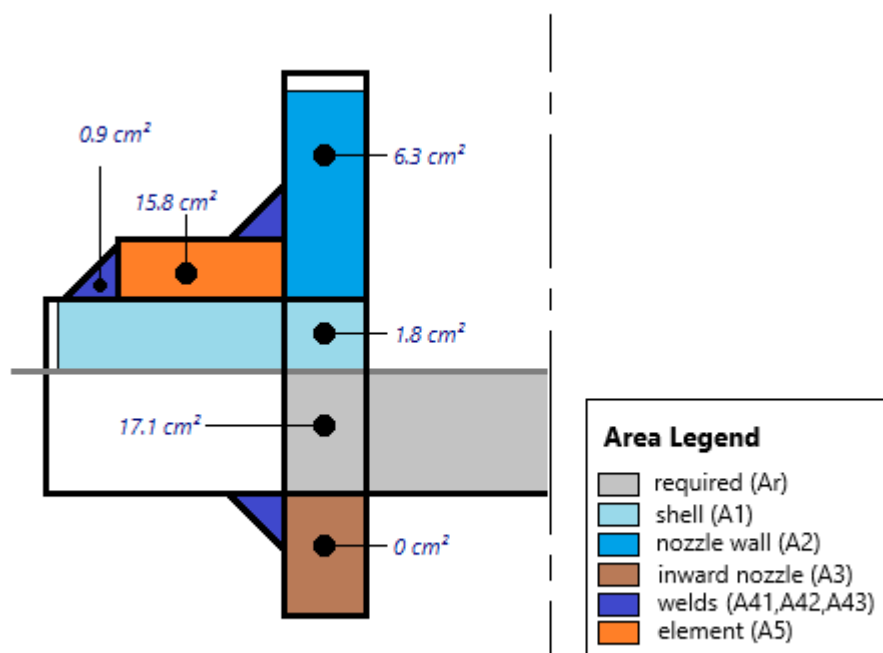


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	17.146	NA	NA
Area in Shell	A1	1.782	NA	NA
Area in Nozzle Wall	A2	6.343	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	0.855	NA	NA
Area in Element	A5	15.781	NA	NA
TOTAL AREA AVAILABLE	Atot	24.761	NA	NA

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Insufficient.
The area available with the given pad is Sufficient.

SELECTION OF POSSIBLE REINFORCING PADS: Diameter Thickness
Based on given Pad Thickness: 118.0656 28.0000 mm.
Based on given Pad Diameter: 190.0000 14.4892 mm.
Based on the Estimated Diameter Limit: 143.6743 14.9091 mm.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1))$ UG-37(c)
 $= (72.6309 \cdot 22.8648 \cdot 1.0 + 2 \cdot 8.1346 \cdot 22.8648 \cdot 1.0 \cdot (1 - 0.86))$
 $= 17.146 \text{ cm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 72.631(1.0 \cdot 25.4 - 1.0 \cdot 22.865) - 2 \cdot 8.135$
 $(1.0 \cdot 25.4 - 1.0 \cdot 22.8648) \cdot (1 - 0.855)$
 $= 1.782 \text{ cm}^2$

Area Available in Nozzle Wall Projecting Outward [A2]:
 $= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2$



$$= (2 \cdot 48.34) \cdot (8.13 - 0.46) \cdot 0.8551$$
$$= 6.343 \text{ cm}^2$$

Area Available in Welds [A41 + A42 + A43]:

$$= W_o^2 \cdot fr3 + (W_i - can / 0.707)^2 \cdot fr2 + W_p^2 \cdot fr4$$
$$= 10.0^2 \cdot 0.86 + (0.0)^2 \cdot 0.86 + 0.0^2 \cdot 1.0$$
$$= 0.855 \text{ cm}^2$$

Area Available in Element [A5]:

$$= (\min(D_p, D_L) - (\text{Nozzle OD})) \cdot (\min(t_p, T_{lwp}, t_e)) \cdot fr4$$
$$= (145.2618 - 88.9) \cdot 28.0 \cdot 1.0$$
$$= 15.781 \text{ cm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	$t_a = 2.0610 \text{ mm.}$
Wall Thickness per UG16(b),	$tr_{16b} = 4.1000 \text{ mm.}$
Wall Thickness, shell/head, internal pressure	$trb1 = 27.0000 \text{ mm.}$
Wall Thickness	$tb1 = \max(trb1, tr_{16b}) = 27.0000 \text{ mm.}$
Wall Thickness	$tb2 = \max(trb2, tr_{16b}) = 4.1000 \text{ mm.}$
Wall Thickness per table UG-45	$tb3 = 6.4000 \text{ mm.}$

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb3, \max(tb1, tb2)]$$
$$= \min[6.4, \max(27.0, 4.1)]$$
$$= 6.4000 \text{ mm.}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(t_a, t_b)$$
$$= \max(2.061, 6.4)$$
$$= 6.4000 \text{ mm.}$$

Available Nozzle Neck Thickness = 9.7345 mm. -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 38.5,	Allowable	: 118.0 N./mm ²	Passed
Expansion	: 0.0,	Allowable	: 256.5 N./mm ²	Passed
Occasional	: 2.5,	Allowable	: 156.9 N./mm ²	Passed
Shear	: 25.1,	Allowable	: 82.6 N./mm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

$$\text{Govrn. thk, } t_g = 9.735, tr = 0.461, c = 1.6 \text{ mm., } E^* = 1.0$$
$$\text{Thickness Ratio} = tr \cdot E^* / (t_g - c) = 0.057, \text{ Temp. Reduction} = 78 \text{ }^\circ\text{C}$$

Min Metal Temp. w/o impact per UCS-66, Curve A	-8 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle Neck to Pad Weld for the Nozzle, Curve: B

$$\text{Govrn. thk, } t_g = 9.735, tr = 0.461, c = 1.6 \text{ mm., } E^* = 1.0$$
$$\text{Thickness Ratio} = tr \cdot E^* / (t_g - c) = 0.057, \text{ Temp. Reduction} = 78 \text{ }^\circ\text{C}$$

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: B

$$\text{Govrn. thk, } t_g = 9.735, tr = 0.461, c = 1.6 \text{ mm., } E^* = 1.0$$
$$\text{Thickness Ratio} = tr \cdot E^* / (t_g - c) = 0.057, \text{ Temp. Reduction} = 78 \text{ }^\circ\text{C}$$

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 °C
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Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Shell to Pad Weld Junction at Pad OD, min(Curve:B, Curve:D)

Govrn. thk, tg = 27.0, tr = 22.865, c = 1.6 mm., E* = 1.0
 Thickness Ratio = $tr \cdot E^*/(tg - c) = 0.9$, Temp. Reduction = 6 °C

Min Metal Temp. w/o impact per UCS-66, Curve B 1 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -4 °C

Nozzle-Shell/Head Weld (UCS-66(a)(b)), min(Curve:B, Curve:D)

Govrn. thk, tg = 9.735, tr = 0.461, c = 1.6 mm., E* = 1.0
 Thickness Ratio = $tr \cdot E^*/(tg - c) = 0.057$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -29 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Gov. MDMT of the Nozzle : -104 °C
 Gov. MDMT of the Reinforcement Pad : -4 °C
 Gov. MDMT of the nozzle to shell joint welded assembly : -4 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
 Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = $12.29/51.00 = 0.241$

Weld Size Calculations, Description: N2-3"

Intermediate Calc. for nozzle/shell Welds Tmin 8.1346 mm.
 Intermediate Calc. for pad/shell Welds TminPad 19.0000 mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$5.6942 = 0.7 \cdot t_{min}$	$7.0700 = 0.7 \cdot W_o$ mm.
Pad Weld	$9.5000 = 0.5 \cdot T_{minPad}$	$14.1400 = 0.7 \cdot W_p$ mm.

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A-A1+2 \cdot t_n \cdot f r_1 \cdot (E1 \cdot t - tr)) \cdot S_v)$
 $= \max(0, (17.1461 - 1.7815 + 2 \cdot 8.1346 \cdot 0.8551 \cdot (1.0 \cdot 25.4 - 22.8648)) \cdot 138)$
 $= 22116.96 \text{ Kg}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2+A5+A4-(W_i-Can/.707)^2 \cdot f r_2) \cdot S_v$
 $= (6.3431 + 15.7813 + 0.8551 - 0.0 \cdot 0.86) \cdot 138$
 $= 32336.30 \text{ Kg}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f r_1)) \cdot S_v$
 $= (6.3431 + 0.0 + 0.8551 + (3.5335)) \cdot 138$
 $= 15101.34 \text{ Kg}$

Weld Load [W3]:
 $= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f r_1)) \cdot S$
 $= (6.3431 + 0.0 + 0.8551 + 15.7813 + (3.5335)) \cdot 138$
 $= 37308.52 \text{ Kg}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$



$$= (3.1416/2.0) \cdot 88.9 \cdot 10.0 \cdot 0.49 \cdot 118$$

$$= 8233. \text{ Kg}$$

Shear, Pad Element Weld [Spew]:

$$= (\pi/2) \cdot DP \cdot WP \cdot 0.49 \cdot SEW$$

$$= (3.1416/2.0) \cdot 190.0 \cdot 20.0 \cdot 0.49 \cdot 138$$

$$= 41158. \text{ Kg}$$

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (Dlr + Dlo) / 4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn$$

$$= (3.1416 \cdot 40.3827) \cdot (9.7345 - 1.6) \cdot 0.7 \cdot 118$$

$$= 8692. \text{ Kg}$$

Tension, Pad Groove Weld [Tpgw]:

$$= (\pi/2) \cdot Dlo \cdot Wgpn \cdot 0.74 \cdot Seg$$

$$= (3.1416/2) \cdot 88.9 \cdot 28.0 \cdot 0.74 \cdot 138$$

$$= 40716. \text{ Kg}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot Dlo \cdot (T - cas) \cdot 0.74 \cdot Sng$$

$$= (3.1416/2.0) \cdot 88.9 \cdot (27.0 - 1.6) \cdot 0.74 \cdot 138$$

$$= 36935. \text{ Kg}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SPEW + SNW) = (41158 + 8692) = 49850 \text{ Kg}$$

Failure Path [Path 2-2]:

$$= (Sonw + Tpgw + Tngw + Sinw)$$

$$= (8233 + 40716 + 36935 + 0) = 85884 \text{ Kg}$$

Failure Path [Path 3-3]:

$$= (Spew + Tngw + Sinw)$$

$$= (41158 + 36935 + 0) = 78093 \text{ Kg}$$

Summary of Failure Path Calculations:

Path 1-1 = 49849	Kgf, must exceed W = 22116 Kg	or W1 = 32336 Kg
Path 2-2 = 85883	Kgf, must exceed W = 22116 Kg	or W2 = 15101 Kg
Path 3-3 = 78092	Kgf, must exceed W = 22116 Kg	or W3 = 37308 Kg

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 12 bars

Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 0.1918 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 227.1917 mm.

Input Echo, WRC107/537 Item 1, Description: N2-3" :

Diameter Basis for Vessel	Vbasis	ID	
Cylindrical or Spherical Vessel	Cylsph	Spherical	
Internal Corrosion Allowance	Cas	1.6000	mm.
Vessel Diameter	Dv	10260.000	mm.
Vessel Thickness	Tv	27.000	mm.
Design Temperature	T1	82.0	C
Vessel Material		SA-516 70	
Vessel UNS Number		K02700	
Vessel Cold S.I. Allowable	Smc	138.00	N./mm ²
Vessel Hot S.I. Allowable	Smh	138.00	N./mm ²

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Attachment Type	Type	Round
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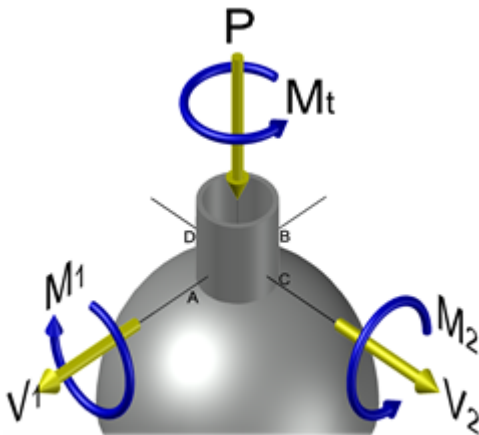
WRC107 Attachment Classification	Holsol	Hollow
Diameter Basis for Nozzle	Nbasis	OD
Corrosion Allowance for Nozzle	Can	1.6000 mm.
Nozzle Diameter	Dn	88.900 mm.
Nozzle Thickness	Tn	9.735 mm.
Nozzle Material		SA-106 B
Nozzle UNS Number		K03006
Nozzle Cold S.I. Allowable	SNmc	118.00 N./mm ²
Nozzle Hot S.I. Allowable	SNmh	118.00 N./mm ²
Thickness of Reinforcing Pad	Tpad	28.000 mm.
Diameter of Reinforcing Pad	Dpad	190.000 mm.
Design Internal Pressure	Dp	12.290 bars
Include Pressure Thrust		No

External Forces and Moments in WRC 107/537 Convention:

Radial Load (SUS)	P	611.9	Kgf
Longitudinal Shear (SUS)	(V1) V1	407.9	Kgf
Circumferential Shear (SUS)	(Vc) V2	407.9	Kgf
Circumferential Moment (SUS)	(Mc) M1	91.3	Kg-m.
Longitudinal Moment (SUS)	(Ml) M2	91.3	Kg-m.
Torsional Moment (SUS)	Mt	130.5	Kg-m.

Use Interactive Control	No
WRC107 Version	Version March 1979
Include Pressure Stress Indices per Div. 2	No
Compute Pressure Stress per WRC-368	No
Local Loads applied at end of Nozzle/Attachment	No

WRC Bulletin 537 provides equations for the dimensionless curves found in bulletin 107. As noted in the foreword to bulletin 537, "537 is equivalent to WRC 107". Where 107 is printed in the results below, "537" can be interchanged with "107".



Error/Warning Messages for WRC 107/537 Analysis: 1
Warning - The ratio Ro/Rm, 0.0086, must be > 0.0571 and < 0.571.

WRC 107/537 has dimensional limitations that can be found in the bulletin's Foreword and which can also be inferred by review of the non-dimensional curve limits.

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
= NozzleOD + 2 • 1.65 • $\sqrt{R_{mean}(t - ca)}$
= 88.9 + 2 • 1.65 • $\sqrt{5144.3 (27.0 - 1.6)}$
= 1281.772 mm.

WRC 107 Stress Calculation for SUStained loads:
Radial Load P 611.9 Kgf

Circumferential Shear	(VC) V2	407.9	Kgf
Longitudinal Shear	(VL) V1	407.9	Kgf
Circumferential Moment	(MC) M1	91.3	Kg-m.
Longitudinal Moment	(ML) M2	91.3	Kg-m.
Torsional Moment	MT	130.5	Kg-m.

Unitless Prm: U = 0.08 TAU = 5.00 (4.96) RHO = 4.00 (6.56)

Dimensionless Loads for Spherical Shells at Attachment Junction:

Curves read for 1979	Figure	Value	Location
N(x) * T / P	SP 4	0.11815	(A,B,C,D)
M(x) / P	SP 4	0.04494	(A,B,C,D)
N(x) * T * SQRT(Rm * T) / MC	SM 4	0.62112	(A,B,C,D)
M(x) * SQRT(Rm * T) / MC	SM 4	0.26888	(A,B,C,D)
N(x) * T * SQRT(Rm * T) / ML	SM 4	0.62112	(A,B,C,D)
M(x) * SQRT(Rm * T) / ML	SM 4	0.26888	(A,B,C,D)
N(y) * T / P	SP 4	0.31974	(A,B,C,D)
M(y) / P	SP 4	0.35674	(A,B,C,D)
N(y) * T * SQRT(Rm * T) / MC	SM 4	0.10351	(A,B,C,D)
M(y) * SQRT(Rm * T) / MC	SM 4	2.31825	(A,B,C,D)
N(y) * T * SQRT(Rm * T) / ML	SM 4	0.10351	(A,B,C,D)
M(y) * SQRT(Rm * T) / ML	SM 4	2.31825	(A,B,C,D)

Stress Concentration Factors: Kn = 1.00, Kb = 1.00

Stresses in the Vessel at the Attachment Junction (N./mm²)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad.	Memb. P	-0.2	-0.2	-0.2	-0.2	-0.2	-0.2	-0.2	-0.2
Rad.	Bend. P	-0.6	0.6	-0.6	0.6	-0.6	0.6	-0.6	0.6
Rad.	Memb. MC	0.0	0.0	0.0	0.0	-0.4	-0.4	0.4	0.4
Rad.	Bend. MC	0.0	0.0	0.0	0.0	-1.0	1.0	1.0	-1.0
Rad.	Memb. ML	-0.4	-0.4	0.4	0.4	0.0	0.0	0.0	0.0
Rad.	Bend. ML	-1.0	1.0	1.0	-1.0	0.0	0.0	0.0	0.0
Tot. Rad. Str.		-2.2	0.9	0.5	-0.3	-2.2	0.9	0.5	-0.3
Tang.	Memb. P	-0.7	-0.7	-0.7	-0.7	-0.7	-0.7	-0.7	-0.7
Tang.	Bend. P	-4.5	4.5	-4.5	4.5	-4.5	4.5	-4.5	4.5
Tang.	Memb. MC	0.0	0.0	0.0	0.0	-0.1	-0.1	0.1	0.1
Tang.	Bend. MC	0.0	0.0	0.0	0.0	-8.3	8.3	8.3	-8.3
Tang.	Memb. ML	-0.1	-0.1	0.1	0.1	0.0	0.0	0.0	0.0
Tang.	Bend. ML	-8.3	8.3	8.3	-8.3	0.0	0.0	0.0	0.0
Tot. Tang. Str.		-13.6	12.1	3.2	-4.4	-13.6	12.1	3.2	-4.4
Shear VC		0.5	0.5	-0.5	-0.5	0.0	0.0	0.0	0.0
Shear VL		0.0	0.0	0.0	0.0	-0.5	-0.5	0.5	0.5
Shear MT		1.9	1.9	1.9	1.9	1.9	1.9	1.9	1.9
Tot. Shear		2.5	2.5	1.4	1.4	1.4	1.4	2.5	2.5
Str. Int.		14.1	12.6	3.9	5.0	13.7	12.3	5.6	6.5

Unitless Prm: U = 0.26 TAU = 0.00 (11.18) RHO = 0.00 (3.12)

Dimensionless Loads for Spherical Shells at Pad edge:

Curves read for 1979	Figure	Value	Location
N(x) * T / P	SR 2	0.20435	(A,B,C,D)
M(x) / P	SR 2	0.15833	(A,B,C,D)
N(x) * T * SQRT(Rm * T) / MC	SR 3	0.16259	(A,B,C,D)

$M(x) * T / P$ SR 3 0.56487 (A,B,C,D)
 $N(x) * T * \text{SQRT}(R_m * T) / MC$ SR 3 0.16259 (A,B,C,D)
 $M(x) * T * \text{SQRT}(R_m * T) / ML$ SR 3 0.56487 (A,B,C,D)

 $N(y) * T / P$ SR 2 0.06132 (A,B,C,D)
 $M(y) / P$ SR 2 0.04755 (A,B,C,D)
 $N(y) * T * \text{SQRT}(R_m * T) / MC$ SR 3 0.04968 (A,B,C,D)
 $M(y) * \text{SQRT}(R_m * T) / MC$ SR 3 0.16879 (A,B,C,D)
 $N(y) * T * \text{SQRT}(R_m * T) / ML$ SR 3 0.04968 (A,B,C,D)
 $M(y) * \text{SQRT}(R_m * T) / ML$ SR 3 0.16879 (A,B,C,D)

Stress Concentration Factors: $K_n = 1.00$, $K_b = 1.00$

Stresses in the Vessel at the Edge of Reinforcing Pad (N./mm²)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad.	Memb. P	-1.9	-1.9	-1.9	-1.9	-1.9	-1.9	-1.9	-1.9
Rad.	Bend. P	-8.8	8.8	-8.8	8.8	-8.8	8.8	-8.8	8.8
Rad.	Memb. MC	0.0	0.0	0.0	0.0	-0.6	-0.6	0.6	0.6
Rad.	Bend. MC	0.0	0.0	0.0	0.0	-13.0	13.0	13.0	-13.0
Rad.	Memb. ML	-0.6	-0.6	0.6	0.6	0.0	0.0	0.0	0.0
Rad.	Bend. ML	-13.0	13.0	13.0	-13.0	0.0	0.0	0.0	0.0
Tot. Rad. Str.		-24.4	19.3	2.9	-5.5	-24.4	19.3	2.9	-5.5
Tang.	Memb. P	-0.6	-0.6	-0.6	-0.6	-0.6	-0.6	-0.6	-0.6
Tang.	Bend. P	-2.7	2.7	-2.7	2.7	-2.7	2.7	-2.7	2.7
Tang.	Memb. MC	0.0	0.0	0.0	0.0	-0.2	-0.2	0.2	0.2
Tang.	Bend. MC	0.0	0.0	0.0	0.0	-3.9	3.9	3.9	-3.9
Tang.	Memb. ML	-0.2	-0.2	0.2	0.2	0.0	0.0	0.0	0.0
Tang.	Bend. ML	-3.9	3.9	3.9	-3.9	0.0	0.0	0.0	0.0
Tot. Tang. Str.		-7.3	5.8	0.9	-1.6	-7.3	5.8	0.9	-1.6
Shear VC		0.5	0.5	-0.5	-0.5	0.0	0.0	0.0	0.0
Shear VL		0.0	0.0	0.0	0.0	-0.5	-0.5	0.5	0.5
Shear MT		0.9	0.9	0.9	0.9	0.9	0.9	0.9	0.9
Tot. Shear		1.4	1.4	0.4	0.4	0.4	0.4	1.4	1.4
Str. Int.		24.5	19.5	3.0	5.5	24.4	19.3	3.6	5.9

WRC 107/537 Stress Summations:

Vessel Stress Summation at Attachment Junction (N./mm²)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad.	Pm (SUS)	59.1	59.1	59.1	59.1	59.1	59.1	59.1	59.1
Rad.	Pl (SUS)	-0.6	-0.6	0.1	0.1	-0.6	-0.6	0.1	0.1
Rad.	Q (SUS)	-1.5	1.5	0.4	-0.4	-1.5	1.5	0.4	-0.4
Long.	Pm (SUS)	59.1	59.1	59.1	59.1	59.1	59.1	59.1	59.1
Long.	Pl (SUS)	-0.7	-0.7	-0.6	-0.6	-0.7	-0.7	-0.6	-0.6
Long.	Q (SUS)	-12.8	12.8	3.8	-3.8	-12.8	12.8	3.8	-3.8
Shear	Pm (SUS)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear	Pl (SUS)	0.5	0.5	-0.5	-0.5	-0.5	-0.5	0.5	0.5
Shear	Q (SUS)	1.9	1.9	1.9	1.9	1.9	1.9	1.9	1.9
Pm (SUS)		59.1	59.1	59.1	59.1	59.1	59.1	59.1	59.1
Pm+Pl (SUS)		58.9	58.9	59.5	59.5	58.9	58.9	59.5	59.5

Pm+Pl+Q (Total)	57.4	71.7	62.9	59.2	57.1	71.3	63.7	59.9
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Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	59.06	138.00	Passed
Pm+Pl (SUS)	59.46	207.00	Passed
Pm+Pl+Q (TOTAL)	71.67	414.00	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 107/537 Stress Summations:

Vessel Stress Summation at Reinforcing Pad Edge (N./mm²)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad.	Pm (SUS)	124.2	124.2	124.2	124.2	124.2	124.2	124.2	124.2
Rad.	Pl (SUS)	-2.5	-2.5	-1.3	-1.3	-2.5	-2.5	-1.3	-1.3
Rad.	Q (SUS)	-21.9	21.9	4.2	-4.2	-21.9	21.9	4.2	-4.2
Long.	Pm (SUS)	124.2	124.2	124.2	124.2	124.2	124.2	124.2	124.2
Long.	Pl (SUS)	-0.8	-0.8	-0.4	-0.4	-0.8	-0.8	-0.4	-0.4
Long.	Q (SUS)	-6.5	6.5	1.2	-1.2	-6.5	6.5	1.2	-1.2
Shear	Pm (SUS)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear	Pl (SUS)	0.5	0.5	-0.5	-0.5	-0.5	-0.5	0.5	0.5
Shear	Q (SUS)	0.9	0.9	0.9	0.9	0.9	0.9	0.9	0.9
Pm (SUS)		124.2	124.2	124.2	124.2	124.2	124.2	124.2	124.2
Pm+Pl (SUS)		123.5	123.5	124.0	124.0	123.5	123.5	124.0	124.0
Pm+Pl+Q (Total)		117.0	143.6	127.1	122.6	116.9	143.5	127.8	123.0

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	124.15	138.00	Passed
Pm+Pl (SUS)	124.02	207.00	Passed
Pm+Pl+Q (TOTAL)	143.63	414.00	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

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Input, Nozzle Desc: K1B-2"

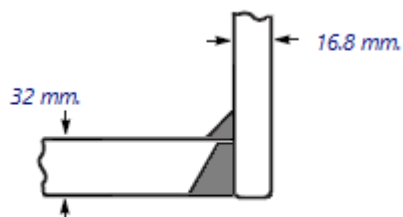
From: 40

Pressure for Reinforcement Calculations	P	12.289	bars
Temperature for Internal Pressure	Temp	82	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	5700.00	mm.
Shell Finished (Minimum) Thickness	t	32.0000	mm.
Shell Internal Corrosion Allowance	c	1.6000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		650.00	mm.
User Entered Minimum Design Metal Temperature		-1.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-105	
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	N./mm ²
Allowable Stress At Ambient	Sna	138.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		120.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	16.8000	mm.
Flange Material		SA-105	
Flange Type		Long Weld Neck	
Corrosion Allowance	can	1.6000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	206.3750	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	24.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).



Reinforcement CALCULATION, Description: K1B-2"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation 2.000 in.



Actual Thickness Used in Calculation 0.661 in.

Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
= $P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $12.29 \cdot 2851.6001 / (138 \cdot 1.0 - 0.6 \cdot 12.29)$
= 25.5323 mm.

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]
= $R \cdot (\exp([P / (S_n \cdot E)] - 1))$ per Appendix 1-2 (a)(1)
= $27.0 \cdot (\exp([12.29 / (138.0 \cdot 1.0)] - 1))$
= 0.2415 mm.

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	145.2000 mm.
Parallel to Vessel Wall	Rn+tn+t	72.6000 mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	38.0000 mm.

*Taking a UG-36(c)(3)(a) exemption for nozzle: K1B-2".
This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 1.8415 mm.
Wall Thickness per UG16(b),	tr16b	= 4.1000 mm.
Wall Thickness, shell/head, internal pressure	trb1	= 27.1323 mm.
Wall Thickness	tb1 = max(trb1, tr16b)	= 27.1323 mm.
Wall Thickness	tb2 = max(trb2, tr16b)	= 4.1000 mm.
Wall Thickness per table UG-45	tb3	= 6.4000 mm.

Determine Nozzle Thickness candidate [tb]:
= min[tb3, max(tb1, tb2)]
= min[6.4, max(27.1323, 4.1)]
= 6.4000 mm.

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
= max(ta, tb)
= max(1.8415, 6.4)
= 6.4000 mm.

Available Nozzle Neck Thickness = 16.8000 mm. -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)(b)), min(Curve:A, Curve:D)

Govrn. thk, tg = 16.8, tr = 0.242, c = 1.6 mm., E* = 1.0
Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.016$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	7 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-104 °C



Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
Design Pressure/Ambient Rating = $12.29/51.00 = 0.241$

Weld Size Calculations, Description: K1B-2"

Intermediate Calc. for nozzle/shell Welds Tmin 15.2000 mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm.

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 0.3126 mm.
The Cut Length for this Nozzle is, Drop + Ho + H + T : 238.6876 mm.

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Input, Nozzle Desc: N6-2"

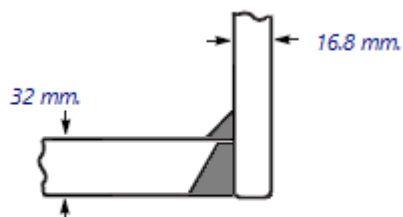
From: 40

Pressure for Reinforcement Calculations	P	12.289	bars
Temperature for Internal Pressure	Temp	82	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	5700.00	mm.
Shell Finished (Minimum) Thickness	t	32.0000	mm.
Shell Internal Corrosion Allowance	c	1.6000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		1500.00	mm.
User Entered Minimum Design Metal Temperature		-1.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-105	
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	N./mm ²
Allowable Stress At Ambient	Sna	138.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		270.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	16.8000	mm.
Flange Material		SA-105	
Flange Type		Long Weld Neck	
Corrosion Allowance	can	1.6000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	206.3750	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	24.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).



Reinforcement CALCULATION, Description: N6-2"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	2.000	in.
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Actual Thickness Used in Calculation 0.661 in.

Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
= $P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $12.29 \cdot 2851.6001 / (138 \cdot 1.0 - 0.6 \cdot 12.29)$
= 25.5320 mm.

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]
= $R \cdot (\exp([P / (S_n \cdot E)] - 1))$ per Appendix 1-2 (a)(1)
= $27.0 \cdot (\exp([12.29 / (138.0 \cdot 1.0)] - 1))$
= 0.2415 mm.

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	Dl	145.2000 mm.
Parallel to Vessel Wall	Rn+tn+t	72.6000 mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	38.0000 mm.

*Taking a UG-36(c)(3)(a) exemption for nozzle: N6-2".
This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 1.8415 mm.
Wall Thickness per UG16(b),	tr16b	= 4.1000 mm.
Wall Thickness, shell/head, internal pressure	trb1	= 27.1320 mm.
Wall Thickness	tb1 = max(trb1, tr16b)	= 27.1320 mm.
Wall Thickness	tb2 = max(trb2, tr16b)	= 4.1000 mm.
Wall Thickness per table UG-45	tb3	= 6.4000 mm.

Determine Nozzle Thickness candidate [tb]:
= min[tb3, max(tb1, tb2)]
= min[6.4, max(27.132, 4.1)]
= 6.4000 mm.

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
= max(ta, tb)
= max(1.8415, 6.4)
= 6.4000 mm.

Available Nozzle Neck Thickness = 16.8000 mm. -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 12.1,	Allowable	: 138.0 N./mm ²	Passed
Expansion	: 0.0,	Allowable	: 332.9 N./mm ²	Passed
Occasional	: 0.9,	Allowable	: 183.5 N./mm ²	Passed
Shear	: 9.9,	Allowable	: 96.6 N./mm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), min(Curve:A, Curve:D)

Govern. thk, tg = 16.8, tr = 0.242, c = 1.6 mm., E* = 1.0
Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.016$, Temp. Reduction = 78 °C



Min Metal Temp. w/o impact per UCS-66, Curve A 7 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
 Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = 12.29/51.00 = 0.241

Weld Size Calculations, Description: N6-2"

Intermediate Calc. for nozzle/shell Welds Tmin 15.2000 mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm.

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 0.3126 mm.
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 238.6876 mm.

Input Echo, WRC297 Item 1, Description: N6-2" :

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	1.6000	mm.
Shell External Corrosion Allowance	0.0000	mm.
Shell Diameter	5700.000	mm.
Shell Thickness	32.0000	mm.
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc 138.00	N./mm ²
Vessel Hot S.I. Allowable	Smh 138.00	N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter Basis for Nozzle	ID	
Nozzle Corrosion Allowance	1.6000	mm.
Nozzle Diameter	50.800	mm.
Nozzle Thickness	16.8000	mm.
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-105	
Nozzle Cold S.I. Allowable	SNmc 138.00	N./mm ²
Nozzle Hot S.I. Allowable	SNmh 138.00	N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Note:

External Forces and Moments in WRC 107/537 Convention:
 These loads are assumed to be SUstained loads.

Design Internal Pressure	Dp	12.29	bars
Radial Load	P	382.42	Kgf
Circumferential Shear	Vc	254.95	Kgf
Longitudinal Shear	Vl	254.95	Kgf
Circumferential Moment	Mc	35.68	Kg-m.
Longitudinal Moment	Ml	35.68	Kg-m.
Torsional Moment	Mt	50.97	Kg-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No

Local Loads applied at end of Nozzle/Attachment No

*Warning - The ratio, Dn/Tv, (2.637) is < 5
 check the Limitations of WRC bulletin 297.*
*Warning - The ratio, Dn/Tn (5.024) is < 10
 check the Limitations of WRC bulletin 297.*

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - ca))}$
 $= 84.4 + 2 \cdot 1.65 \cdot \sqrt{(2866.8(32.0 - 1.6))}$
 $= 1058.603 \text{ mm.}$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter (D) = 5733.600 mm.
 Nozzle Outside Diameter (d) = 84.400 mm.
 Vessel Thickness used (T) = 30.400 mm.
 Nozzle Thickness used (t) = 15.200 mm.
 T / t = 2.000
 d / t = 10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 0.202$$

Nr/P = 0.288
 Mr/P = 0.290
 M0/P = 0.191
 N0/P = 0.167
 MrD/Mc = 0.232
 NrDL/Mc = 0.201
 M0d/Mc = 0.183
 N0DL/Mc = -0.132
 MrD/Ml = 0.231
 NrDL/Ml = 0.205
 M0D/Ml = 0.182
 N0DL/Ml = -0.137

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	0	0	0	0
Outplane Bending (P)	-4	4	-4	4
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	-4	4	4	-4
Normal Pressure Stress	114	115	114	115
Outplane Stress Summary	105	125	113	114

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	-1	-1	-1	-1
Inplane Bending (P)	-7	7	-7	7
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	-6	6	6	-6
Inplane Pressure Stress	57	57	57	57
Inplane Stress Summary	41	68	56	57

Vessel Stresses (N./mm²):



LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)			Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)			0	0	0	0
Outplane Shear (Vl)			0	0	0	0
Outplane Shear (Mt)			1	1	1	1
Shear Stress Summary			2	2	0	0

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)			Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress			105	125	113	114

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)			Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Membrane (P)			0	0	0	0
Outplane Bending (P)			-4	4	-4	4
Outplane Membrane (Mc)			0	0	0	0
Outplane Bending (Mc)			-4	4	4	-4
Outplane Membrane (ML)			0	0	0	0
Outplane Bending (ML)			0	0	0	0
Normal Pressure Stress			57	57	57	57
Outplane Stress Summary			47	66	56	55

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)			Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Inplane Membrane (P)			-1	-1	-1	-1
Inplane Bending (P)			-7	7	-7	7
Inplane Membrane (Mc)			0	0	0	0
Inplane Bending (Mc)			-6	6	6	-6
Inplane Membrane (ML)			0	0	0	0
Inplane Bending (ML)			0	0	0	0
Inplane Pressure Stress			114	115	114	115
Inplane Stress Summary			99	127	113	116

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)			Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Shear (Vc)			0	0	0	0
Outplane Shear (Vl)			0	0	0	0
Torsional Shear (Mt)			1	1	1	1
Shear Stress Summary			0	0	2	2

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)			Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Two * Max Shear Stress			99	127	113	116

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses in the hoop direction)			Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
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Hoop Membrane (P)	0	0	0	0
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	2	3	2	3

Hoop Stress Summary	1	3	0	2
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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses Normal to	Top	Top	Bottom	Bottom
pipe cross-section)	Outside	Inside	Outside	Inside

Axial Membrane (P)	-1	-1	-1	-1
Axial Bending (P)	-14	14	-14	14
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-7	-7	7	7
Axial Bending (ML)	-13	13	13	-13
Axial Pressure Stress	1	1	1	1

Axial Stress Summary	-35	20	6	7
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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Shear stress)	Outside	Inside	Outside	Inside

Shear due to (Vc)	1	1	-1	-1
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	2	2	2	2

Shear Stress Summary	4	4	1	1
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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stress Intensities	Outside	Inside	Outside	Inside

Two * Max Shear Stress	38	21	7	7
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Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses in the	Left	Left	Right	Right
hoop direction)	Outside	Inside	Outside	Inside

Hoop Membrane (P)	0	0	0	0
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	2	3	2	3

Hoop Stress Summary	1	3	0	2
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Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses Normal to	Left	Left	Right	Right
pipe cross-section)	Outside	Inside	Outside	Inside

Axial Membrane (P)	-1	-1	-1	-1
Axial Bending (P)	-14	14	-14	14
Axial Membrane (Mc)	-7	-7	7	7
Axial Bending (Mc)	-14	14	14	-14
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	1	1	1	1



Axial Stress Summary -35 21 6 7

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE
(Shear stress) Cu Cl Du D1
Outside Inside Outside Inside

Shear due to (Vc) 0 0 0 0
Shear due to (Vl) -1 -1 1 1
Shear due to Torsion 2 2 2 2

Shear Stress Summary 1 1 4 4

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE
(Stress Intensities) Cu Cl Du D1
Outside Inside Outside Inside

Two * Max Shear Stress 37 21 10 9

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	114	115	114	115	114	115	114	115
Circ. Pl (SUS)	0	0	0	0	-1	-1	-1	-1
Circ. Q (SUS)	-8	8	0	0	-13	13	-1	1
Long. Pm (SUS)	57	57	57	57	57	57	57	57
Long. Pl (SUS)	-1	-1	-1	-1	0	0	0	0
Long. Q (SUS)	-13	13	-1	1	-8	8	0	0
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	1	1	1	1	1	1	1	1
Pm (SUS)	114.0	115.0	114.0	115.0	114.0	115.0	114.0	115.0
Pm+Pl (SUS)	114.0	115.0	114.0	115.0	113.0	114.0	113.0	114.0
Pm+Pl+Q (Total)	106.0	123.0	114.0	115.0	100.0	127.0	112.0	115.0

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	115.00	138.00	Passed
Pm+Pl (SUS)	115.00	207.00	Passed
Pm+Pl+Q (TOTAL)	127.02	414.00	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	2	3	2	3	2	3	2	3
Circ. Pl (SUS)	0	0	0	0	0	0	0	0
Circ. Q (SUS)	0	0	0	0	0	0	0	0



Long. Pm (SUS)	1	1	1	1	1	1	1	1
Long. Pl (SUS)	-8	-8	6	6	-8	-8	6	6
Long. Q (SUS)	-27	27	-1	1	-28	28	0	0
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	2	2	2	2	2	2	2	2
Pm (SUS)	2.0	3.0	2.0	3.0	2.0	3.0	2.0	3.0
Pm+Pl (SUS)	9.2	10.2	7.2	7.2	9.2	10.2	7.2	7.2
Pm+Pl+Q (Total)	36.5	20.5	6.2	8.2	37.1	21.1	8.4	8.6

Nozzle Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	3.00	138.00	Passed
Pm+Pl (SUS)	10.20	207.00	Passed
Pm+Pl+Q (TOTAL)	37.05	414.00	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

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Input, Nozzle Desc: K2B-2"

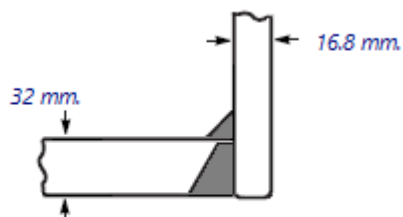
From: 40

Pressure for Reinforcement Calculations	P	12.289	bars
Temperature for Internal Pressure	Temp	82	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	5700.00	mm.
Shell Finished (Minimum) Thickness	t	32.0000	mm.
Shell Internal Corrosion Allowance	c	1.6000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		650.00	mm.
User Entered Minimum Design Metal Temperature		-1.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-105	
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	N./mm ²
Allowable Stress At Ambient	Sna	138.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		90.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	16.8000	mm.
Flange Material		SA-105	
Flange Type		Long Weld Neck	
Corrosion Allowance	can	1.6000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	206.3750	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	24.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).



Reinforcement CALCULATION, Description: K2B-2"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation 2.000 in.



Actual Thickness Used in Calculation 0.661 in.

Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
= $P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $12.29 \cdot 2851.6001 / (138 \cdot 1.0 - 0.6 \cdot 12.29)$
= 25.5323 mm.

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]
= $R \cdot (\exp([P / (S_n \cdot E)] - 1))$ per Appendix 1-2 (a)(1)
= $27.0 \cdot (\exp([12.29 / (138.0 \cdot 1.0)] - 1))$
= 0.2415 mm.

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	145.2000 mm.
Parallel to Vessel Wall	Rn+tn+t	72.6000 mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	38.0000 mm.

*Taking a UG-36(c)(3)(a) exemption for nozzle: K2B-2".
This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 1.8415 mm.
Wall Thickness per UG16(b),	tr16b	= 4.1000 mm.
Wall Thickness, shell/head, internal pressure	trb1	= 27.1323 mm.
Wall Thickness	tb1 = max(trb1, tr16b)	= 27.1323 mm.
Wall Thickness	tb2 = max(trb2, tr16b)	= 4.1000 mm.
Wall Thickness per table UG-45	tb3	= 6.4000 mm.

Determine Nozzle Thickness candidate [tb]:
= min[tb3, max(tb1, tb2)]
= min[6.4, max(27.1323, 4.1)]
= 6.4000 mm.

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
= max(ta, tb)
= max(1.8415, 6.4)
= 6.4000 mm.

Available Nozzle Neck Thickness = 16.8000 mm. -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), min(Curve:A, Curve:D)

Govrn. thk, tg = 16.8, tr = 0.242, c = 1.6 mm., E* = 1.0
Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.016$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	7 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-104 °C



Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
Design Pressure/Ambient Rating = $12.29/51.00 = 0.241$

Weld Size Calculations, Description: K2B-2"

Intermediate Calc. for nozzle/shell Welds Tmin 15.2000 mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$6.0000 = \text{Min per Code}$	$7.0700 = 0.7 * W_o \text{ mm.}$

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 0.3126 mm.
The Cut Length for this Nozzle is, Drop + Ho + H + T : 238.6876 mm.

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Input, Nozzle Desc: K1A-2"

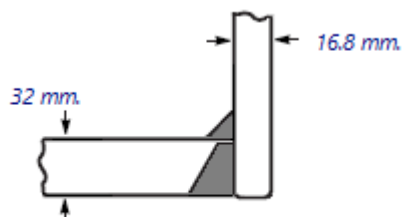
From: 40

Pressure for Reinforcement Calculations	P	12.289	bars
Temperature for Internal Pressure	Temp	82	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	5700.00	mm.
Shell Finished (Minimum) Thickness	t	32.0000	mm.
Shell Internal Corrosion Allowance	c	1.6000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		1500.00	mm.
User Entered Minimum Design Metal Temperature		-1.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-105	
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	N./mm ²
Allowable Stress At Ambient	Sna	138.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		90.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	16.8000	mm.
Flange Material		SA-105	
Flange Type		Long Weld Neck	
Corrosion Allowance	can	1.6000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	206.3750	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	24.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).



Reinforcement CALCULATION, Description: K1A-2"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	2.000	in.
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Actual Thickness Used in Calculation 0.661 in.

Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
= $P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $12.29 \cdot 2851.6001 / (138 \cdot 1.0 - 0.6 \cdot 12.29)$
= 25.5320 mm.

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]
= $R \cdot (\exp([P / (S_n \cdot E)] - 1))$ per Appendix 1-2 (a)(1)
= $27.0 \cdot (\exp([12.29 / (138.0 \cdot 1.0)] - 1))$
= 0.2415 mm.

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	145.2000 mm.
Parallel to Vessel Wall	Rn+tn+t	72.6000 mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	38.0000 mm.

*Taking a UG-36(c)(3)(a) exemption for nozzle: K1A-2".
This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 1.8415 mm.
Wall Thickness per UG16(b),	tr16b	= 4.1000 mm.
Wall Thickness, shell/head, internal pressure	trb1	= 27.1320 mm.
Wall Thickness	tb1 = max(trb1, tr16b)	= 27.1320 mm.
Wall Thickness	tb2 = max(trb2, tr16b)	= 4.1000 mm.
Wall Thickness per table UG-45	tb3	= 6.4000 mm.

Determine Nozzle Thickness candidate [tb]:
= min[tb3, max(tb1, tb2)]
= min[6.4, max(27.132, 4.1)]
= 6.4000 mm.

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
= max(ta, tb)
= max(1.8415, 6.4)
= 6.4000 mm.

Available Nozzle Neck Thickness = 16.8000 mm. -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), min(Curve:A, Curve:D)

Govrn. thk, tg = 16.8, tr = 0.242, c = 1.6 mm., E* = 1.0
Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.016$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	7 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-104 °C



Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
Design Pressure/Ambient Rating = $12.29/51.00 = 0.241$

Weld Size Calculations, Description: K1A-2"

Intermediate Calc. for nozzle/shell Welds Tmin 15.2000 mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$6.0000 = \text{Min per Code}$	$7.0700 = 0.7 * W_o \text{ mm.}$

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 0.3126 mm.
The Cut Length for this Nozzle is, Drop + Ho + H + T : 238.6876 mm.

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Input, Nozzle Desc: K2A-2"

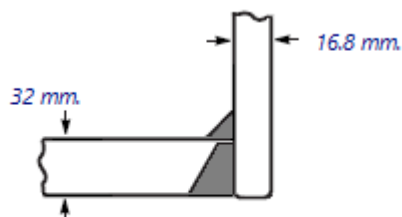
From: 40

Pressure for Reinforcement Calculations	P	12.289	bars
Temperature for Internal Pressure	Temp	82	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	5700.00	mm.
Shell Finished (Minimum) Thickness	t	32.0000	mm.
Shell Internal Corrosion Allowance	c	1.6000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		1500.00	mm.
User Entered Minimum Design Metal Temperature		-1.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-105	
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	N./mm ²
Allowable Stress At Ambient	Sna	138.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		120.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	16.8000	mm.
Flange Material		SA-105	
Flange Type		Long Weld Neck	
Corrosion Allowance	can	1.6000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	206.3750	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	24.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).



Reinforcement CALCULATION, Description: K2A-2"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	2.000	in.
--	-------	-----



Actual Thickness Used in Calculation 0.661 in.

Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
= $P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $12.29 \cdot 2851.6001 / (138 \cdot 1.0 - 0.6 \cdot 12.29)$
= 25.5320 mm.

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]
= $R \cdot (\exp([P / (S_n \cdot E)] - 1))$ per Appendix 1-2 (a)(1)
= $27.0 \cdot (\exp([12.29 / (138.0 \cdot 1.0)] - 1))$
= 0.2415 mm.

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	145.2000 mm.
Parallel to Vessel Wall	Rn+tn+t	72.6000 mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	38.0000 mm.

*Taking a UG-36(c)(3)(a) exemption for nozzle: K2A-2".
This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 1.8415 mm.
Wall Thickness per UG16(b),	tr16b	= 4.1000 mm.
Wall Thickness, shell/head, internal pressure	trb1	= 27.1320 mm.
Wall Thickness	tb1 = max(trb1, tr16b)	= 27.1320 mm.
Wall Thickness	tb2 = max(trb2, tr16b)	= 4.1000 mm.
Wall Thickness per table UG-45	tb3	= 6.4000 mm.

Determine Nozzle Thickness candidate [tb]:
= min[tb3, max(tb1, tb2)]
= min[6.4, max(27.132, 4.1)]
= 6.4000 mm.

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
= max(ta, tb)
= max(1.8415, 6.4)
= 6.4000 mm.

Available Nozzle Neck Thickness = 16.8000 mm. -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), min(Curve:A, Curve:D)

Govrn. thk, tg = 16.8, tr = 0.242, c = 1.6 mm., E* = 1.0
Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.016$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	7 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-104 °C



Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
Design Pressure/Ambient Rating = $12.29/51.00 = 0.241$

Weld Size Calculations, Description: K2A-2"

Intermediate Calc. for nozzle/shell Welds Tmin 15.2000 mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$6.0000 = \text{Min per Code}$	$7.0700 = 0.7 * W_o \text{ mm.}$

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 0.3126 mm.
The Cut Length for this Nozzle is, Drop + Ho + H + T : 238.6876 mm.

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Input, Nozzle Desc: M1-24"

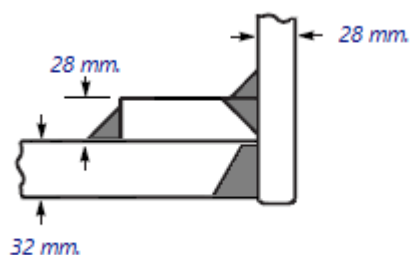
From: 60

Pressure for Reinforcement Calculations	P	12.288	bars
Temperature for Internal Pressure	Temp	82	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	5700.00	mm.
Shell Finished (Minimum) Thickness	t	32.0000	mm.
Shell Internal Corrosion Allowance	c	1.6000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		8550.00	mm.
User Entered Minimum Design Metal Temperature		-1.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-516 70	
Material UNS Number		K02700	
Material Specification/Type		Plate	
Allowable Stress at Temperature	Sn	138.00	N./mm ²
Allowable Stress At Ambient	Sna	138.00	N./mm ²
Diameter Basis (for tr calc only)		Outside	
Layout Angle		90.00	deg
Diameter		24.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	28.0000	mm.
Flange Material		SA-105	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	1.6000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	250.0000	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	26.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Pad Material		SA-516 70	
Pad Allowable Stress at Temperature	Sp	138.00	N./mm ²
Pad Allowable Stress At Ambient	Spa	138.00	N./mm ²
Diameter of Pad along vessel surface	Dp	1009.6000	mm.
Thickness of Pad	te	28.0000	mm.
Weld leg size between Pad and Shell	Wp	20.0000	mm.
Groove weld depth between Pad and Nozzle	Wgpn	28.0000	mm.
Reinforcing Pad Width		200.0000	mm.
This is a Manway, Coupling or Access Opening.			
Flange Class		150	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).



Reinforcement CALCULATION, Description: M1-24"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	24.000 in.
Actual Thickness Used in Calculation	1.102 in.

Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 12.29 \cdot 2851.6001 / (138 \cdot 1.0 - 0.6 \cdot 12.29)$
 $= 25.5302 \text{ mm.}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 12.29 \cdot 304.8 / (138 \cdot 1.0 + 0.4 \cdot 12.29)$
 $= 2.7046 \text{ mm.}$

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	1113.6000 mm.
Parallel to Vessel Wall, opening length	d	556.8000 mm.
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		76.0000 mm.

Weld Strength Reduction Factor [fr1]:

$= \min(1, S_n / S_v)$
 $= \min(1, 138.0 / 138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:

$= \min(1, S_n / S_v)$
 $= \min(1, 138.0 / 138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr4]:

$= \min(1, S_p / S_v)$
 $= \min(1, 138.0 / 138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:

$= \min(fr2, fr4)$
 $= \min(1.0, 1.0)$
 $= 1.000$

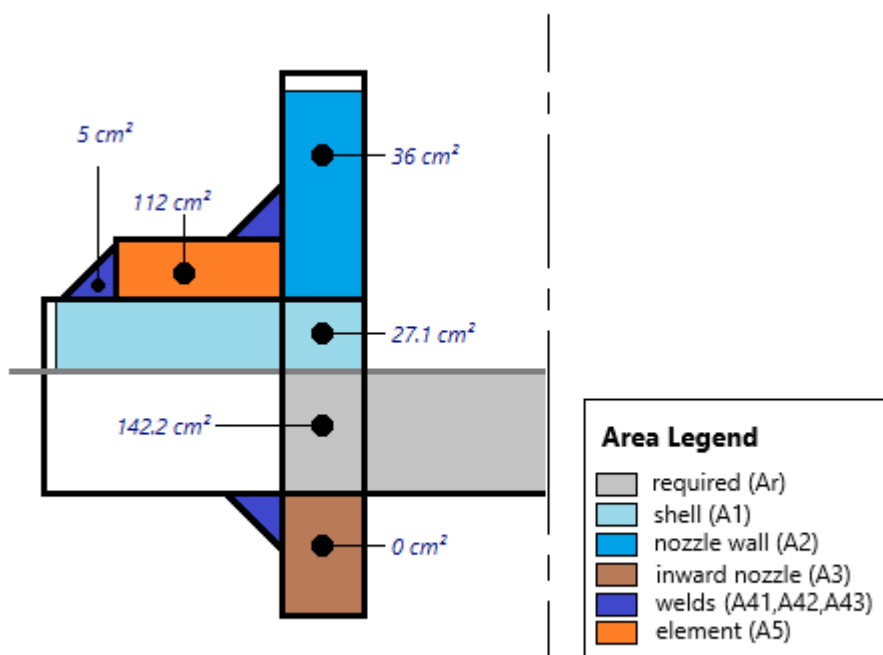


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	142.152	NA	NA
Area in Shell	A1	27.115	NA	NA
Area in Nozzle Wall	A2	36.017	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	5.000	NA	NA
Area in Element	A5	112.000	NA	NA
TOTAL AREA AVAILABLE	Atot	180.132	NA	NA

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Insufficient.
The area available with the given pad is Sufficient.

SELECTION OF POSSIBLE REINFORCING PADS: Diameter Thickness
Based on given Pad Thickness: 873.9559 28.0000 mm.
Based on given Pad Diameter: 1009.5999 18.5049 mm.
Based on Shell or Nozzle Thickness: 873.9559 28.0000 mm.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1))$ UG-37(c)
 $= (556.8 \cdot 25.5302 \cdot 1.0 + 2 \cdot 26.4 \cdot 25.5302 \cdot 1.0 \cdot (1 - 1.0))$
 $= 142.152 \text{ cm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 556.8(1.0 \cdot 30.4 - 1.0 \cdot 25.53) - 2 \cdot 26.4$
 $(1.0 \cdot 30.4 - 1.0 \cdot 25.5302) \cdot (1 - 1.0)$
 $= 27.115 \text{ cm}^2$

Area Available in Nozzle Wall Projecting Outward [A2]:
 $= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2$



$$= (2 \cdot 76.0) \cdot (26.4 - 2.7) \cdot 1.0$$
$$= 36.017 \text{ cm}^2$$

Area Available in Welds [A41 + A42 + A43]:

$$= W_o^2 \cdot fr_3 + (W_i - can/0.707)^2 \cdot fr_2 + W_p^2 \cdot fr_4$$
$$= 10.0^2 \cdot 1.0 + (0.0)^2 \cdot 1.0 + 20.0^2 \cdot 1.0$$
$$= 5.000 \text{ cm}^2$$

Area Available in Element [A5]:

$$= (\min(D_p, D_L) - (\text{Nozzle OD})) \cdot (\min(t_p, T_{lwp}, t_e)) \cdot fr_4$$
$$= (1009.5999 - 609.6) \cdot 28.0 \cdot 1.0$$
$$= 112.000 \text{ cm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement:

Wall Thickness for Internal/External pressures $t_a = 4.3046 \text{ mm.}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= t_a$$
$$= 4.3046 \text{ mm.}$$

Available Nozzle Neck Thickness = 28.0000 mm. -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:D, Curve:A)

Govrn. thk, $t_g = 28.0$, $t_r = 2.705$, $c = 1.6 \text{ mm.}$, $E^* = 1.0$
Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.102$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	22 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Nozzle Neck to Pad Weld for the Nozzle, Curve: D

Govrn. thk, $t_g = 28.0$, $t_r = 2.705$, $c = 1.6 \text{ mm.}$, $E^* = 1.0$
Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.102$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve D	-32 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: D

Govrn. thk, $t_g = 28.0$, $t_r = 2.705$, $c = 1.6 \text{ mm.}$, $E^* = 1.0$
Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.102$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve D	-32 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Shell to Pad Weld Junction at Pad OD, Curve: D

Govrn. thk, $t_g = 28.0$, $c = 1.6 \text{ mm.}$, $E^* = 1.0$
Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.84$, Temp. Reduction = 9 °C
Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve D	-32 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-41 °C

Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: D

Govrn. thk, $t_g = 28.0$, $t_r = 2.705$, $c = 1.6 \text{ mm.}$, $E^* = 1.0$
Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.102$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve D	-32 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Gov. MDMT of the Nozzle	: -104 °C
Gov. MDMT of the Reinforcement Pad	: -41 °C
Gov. MDMT of the nozzle to shell joint welded assembly	: -41 °C



Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
 Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -39 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = 12.29/19.60 = 0.627

Weld Size Calculations, Description: M1-24"

Intermediate Calc. for nozzle/shell Welds Tmin 19.0000 mm.
 Intermediate Calc. for pad/shell Welds TminPad 19.0000 mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm.
Pad Weld	9.5000 = 0.5*TminPad	14.1400 = 0.7 * Wp mm.

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A-A1+2 \cdot t_n \cdot f_{r1} \cdot (E1 \cdot t - t_r)) \cdot S_v)$
 $= \max(0, (142.1519 - 27.1153 + 2 \cdot 26.4 \cdot 1.0 \cdot (1.0 \cdot 30.4 - 25.5302)) \cdot 138)$
 $= 165495.72 \text{ Kgf}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2+A5+A4-(W_i-\text{Can}/.707)^2 \cdot f_{r2}) \cdot S_v$
 $= (36.017 + 112.0 + 5.0 - 0.0 \cdot 1.0) \cdot 138$
 $= 215322.73 \text{ Kgf}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S_v$
 $= (36.017 + 0.0 + 1.0 + (16.0512)) \cdot 138$
 $= 74676.58 \text{ Kgf}$

Weld Load [W3]:
 $= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S$
 $= (36.017 + 0.0 + 5.0 + 112.0 + (16.0512)) \cdot 138$
 $= 237909.70 \text{ Kgf}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$
 $= (3.1416/2.0) \cdot 609.6 \cdot 10.0 \cdot 0.49 \cdot 138$
 $= 66025. \text{ Kgf}$

Shear, Pad Element Weld [Spew]:
 $= (\pi/2) \cdot D_P \cdot W_P \cdot 0.49 \cdot S_{EW}$
 $= (3.1416/2.0) \cdot 1009.6 \cdot 20.0 \cdot 0.49 \cdot 138$
 $= 218698. \text{ Kgf}$

Shear, Nozzle Wall [Snw]:
 $= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - \text{Can}) \cdot 0.7 \cdot S_n$
 $= (3.1416 \cdot 291.6) \cdot (28.0 - 1.6) \cdot 0.7 \cdot 138$
 $= 238226. \text{ Kgf}$

Tension, Pad Groove Weld [Tpgw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_{gpn} \cdot 0.74 \cdot S_{eg}$
 $= (3.1416/2) \cdot 609.6 \cdot 28.0 \cdot 0.74 \cdot 138$
 $= 279193. \text{ Kgf}$

Tension, Shell Groove Weld [Tngw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_{gnvi} \cdot 0.74 \cdot S_{ng}$
 $= (3.1416/2.0) \cdot 609.6 \cdot (26.0 - 0.0) \cdot 0.74 \cdot 138$
 $= 259251. \text{ Kgf}$



See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (\text{SPEW} + \text{SNW}) = (218698 + 238226) = 456925 \text{ Kgf}$$

Failure Path [Path 2-2]:

$$= (\text{Sonw} + \text{Tpgw} + \text{Tngw} + \text{Sinw}) \\ = (66025 + 279193 + 259251 + 0) = 604470 \text{ Kgf}$$

Failure Path [Path 3-3]:

$$= (\text{Spew} + \text{Tngw} + \text{Sinw}) \\ = (218698 + 259251 + 0) = 477949 \text{ Kgf}$$

Summary of Failure Path Calculations:

Path 1-1 = 456924 Kgf, must exceed W = 165495 Kgf or W1 = 215322 Kgf

Path 2-2 = 604469 Kgf, must exceed W = 165495 Kgf or W2 = 74676 Kgf

Path 3-3 = 477949 Kgf, must exceed W = 165495 Kgf or W3 = 237909 Kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 12 bars

Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 16.3457 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 298.3457 mm.

Percent Elongation Calculations:

% Elongation per Table UG-79-1 ($50 \cdot t_{nom} / R_f \cdot (1 - R_f / R_o)$) 4.814 %

This nozzle or its diameter limit intersects weld seam # [1] on this element. The distance between the nozzle and seam is, 0.00 mm., while the distance from the center of the nozzle to the edge of the diameter limit is: 556.80 mm.

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Input, Nozzle Desc: M2-24"

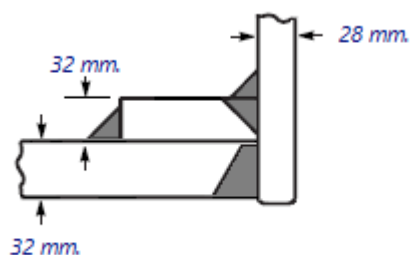
From: 60

Pressure for Reinforcement Calculations	P	12.288	bars
Temperature for Internal Pressure	Temp	82	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	5700.00	mm.
Shell Finished (Minimum) Thickness	t	32.0000	mm.
Shell Internal Corrosion Allowance	c	1.6000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		7550.00	mm.
User Entered Minimum Design Metal Temperature		-1.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-516 70	
Material UNS Number		K02700	
Material Specification/Type		Plate	
Allowable Stress at Temperature	Sn	138.00	N./mm ²
Allowable Stress At Ambient	Sna	138.00	N./mm ²
Diameter Basis (for tr calc only)		Outside	
Layout Angle		220.00	deg
Diameter		24.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	28.0000	mm.
Flange Material		SA-105	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	1.6000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	250.0000	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	26.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Pad Material		SA-516 70	
Pad Allowable Stress at Temperature	Sp	138.00	N./mm ²
Pad Allowable Stress At Ambient	Spa	138.00	N./mm ²
Diameter of Pad along vessel surface	Dp	1009.6000	mm.
Thickness of Pad	te	32.0000	mm.
Weld leg size between Pad and Shell	Wp	20.0000	mm.
Groove weld depth between Pad and Nozzle	Wgpn	32.0000	mm.
Reinforcing Pad Width		200.0000	mm.
This is a Manway, Coupling or Access Opening.			
Flange Class		150	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).



Reinforcement CALCULATION, Description: M2-24"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	24.000 in.
Actual Thickness Used in Calculation	1.102 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 12.29 \cdot 2851.6001 / (138 \cdot 1.0 - 0.6 \cdot 12.29)$
 $= 25.5304 \text{ mm.}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 12.29 \cdot 304.8 / (138 \cdot 1.0 + 0.4 \cdot 12.29)$
 $= 2.7047 \text{ mm.}$

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	1113.6000 mm.
Parallel to Vessel Wall, opening length	d	556.8000 mm.
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		76.0000 mm.

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138.0 / 138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138.0 / 138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr4]:
 $= \min(1, S_p / S_v)$
 $= \min(1, 138.0 / 138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(1.0, 1.0)$
 $= 1.000$

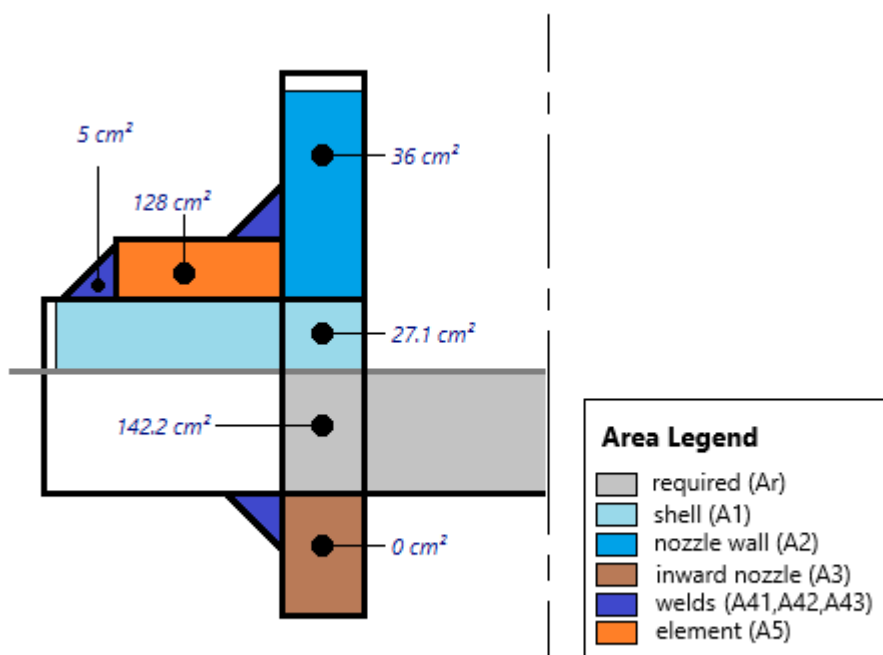


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	142.153	NA	NA
Area in Shell	A1	27.114	NA	NA
Area in Nozzle Wall	A2	36.017	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	5.000	NA	NA
Area in Element	A5	128.000	NA	NA
TOTAL AREA AVAILABLE	Atot	196.131	NA	NA

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Insufficient.
The area available with the given pad is Sufficient.

SELECTION OF POSSIBLE REINFORCING PADS: Diameter Thickness
Based on given Pad Thickness: 840.9208 32.0000 mm.
Based on given Pad Diameter: 1009.5999 18.5057 mm.
Based on Shell or Nozzle Thickness: 873.9666 28.0000 mm.

Area Required [A]:
= (d • tr • F + 2 • tn • tr • F • (1-fr1)) UG-37(c)
= (556.8 • 25.5304 • 1.0 + 2 • 26.4 • 25.5304 • 1.0 • (1-1.0))
= 142.153 cm²

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
= d (E1 • t - F • tr) - 2 • tn (E1 • t - F • tr) • (1 - fr1)
= 556.8 (1.0 • 30.4 - 1.0 • 25.53) - 2 • 26.4
(1.0 • 30.4 - 1.0 • 25.5304) • (1 - 1.0)
= 27.114 cm²

Area Available in Nozzle Wall Projecting Outward [A2]:
= (2 • Tlwp) • (tn - trn) • fr2



$$= (2 \cdot 76.0) \cdot (26.4 - 2.7) \cdot 1.0$$

$$= 36.017 \text{ cm}^2$$

Area Available in Welds [A41 + A42 + A43]:

$$= W_o^2 \cdot fr_3 + (W_i - can/0.707)^2 \cdot fr_2 + W_p^2 \cdot fr_4$$

$$= 10.0^2 \cdot 1.0 + (0.0)^2 \cdot 1.0 + 20.0^2 \cdot 1.0$$

$$= 5.000 \text{ cm}^2$$

Area Available in Element [A5]:

$$= (\min(D_p, D_L) - (\text{Nozzle OD})) \cdot (\min(t_p, T_{lwp}, t_e)) \cdot fr_4$$

$$= (1009.5999 - 609.6) \cdot 32.0 \cdot 1.0$$

$$= 128.000 \text{ cm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement:

Wall Thickness for Internal/External pressures $t_a = 4.3047 \text{ mm.}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= t_a$$

$$= 4.3047 \text{ mm.}$$

Available Nozzle Neck Thickness = 28.0000 mm. -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:D, Curve:A)

Govrn. thk, $t_g = 28.0$, $t_r = 2.705$, $c = 1.6 \text{ mm.}$, $E^* = 1.0$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.102$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	22 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Nozzle Neck to Pad Weld for the Nozzle, Curve: D

Govrn. thk, $t_g = 28.0$, $t_r = 2.705$, $c = 1.6 \text{ mm.}$, $E^* = 1.0$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.102$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve D	-32 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: D

Govrn. thk, $t_g = 28.0$, $t_r = 2.705$, $c = 1.6 \text{ mm.}$, $E^* = 1.0$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.102$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve D	-32 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Shell to Pad Weld Junction at Pad OD, Curve: D

Govrn. thk, $t_g = 32.0$, $t_r = 25.53$, $c = 1.6 \text{ mm.}$, $E^* = 1.0$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.84$, Temp. Reduction = 9 °C

Min Metal Temp. w/o impact per UCS-66, Curve D	-29 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-38 °C

Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: D

Govrn. thk, $t_g = 28.0$, $t_r = 2.705$, $c = 1.6 \text{ mm.}$, $E^* = 1.0$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.102$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve D	-32 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Gov. MDMT of the Nozzle	: -104 °C
Gov. MDMT of the Reinforcement Pad	: -38 °C
Gov. MDMT of the nozzle to shell joint welded assembly	: -38 °C

Flange MDMT including Temperature reduction per UCS-66.1:



Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
 Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -39 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = 12.29/19.60 = 0.627

Weld Size Calculations, Description: M2-24"

Intermediate Calc. for nozzle/shell Welds Tmin 19.0000 mm.
 Intermediate Calc. for pad/shell Welds TminPad 19.0000 mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm.
Pad Weld	9.5000 = 0.5*TminPad	14.1400 = 0.7 * Wp mm.

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A-A1+2 \cdot t_n \cdot f_{r1} \cdot (E1 \cdot t - t_r)) \cdot S_v)$
 $= \max(0, (142.1534 - 27.1138 + 2 \cdot 26.4 \cdot 1.0 \cdot (1.0 \cdot 30.4 - 25.5304)) \cdot 138)$
 $= 165499.70 \text{ Kgf}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2+A5+A4-(W_i \cdot \text{Can}/.707)^2 \cdot f_{r2}) \cdot S_v$
 $= (36.0169 + 128.0 + 5.0 - 0.0 \cdot 1.0) \cdot 138$
 $= 237837.56 \text{ Kgf}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S_v$
 $= (36.0169 + 0.0 + 1. + (16.0512)) \cdot 138$
 $= 74676.52 \text{ Kgf}$

Weld Load [W3]:
 $= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S$
 $= (36.0169 + 0.0 + 5.0 + 128.0 + (16.0512)) \cdot 138$
 $= 260424.55 \text{ Kgf}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$
 $= (3.1416/2.0) \cdot 609.6 \cdot 10.0 \cdot 0.49 \cdot 138$
 $= 66025. \text{ Kgf}$

Shear, Pad Element Weld [Spew]:
 $= (\pi/2) \cdot D_P \cdot W_P \cdot 0.49 \cdot S_{EW}$
 $= (3.1416/2.0) \cdot 1009.6 \cdot 20.0 \cdot 0.49 \cdot 138$
 $= 218698. \text{ Kgf}$

Shear, Nozzle Wall [Snw]:
 $= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - \text{Can}) \cdot 0.7 \cdot S_n$
 $= (3.1416 \cdot 291.6) \cdot (28.0 - 1.6) \cdot 0.7 \cdot 138$
 $= 238226. \text{ Kgf}$

Tension, Pad Groove Weld [Tpgw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_{gpn} \cdot 0.74 \cdot S_{eg}$
 $= (3.1416/2) \cdot 609.6 \cdot 32.0 \cdot 0.74 \cdot 138$
 $= 319078. \text{ Kgf}$

Tension, Shell Groove Weld [Tngw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_{gnvi} \cdot 0.74 \cdot S_{ng}$
 $= (3.1416/2.0) \cdot 609.6 \cdot (26.0 - 0.0) \cdot 0.74 \cdot 138$
 $= 259251. \text{ Kgf}$



See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:
 $= (\text{SPEW} + \text{SNW}) = (218698 + 238226) = 456925 \text{ Kgf}$

Failure Path [Path 2-2]:
 $= (\text{Sonw} + \text{Tpgw} + \text{Tngw} + \text{Sinw})$
 $= (66025 + 319078 + 259251 + 0) = 644354 \text{ Kgf}$

Failure Path [Path 3-3]:
 $= (\text{Spew} + \text{Tngw} + \text{Sinw})$
 $= (218698 + 259251 + 0) = 477949 \text{ Kgf}$

Summary of Failure Path Calculations:

Path 1-1 = 456924 Kgf, must exceed W = 165499 Kgf or W1 = 237837 Kgf
Path 2-2 = 644354 Kgf, must exceed W = 165499 Kgf or W2 = 74676 Kgf
Path 3-3 = 477949 Kgf, must exceed W = 165499 Kgf or W3 = 260424 Kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 12 bars

Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 16.3457 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 298.3457 mm.

Percent Elongation Calculations:

% Elongation per Table UG-79-1 $(50 * t_{nom} / R_f * (1 - R_f / R_o))$ 4.814 %

This nozzle or its diameter limit intersects weld seam # [3] on this element. The distance between the nozzle and seam is, 155.74 mm., while the distance from the center of the nozzle to the edge of the diameter limit is: 556.80 mm.

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Input, Nozzle Desc: K5A-2"

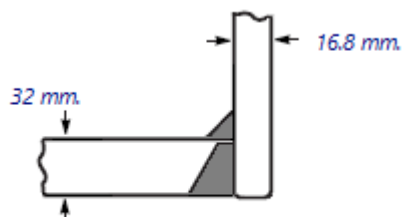
From: 70

Pressure for Reinforcement Calculations	P	12.288	bars
Temperature for Internal Pressure	Temp	82	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	5700.00	mm.
Shell Finished (Minimum) Thickness	t	32.0000	mm.
Shell Internal Corrosion Allowance	c	1.6000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		11000.00	mm.
User Entered Minimum Design Metal Temperature		-1.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-105	
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	N./mm ²
Allowable Stress At Ambient	Sna	138.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		120.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	16.8000	mm.
Flange Material		SA-105	
Flange Type		Long Weld Neck	
Corrosion Allowance	can	1.6000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	206.3750	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	24.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).



Reinforcement CALCULATION, Description: K5A-2"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation 2.000 in.



Actual Thickness Used in Calculation 0.661 in.

Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 12.29 \cdot 2851.6001 / (138 \cdot 1.0 - 0.6 \cdot 12.29)$
 $= 25.5295 \text{ mm.}$

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]
 $= R \cdot (\exp([P / (S_n \cdot E)] - 1))$ per Appendix 1-2 (a)(1)
 $= 27.0 \cdot (\exp([12.29 / (138.0 \cdot 1.0)] - 1))$
 $= 0.2415 \text{ mm.}$

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	145.2000 mm.
Parallel to Vessel Wall	Rn+tn+t	72.6000 mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	38.0000 mm.

*Taking a UG-36(c)(3)(a) exemption for nozzle: K5A-2".
 This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 1.8415 mm.
Wall Thickness per UG16(b),	tr16b	= 4.1000 mm.
Wall Thickness, shell/head, internal pressure	trb1	= 27.1295 mm.
Wall Thickness	tb1 = max(trb1, tr16b)	= 27.1295 mm.
Wall Thickness	tb2 = max(trb2, tr16b)	= 4.1000 mm.
Wall Thickness per table UG-45	tb3	= 6.4000 mm.

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[6.4, \max(27.1295, 4.1)]$
 $= 6.4000 \text{ mm.}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(1.8415, 6.4)$
 $= 6.4000 \text{ mm.}$

Available Nozzle Neck Thickness = 16.8000 mm. -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)(1)(b)), min(Curve:A, Curve:D)

Govrn. thk, tg = 16.8, tr = 0.242, c = 1.6 mm., E* = 1.0
 Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.016$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	7 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-104 °C



Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
Design Pressure/Ambient Rating = $12.29/51.00 = 0.241$

Weld Size Calculations, Description: K5A-2"

Intermediate Calc. for nozzle/shell Welds Tmin 15.2000 mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$6.0000 = \text{Min per Code}$	$7.0700 = 0.7 * W_o \text{ mm.}$

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 0.3126 mm.
The Cut Length for this Nozzle is, Drop + Ho + H + T : 238.6876 mm.

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Input, Nozzle Desc: K5B-2"

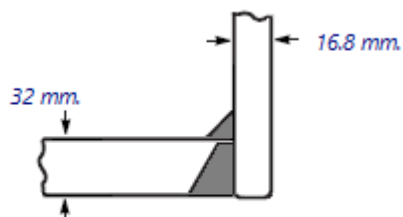
From: 70

Pressure for Reinforcement Calculations	P	12.288	bars
Temperature for Internal Pressure	Temp	82	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	5700.00	mm.
Shell Finished (Minimum) Thickness	t	32.0000	mm.
Shell Internal Corrosion Allowance	c	1.6000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		11000.00	mm.
User Entered Minimum Design Metal Temperature		-1.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-105	
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	N./mm ²
Allowable Stress At Ambient	Sna	138.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		90.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	16.8000	mm.
Flange Material		SA-105	
Flange Type		Long Weld Neck	
Corrosion Allowance	can	1.6000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	206.3750	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	24.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).



Reinforcement CALCULATION, Description: K5B-2"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	2.000	in.
--	-------	-----



Actual Thickness Used in Calculation 0.661 in.

Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
= $P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $12.29 \cdot 2851.6001 / (138 \cdot 1.0 - 0.6 \cdot 12.29)$
= 25.5295 mm.

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]
= $R \cdot (\exp([P / (S_n \cdot E)] - 1))$ per Appendix 1-2 (a)(1)
= $27.0 \cdot (\exp([12.29 / (138.0 \cdot 1.0)] - 1))$
= 0.2415 mm.

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	Dl	145.2000 mm.
Parallel to Vessel Wall	Rn+tn+t	72.6000 mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	38.0000 mm.

*Taking a UG-36(c)(3)(a) exemption for nozzle: K5B-2".
This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 1.8415 mm.
Wall Thickness per UG16(b),	tr16b	= 4.1000 mm.
Wall Thickness, shell/head, internal pressure	trb1	= 27.1295 mm.
Wall Thickness	tb1 = max(trb1, tr16b)	= 27.1295 mm.
Wall Thickness	tb2 = max(trb2, tr16b)	= 4.1000 mm.
Wall Thickness per table UG-45	tb3	= 6.4000 mm.

Determine Nozzle Thickness candidate [tb]:
= min[tb3, max(tb1, tb2)]
= min[6.4, max(27.1295, 4.1)]
= 6.4000 mm.

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
= max(ta, tb)
= max(1.8415, 6.4)
= 6.4000 mm.

Available Nozzle Neck Thickness = 16.8000 mm. -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)(b)), min(Curve:A, Curve:D)

Govrn. thk, tg = 16.8, tr = 0.242, c = 1.6 mm., E* = 1.0
Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.016$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	7 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-104 °C



Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
Design Pressure/Ambient Rating = $12.29/51.00 = 0.241$

Weld Size Calculations, Description: K5B-2"

Intermediate Calc. for nozzle/shell Welds Tmin 15.2000 mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$6.0000 = \text{Min per Code}$	$7.0700 = 0.7 * W_o \text{ mm.}$

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 0.3126 mm.
The Cut Length for this Nozzle is, Drop + Ho + H + T : 238.6876 mm.

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Input, Nozzle Desc: K3-2"

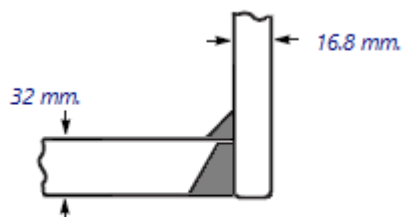
From: 70

Pressure for Reinforcement Calculations	P	12.288	bars
Temperature for Internal Pressure	Temp	82	°C
Parent Material		SA-516	70
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	5700.00	mm.
Shell Finished (Minimum) Thickness	t	32.0000	mm.
Shell Internal Corrosion Allowance	c	1.6000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		10700.00	mm.
User Entered Minimum Design Metal Temperature		-1.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-105	
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	N./mm ²
Allowable Stress At Ambient	Sna	138.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		90.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	16.8000	mm.
Flange Material		SA-105	
Flange Type		Long Weld Neck	
Corrosion Allowance	can	1.6000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	206.3750	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	24.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).



Reinforcement CALCULATION, Description: K3-2"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	2.000	in.
--	-------	-----



Actual Thickness Used in Calculation 0.661 in.

Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 12.29 \cdot 2851.6001 / (138 \cdot 1.0 - 0.6 \cdot 12.29)$
 $= 25.5296 \text{ mm.}$

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]
 $= R \cdot (\exp([P / (S_n \cdot E)] - 1))$ per Appendix 1-2 (a)(1)
 $= 27.0 \cdot (\exp([12.29 / (138.0 \cdot 1.0)] - 1))$
 $= 0.2415 \text{ mm.}$

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	Dl	145.2000 mm.
Parallel to Vessel Wall	Rn+tn+t	72.6000 mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	38.0000 mm.

*Taking a UG-36(c)(3)(a) exemption for nozzle: K3-2".
 This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 1.8415 mm.
Wall Thickness per UG16(b),	tr16b	= 4.1000 mm.
Wall Thickness, shell/head, internal pressure	trb1	= 27.1296 mm.
Wall Thickness	tb1 = max(trb1, tr16b)	= 27.1296 mm.
Wall Thickness	tb2 = max(trb2, tr16b)	= 4.1000 mm.
Wall Thickness per table UG-45	tb3	= 6.4000 mm.

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[6.4, \max(27.1296, 4.1)]$
 $= 6.4000 \text{ mm.}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(1.8415, 6.4)$
 $= 6.4000 \text{ mm.}$

Available Nozzle Neck Thickness = 16.8000 mm. -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)(1)(b)), min(Curve:A, Curve:D)

Govrn. thk, tg = 16.8, tr = 0.242, c = 1.6 mm., E* = 1.0
 Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.016$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	7 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-104 °C



Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
Design Pressure/Ambient Rating = $12.29/51.00 = 0.241$

Weld Size Calculations, Description: K3-2"

Intermediate Calc. for nozzle/shell Welds Tmin 15.2000 mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$6.0000 = \text{Min per Code}$	$7.0700 = 0.7 * W_o \text{ mm.}$

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 0.3126 mm.
The Cut Length for this Nozzle is, Drop + Ho + H + T : 238.6876 mm.

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Input, Nozzle Desc: N1-14"

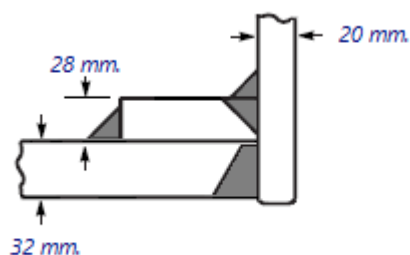
From: 80

Pressure for Reinforcement Calculations	P	12.288	bars
Temperature for Internal Pressure	Temp	82	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	5700.00	mm.
Shell Finished (Minimum) Thickness	t	32.0000	mm.
Shell Internal Corrosion Allowance	c	1.6000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		12550.00	mm.
User Entered Minimum Design Metal Temperature		-1.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-516 70	
Material UNS Number		K02700	
Material Specification/Type		Plate	
Allowable Stress at Temperature	Sn	138.00	N./mm ²
Allowable Stress At Ambient	Sna	138.00	N./mm ²
Diameter Basis (for tr calc only)		Outside	
Layout Angle		270.00	deg
Diameter		14.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	20.0000	mm.
Flange Material		SA-105	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	1.6000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	200.0000	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	26.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Pad Material		SA-516 70	
Pad Allowable Stress at Temperature	Sp	138.00	N./mm ²
Pad Allowable Stress At Ambient	Spa	138.00	N./mm ²
Diameter of Pad along vessel surface	Dp	555.6000	mm.
Thickness of Pad	te	28.0000	mm.
Weld leg size between Pad and Shell	Wp	20.0000	mm.
Groove weld depth between Pad and Nozzle	Wgpn	28.0000	mm.
Reinforcing Pad Width		100.0000	mm.
Flange Class		150	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).



Reinforcement CALCULATION, Description: N1-14"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	14.000 in.
Actual Thickness Used in Calculation	0.787 in.

Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 12.29 \cdot 2851.6001 / (138 \cdot 1.0 - 0.6 \cdot 12.29)$
 $= 25.5291$ mm.

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 12.29 \cdot 177.8 / (138 \cdot 1.0 + 0.4 \cdot 12.29)$
 $= 1.5776$ mm.

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	637.6000	mm.
Parallel to Vessel Wall, opening length	d	318.8000	mm.
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		74.0000	mm.

Weld Strength Reduction Factor [fr1]:

$= \min(1, S_n / S_v)$
 $= \min(1, 138.0 / 138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:

$= \min(1, S_n / S_v)$
 $= \min(1, 138.0 / 138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr4]:

$= \min(1, S_p / S_v)$
 $= \min(1, 138.0 / 138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:

$= \min(fr2, fr4)$
 $= \min(1.0, 1.0)$
 $= 1.000$

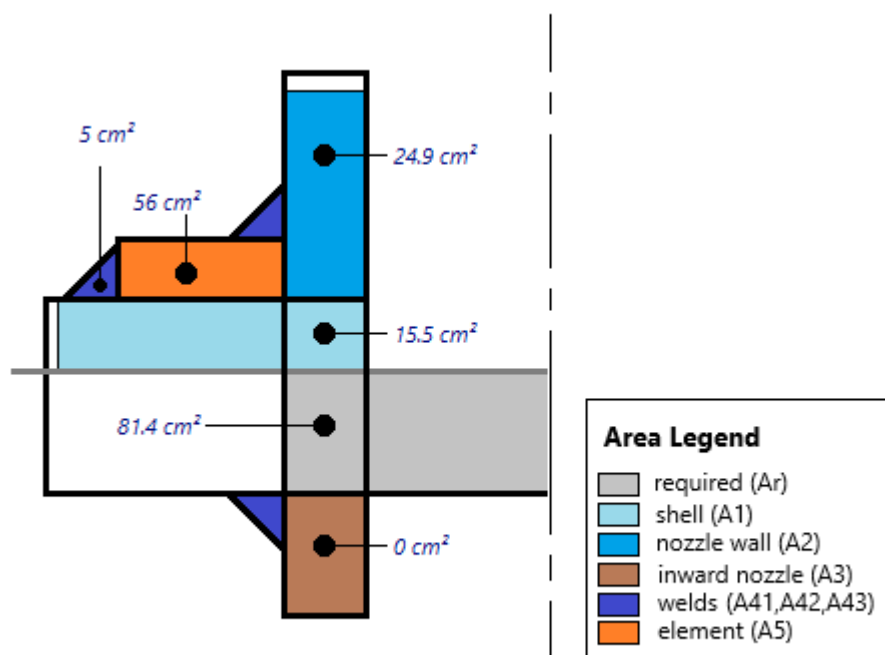


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	81.387	NA	NA
Area in Shell	A1	15.528	NA	NA
Area in Nozzle Wall	A2	24.897	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	5.000	NA	NA
Area in Element	A5	56.000	NA	NA
TOTAL AREA AVAILABLE	Atot	101.426	NA	NA

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Insufficient.
The area available with the given pad is Sufficient.

SELECTION OF POSSIBLE REINFORCING PADS: Diameter Thickness
Based on given Pad Thickness: 484.0327 28.0000 mm.
Based on given Pad Diameter: 555.6000 17.9806 mm.
Based on Shell or Nozzle Thickness: 535.4058 20.0000 mm.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1))$ UG-37(c)
 $= (318.8 \cdot 25.5291 \cdot 1.0 + 2 \cdot 18.4 \cdot 25.5291 \cdot 1.0 \cdot (1 - 1.0))$
 $= 81.387 \text{ cm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 318.8(1.0 \cdot 30.4 - 1.0 \cdot 25.5291) - 2 \cdot 18.4$
 $(1.0 \cdot 30.4 - 1.0 \cdot 25.5291) \cdot (1 - 1.0)$
 $= 15.528 \text{ cm}^2$

Area Available in Nozzle Wall Projecting Outward [A2]:
 $= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2$

$$= (2 \cdot 74.0) \cdot (18.4 - 1.58) \cdot 1.0$$
$$= 24.897 \text{ cm}^2$$

Area Available in Welds [A41 + A42 + A43]:

$$= W_o^2 \cdot fr3 + (W_i - c_n / 0.707)^2 \cdot fr2 + W_p^2 \cdot fr4$$
$$= 10.0^2 \cdot 1.0 + (0.0)^2 \cdot 1.0 + 20.0^2 \cdot 1.0$$
$$= 5.000 \text{ cm}^2$$

Area Available in Element [A5]:

$$= (\min(D_p, D_L) - (\text{Nozzle OD})) \cdot (\min(t_p, t_{lp}, t_e)) \cdot fr4$$
$$= (555.6 - 355.6) \cdot 28.0 \cdot 1.0$$
$$= 56.000 \text{ cm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	$t_a = 3.1776 \text{ mm.}$
Wall Thickness per UG16(b),	$tr_{16b} = 4.1000 \text{ mm.}$
Wall Thickness, shell/head, internal pressure	$trb1 = 27.1291 \text{ mm.}$
Wall Thickness	$tb1 = \max(trb1, tr_{16b}) = 27.1291 \text{ mm.}$
Wall Thickness	$tb2 = \max(trb2, tr_{16b}) = 4.1000 \text{ mm.}$
Wall Thickness per table UG-45	$tb3 = 9.9312 \text{ mm.}$

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb3, \max(tb1, tb2)]$$
$$= \min[9.931, \max(27.1291, 4.1)]$$
$$= 9.9312 \text{ mm.}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(t_a, t_b)$$
$$= \max(3.1776, 9.9312)$$
$$= 9.9312 \text{ mm.}$$

Available Nozzle Neck Thickness = 20.0000 mm. -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 21.9,	Allowable	: 138.0 N./mm ²	Passed
Expansion	: 0.0,	Allowable	: 323.1 N./mm ²	Passed
Occasional	: 5.0,	Allowable	: 183.5 N./mm ²	Passed
Shear	: 11.0,	Allowable	: 96.6 N./mm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:D, Curve:A)

$$\text{Govrn. thk, } t_g = 20.0, tr = 1.578, c = 1.6 \text{ mm., } E^* = 1.0$$
$$\text{Thickness Ratio} = tr \cdot E^* / (t_g - c) = 0.086, \text{Temp. Reduction} = 78 \text{ }^\circ\text{C}$$

Min Metal Temp. w/o impact per UCS-66, Curve A	12 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Nozzle Neck to Pad Weld for the Nozzle, Curve: D

$$\text{Govrn. thk, } t_g = 20.0, tr = 1.578, c = 1.6 \text{ mm., } E^* = 1.0$$
$$\text{Thickness Ratio} = tr \cdot E^* / (t_g - c) = 0.086, \text{Temp. Reduction} = 78 \text{ }^\circ\text{C}$$

Min Metal Temp. w/o impact per UCS-66, Curve D	-40 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: D

$$\text{Govrn. thk, } t_g = 20.0, tr = 1.578, c = 1.6 \text{ mm., } E^* = 1.0$$
$$\text{Thickness Ratio} = tr \cdot E^* / (t_g - c) = 0.086, \text{Temp. Reduction} = 78 \text{ }^\circ\text{C}$$

Min Metal Temp. w/o impact per UCS-66, Curve D	-40 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C



Shell to Pad Weld Junction at Pad OD, Curve: D

Govrn. thk, tg = 28.0, c = 1.6 mm., E* = 1.0
 Thickness Ratio = $tr \cdot E^*/(tg - c) = 0.84$, Temp. Reduction = 9 °C
 Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve D	-32 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-41 °C

Nozzle-Shell/Head Weld (UCS-66(a)(1)(b)), Curve: D

Govrn. thk, tg = 20.0, tr = 1.578, c = 1.6 mm., E* = 1.0
 Thickness Ratio = $tr \cdot E^*/(tg - c) = 0.086$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve D	-40 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Gov. MDMT of the Nozzle	: -104 °C
Gov. MDMT of the Reinforcement Pad	: -41 °C
Gov. MDMT of the nozzle to shell joint welded assembly	: -41 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-39 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = $12.29/19.60 = 0.627$

Weld Size Calculations, Description: N1-14"

Intermediate Calc. for nozzle/shell Welds	Tmin	18.4000 mm.
Intermediate Calc. for pad/shell Welds	TminPad	19.0000 mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$6.0000 = \text{Min per Code}$	$7.0700 = 0.7 \cdot W_o \text{ mm.}$
Pad Weld	$9.5000 = 0.5 \cdot T_{minPad}$	$14.1400 = 0.7 \cdot W_p \text{ mm.}$

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A - A1 + 2 \cdot t_n \cdot f_{r1} \cdot (E1 \cdot t - tr)) \cdot S_v)$
 $= \max(0, (81.3867 - 15.5285 + 2 \cdot 18.4 \cdot 1.0 \cdot (1.0 \cdot 30.4 - 25.5291)) \cdot 138)$
 $= 95196.94 \text{ Kg}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2 + A5 + A4 - (W_i - Can / .707)^2 \cdot f_{r2}) \cdot S_v$
 $= (24.8971 + 56.0 + 5.0 - 0.0 \cdot 1.0) \cdot 138$
 $= 120872.89 \text{ Kg}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S_v$
 $= (24.8971 + 0.0 + 1.0 + (11.1872)) \cdot 138$
 $= 52184.36 \text{ Kg}$

Weld Load [W3]:
 $= (A2 + A3 + A4 + A5 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S$
 $= (24.8971 + 0.0 + 5.0 + 56.0 + (11.1872)) \cdot 138$
 $= 136615.31 \text{ Kg}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$



$$= (3.1416/2.0) \cdot 355.6 \cdot 10.0 \cdot 0.49 \cdot 138$$

$$= 38515. \text{ Kg}$$

Shear, Pad Element Weld [Spew]:

$$= (\pi/2) \cdot DP \cdot WP \cdot 0.49 \cdot SEW$$

$$= (3.1416/2.0) \cdot 555.6 \cdot 20.0 \cdot 0.49 \cdot 138$$

$$= 120353. \text{ Kg}$$

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (Dlr + Dlo) / 4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn$$

$$= (3.1416 \cdot 168.6) \cdot (20.0 - 1.6) \cdot 0.7 \cdot 138$$

$$= 96001. \text{ Kg}$$

Tension, Pad Groove Weld [Tpgw]:

$$= (\pi/2) \cdot Dlo \cdot Wgpn \cdot 0.74 \cdot Seg$$

$$= (3.1416/2) \cdot 355.6 \cdot 28.0 \cdot 0.74 \cdot 138$$

$$= 162863. \text{ Kg}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot Dlo \cdot Wgnvi \cdot 0.74 \cdot Sng$$

$$= (3.1416/2.0) \cdot 355.6 \cdot (26.0 - 0.0) \cdot 0.74 \cdot 138$$

$$= 151230. \text{ Kg}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SPEW + SNW) = (120353 + 96001) = 216354 \text{ Kg}$$

Failure Path [Path 2-2]:

$$= (Sonw + Tpgw + Tngw + Sinw)$$

$$= (38515 + 162863 + 151230 + 0) = 352607 \text{ Kg}$$

Failure Path [Path 3-3]:

$$= (Spew + Tngw + Sinw)$$

$$= (120353 + 151230 + 0) = 271583 \text{ Kg}$$

Summary of Failure Path Calculations:

Path 1-1 = 216354 Kg, must exceed W = 95196 Kg	or W1 = 120872 Kg
Path 2-2 = 352607 Kg, must exceed W = 95196 Kg	or W2 = 52184 Kg
Path 3-3 = 271583 Kg, must exceed W = 95196 Kg	or W3 = 136615 Kg

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 12 bars

Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 5.5516 mm.
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 237.5516 mm.

Percent Elongation Calculations:

% Elongation per Table UG-79-1 ($50 \cdot t_{nom} / R_f \cdot (1 - R_f / R_o)$) 5.959 %
 Note: Please Check Requirements of UCS-79 as Elongation is > 5%.

Input Echo, WRC297 Item 1, Description: N1-14" :

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	1.6000	mm.
Shell External Corrosion Allowance	0.0000	mm.
Shell Diameter	5700.000	mm.
Shell Thickness	32.0000	mm.
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc	138.00 N./mm ²
Vessel Hot S.I. Allowable	Smh	138.00 N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.



Diameter Basis for Nozzle	OD	
Nozzle Corrosion Allowance	1.6000	mm.
Nozzle Diameter	355.600	mm.
Nozzle Thickness	20.0000	mm.
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-516 70	
Nozzle Cold S.I. Allowable	SNmc	138.00 N./mm ²
Nozzle Hot S.I. Allowable	SNmh	138.00 N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter of Reinforcing Pad	555.600	mm.
Thickness of Reinforcing Pad	28.0000	mm.

Note:

*External Forces and Moments in WRC 107/537 Convention:
 These loads are assumed to be SUSTAINED loads.*

Design Internal Pressure	Dp	12.29	bars
Radial Load	P	2676.93	Kgf
Circumferential Shear	Vc	1784.62	Kgf
Longitudinal Shear	Vl	1784.62	Kgf
Circumferential Moment	Mc	1748.26	Kg-m.
Longitudinal Moment	Ml	1748.26	Kg-m.
Torsional Moment	Mt	2497.50	Kg-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - ca))}$
 $= 355.6 + 2 \cdot 1.65 \cdot \sqrt{(2880.8(32.0 - 1.6))}$
 $= 1332.179 \text{ mm.}$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	5761.600	mm.
Nozzle Outside Diameter	(d) =	355.600	mm.
Vessel Thickness used	(T) =	58.400	mm.
Nozzle Thickness used	(t) =	18.400	mm.
T / t	=	3.174	
d / t	=	19.326	

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 0.613$$

Nr/P	=	0.124
Mr/P	=	0.133
M0/P	=	0.134
N0/P	=	0.253
MrD/Mc	=	0.178
NrDL/Mc	=	0.145
M0d/Mc	=	0.217
N0DL/Mc	=	0.124
MrD/Ml	=	0.165
NrDL/Ml	=	0.137
M0D/Ml	=	0.196
N0DL/Ml	=	0.162

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses Normal to	Top	Top	Bottom	Bottom
longitudinal plane)	Outside	Inside	Outside	Inside



Outplane Membrane (P)	-1	-1	-1	-1
Outplane Bending (P)	-6	6	-6	6
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-2	-2	2	2
Outplane Bending (ML)	-16	16	16	-16
Normal Pressure Stress	59	60	59	60

Outplane Stress Summary	32	79	70	50
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Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses parallel to	Top	Top	Bottom	Bottom
longitudinal plane)	Outside	Inside	Outside	Inside

Inplane Membrane (P)	0	0	0	0
Inplane Bending (P)	-6	6	-6	6
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-1	-1	1	1
Inplane Bending (ML)	-13	13	13	-13
Inplane Pressure Stress	29	29	29	29

Inplane Stress Summary	6	46	38	22
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Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Shear stress normal to	Top	Top	Bottom	Bottom
longitudinal plane)	Outside	Inside	Outside	Inside

Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	2	2	2	2

Shear Stress Summary	2	2	1	1
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Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stress Intensities	Top	Top	Bottom	Bottom
	Outside	Inside	Outside	Inside

Two * Max Shear Stress	32	79	70	50
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses Normal to	Left	Left	Right	Right
circumferential plane)	Outside	Inside	Outside	Inside

Outplane Membrane (P)	-1	-1	-1	-1
Outplane Bending (P)	-6	6	-6	6
Outplane Membrane (Mc)	-1	-1	1	1
Outplane Bending (Mc)	-18	18	18	-18
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	29	29	29	29

Outplane Stress Summary	1	50	41	17
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses parallel to	Left	Left	Right	Right
circumferential plane)	Outside	Inside	Outside	Inside

Inplane Membrane (P)	0	0	0	0
Inplane Bending (P)	-6	6	-6	6
Inplane Membrane (Mc)	-2	-2	2	2
Inplane Bending (Mc)	-15	15	15	-15
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0



Inplane Pressure Stress	59	60	59	60
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Inplane Stress Summary	35	78	69	52
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Shear stress normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside

Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Torsional Shear (Mt)	2	2	2	2

Shear Stress Summary	1	1	2	2
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stress Intensities)	Left Outside	Left Inside	Right Outside	Right Inside

Two * Max Shear Stress	35	78	69	52
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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses in the hoop direction)	Top Outside	Top Inside	Bottom Outside	Bottom Inside

Hoop Membrane (P)	-1	-1	-1	-1
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-2	-2	2	2
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	12	13	12	13

Hoop Stress Summary	7	9	12	13
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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses Normal to pipe cross-section)	Top Outside	Top Inside	Bottom Outside	Bottom Inside

Axial Membrane (P)	-1	-1	-1	-1
Axial Bending (P)	-32	32	-32	32
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-10	-10	10	10
Axial Bending (ML)	-82	82	82	-82
Axial Pressure Stress	6	6	6	6

Axial Stress Summary	-121	108	64	-33
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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Shear stress)	Outside	Inside	Outside	Inside

Shear due to (Vc)	1	1	-1	-1
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	6	6	6	6

Shear Stress Summary	8	8	4	4
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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stress Intensities)	Outside	Inside	Outside	Inside

Two * Max Shear Stress	130	109	65	48
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Nozzle Stresses (N./mm²):



CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Hoop Membrane (P)	-1	-1	-1	-1
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-1	-1	1	1
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	12	13	12	13
Hoop Stress Summary	8	9	11	13

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Axial Membrane (P)	-1	-1	-1	-1
Axial Bending (P)	-32	32	-32	32
Axial Membrane (Mc)	-10	-10	10	10
Axial Bending (Mc)	-90	90	90	-90
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	6	6	6	6
Axial Stress Summary	-129	117	73	-41

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-1	-1	1	1
Shear due to Torsion	6	6	6	6
Shear Stress Summary	4	4	8	8

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Two * Max Shear Stress	138	117	74	57

Stress Computations at the Edge of the Reinforcing Pad:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
= NozzleOD + 2 • 1.65 • $\sqrt{R_{mean}(t - ca)}$
= 355.6 + 2 • 1.65 • $\sqrt{2866.8(32.0 - 1.6)}$
= 1329.803 mm.

WRC 297 Curve Access Parameters:

Vessel Mean Diameter (D) = 5733.600 mm.
Nozzle Outside Diameter (d) = 555.600 mm.
Vessel Thickness used (T) = 30.400 mm.
Nozzle Thickness used (t) = 118.400 mm.
T / t = 0.257
d / t = 10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 1.331$$

Note: Re-pad thickness is added to nozzle thickness to simulate the Re-pad.

Nr/P = 0.090
Mr/P = 0.191
M0/P = 0.055



NØ/P = 0.104
MrD/Mc = 0.304
NrDL/Mc = 0.082
MØd/Mc = 0.093
NØDL/Mc = 0.058
MrD/Ml = 0.158
NrDL/Ml = 0.094
MØD/Ml = 0.049
NØDL/Ml = 0.116

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	-2	-2	-2	-2
Outplane Bending (P)	-9	9	-9	9
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-3	-3	3	3
Outplane Bending (ML)	-9	9	9	-9
Normal Pressure Stress	114	115	114	115
Outplane Stress Summary	88	128	115	116

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	-2	-2	-2	-2
Inplane Bending (P)	-32	32	-32	32
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-3	-3	3	3
Inplane Bending (ML)	-31	31	31	-31
Inplane Pressure Stress	57	57	57	57
Inplane Stress Summary	-12	115	57	58

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	1	1	1	1
Shear Stress Summary	2	2	0	0

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	101	128	116	116

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Membrane (P)	-2	-2	-2	-2
Outplane Bending (P)	-9	9	-9	9
Outplane Membrane (Mc)	-1	-1	1	1
Outplane Bending (Mc)	-18	18	18	-18
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0

Normal Pressure Stress	57	57	57	57
Outplane Stress Summary	24	80	65	47

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu	Cl	Du	Dl
	Left Outside	Left Inside	Right Outside	Right Inside
Inplane Membrane (P)	-2	-2	-2	-2
Inplane Bending (P)	-32	32	-32	32
Inplane Membrane (Mc)	-2	-2	2	2
Inplane Bending (Mc)	-60	60	60	-60
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	114	115	114	115
Inplane Stress Summary	15	204	143	87

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu	Cl	Du	Dl
	Left Outside	Left Inside	Right Outside	Right Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Torsional Shear (Mt)	1	1	1	1
Shear Stress Summary	0	0	2	2

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu	Cl	Du	Dl
	Left Outside	Left Inside	Right Outside	Right Inside
Two * Max Shear Stress	24	204	143	87

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	59	60	59	60	59	60	59	60
Circ. Pl (SUS)	-3	-3	1	1	-2	-2	2	2
Circ. Q (SUS)	-22	22	10	-10	-21	21	9	-9
Long. Pm (SUS)	29	29	29	29	29	29	29	29
Long. Pl (SUS)	-1	-1	1	1	-2	-2	0	0
Long. Q (SUS)	-19	19	7	-7	-24	24	12	-12
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	2	2	2	2	2	2	2	2
Pm (SUS)	59.0	60.0	59.0	60.0	59.0	60.0	59.0	60.0
Pm+Pl (SUS)	56.0	57.0	60.0	61.0	57.0	58.0	61.0	62.0
Pm+Pl+Q (Total)	34.2	79.1	70.1	51.1	36.1	79.1	70.1	53.1

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
------------------------	-----------	----------------	--------

Pm (SUS)	60.00	138.00	Passed
Pm+Pl (SUS)	62.00	207.00	Passed
Pm+Pl+Q (TOTAL)	79.14	414.00	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Reinforcing Pad Edge (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	114	115	114	115	114	115	114	115
Circ. Pl (SUS)	-5	-5	1	1	-4	-4	0	0
Circ. Q (SUS)	-18	18	0	0	-92	92	28	-28
Long. Pm (SUS)	57	57	57	57	57	57	57	57
Long. Pl (SUS)	-5	-5	1	1	-3	-3	-1	-1
Long. Q (SUS)	-63	63	-1	1	-27	27	9	-9
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	1	1	1	1	1	1	1	1
Pm (SUS)	114.0	115.0	114.0	115.0	114.0	115.0	114.0	115.0
Pm+Pl (SUS)	109.0	110.0	115.0	116.0	110.0	111.0	114.0	115.0
Pm+Pl+Q (Total)	102.0	128.1	115.0	116.0	27.1	203.0	142.0	87.0

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	115.00	138.00	Passed
Pm+Pl (SUS)	116.00	207.00	Passed
Pm+Pl+Q (TOTAL)	203.01	414.00	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	12	13	12	13	12	13	12	13
Circ. Pl (SUS)	-3	-3	1	1	-2	-2	0	0
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	6	6	6	6	6	6	6	6
Long. Pl (SUS)	-11	-11	9	9	-11	-11	9	9
Long. Q (SUS)	-114	114	50	-50	-122	122	58	-58
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	6	6	6	6	6	6	6	6
Pm (SUS)	12.0	13.0	12.0	13.0	12.0	13.0	12.0	13.0
Pm+Pl (SUS)	14.1	15.1	15.4	15.6	15.1	16.1	15.3	15.4
Pm+Pl+Q (Total)	128.8	109.5	65.5	50.0	137.4	117.2	73.8	57.7



Nozzle Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	13.00	138.00	Passed
Pm+Pl (SUS)	16.12	207.00	Passed
Pm+Pl+Q (TOTAL)	137.36	414.00	Passed

*Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.*

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Input, Nozzle Desc: K4-2"

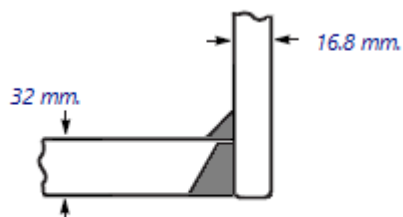
From: 80

Pressure for Reinforcement Calculations	P	12.288	bars
Temperature for Internal Pressure	Temp	82	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	5700.00	mm.
Shell Finished (Minimum) Thickness	t	32.0000	mm.
Shell Internal Corrosion Allowance	c	1.6000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		13900.00	mm.
User Entered Minimum Design Metal Temperature		-1.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-105	
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	N./mm ²
Allowable Stress At Ambient	Sna	138.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		270.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	16.8000	mm.
Flange Material		SA-105	
Flange Type		Long Weld Neck	
Corrosion Allowance	can	1.6000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	206.3750	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	24.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).



Reinforcement CALCULATION, Description: K4-2"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	2.000	in.
--	-------	-----



Actual Thickness Used in Calculation 0.661 in.

Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 12.29 \cdot 2851.6001 / (138 \cdot 1.0 - 0.6 \cdot 12.29)$
 $= 25.5287 \text{ mm.}$

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]
 $= R \cdot (\exp([P / (S_n \cdot E)] - 1))$ per Appendix 1-2 (a)(1)
 $= 27.0 \cdot (\exp([12.29 / (138.0 \cdot 1.0)] - 1))$
 $= 0.2415 \text{ mm.}$

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	145.2000 mm.
Parallel to Vessel Wall	Rn+tn+t	72.6000 mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	38.0000 mm.

*Taking a UG-36(c)(3)(a) exemption for nozzle: K4-2".
 This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 1.8415 mm.
Wall Thickness per UG16(b),	tr16b = 4.1000 mm.
Wall Thickness, shell/head, internal pressure	trb1 = 27.1287 mm.
Wall Thickness	tb1 = max(trb1, tr16b) = 27.1287 mm.
Wall Thickness	tb2 = max(trb2, tr16b) = 4.1000 mm.
Wall Thickness per table UG-45	tb3 = 6.4000 mm.

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[6.4, \max(27.1287, 4.1)]$
 $= 6.4000 \text{ mm.}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(1.8415, 6.4)$
 $= 6.4000 \text{ mm.}$

Available Nozzle Neck Thickness = 16.8000 mm. -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)(1)(b)), min(Curve:A, Curve:D)

Govrn. thk, tg = 16.8, tr = 0.241, c = 1.6 mm., E* = 1.0
 Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.016$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	7 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-104 °C



Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
Design Pressure/Ambient Rating = $12.29/51.00 = 0.241$

Weld Size Calculations, Description: K4-2"

Intermediate Calc. for nozzle/shell Welds Tmin 15.2000 mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$6.0000 = \text{Min per Code}$	$7.0700 = 0.7 * W_o \text{ mm.}$

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 0.3126 mm.
The Cut Length for this Nozzle is, Drop + Ho + H + T : 238.6876 mm.

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Input, Nozzle Desc: N3-14"

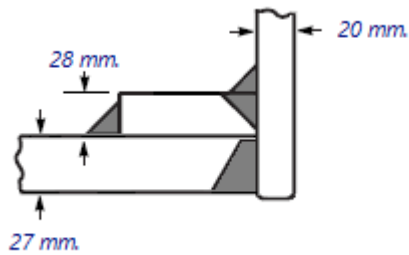
From: 90

Pressure for Reinforcement Calculations	P	12.287	bars
Temperature for Internal Pressure	Temp	82	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Elliptical Head	D	5700.00	mm.
Aspect Ratio of Elliptical Head	Ar	2.00	
Head Finished (Minimum) Thickness	t	27.0000	mm.
Head Internal Corrosion Allowance	c	1.6000	mm.
Head External Corrosion Allowance	co	0.0000	mm.
Distance from Head Centerline	L1	0.0000	mm.
User Entered Minimum Design Metal Temperature		-1.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-516 70	
Material UNS Number		K02700	
Material Specification/Type		Plate	
Allowable Stress at Temperature	Sn	138.00	N./mm ²
Allowable Stress At Ambient	Sna	138.00	N./mm ²
Diameter Basis (for tr calc only)		Outside	
Layout Angle		0.00	deg
Diameter		14.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	20.0000	mm.
Flange Material		SA-105	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	1.6000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	373.0000	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	24.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Pad Material		SA-516 70	
Pad Allowable Stress at Temperature	Sp	138.00	N./mm ²
Pad Allowable Stress At Ambient	Spa	138.00	N./mm ²
Diameter of Pad along vessel surface	Dp	555.6000	mm.
Thickness of Pad	te	28.0000	mm.
Weld leg size between Pad and Shell	Wp	20.0000	mm.
Groove weld depth between Pad and Nozzle	Wgpn	28.0000	mm.
Reinforcing Pad Width		100.0000	mm.
Flange Class		150	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).



Reinforcement CALCULATION, Description: N3-14"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	14.000 in.
Actual Thickness Used in Calculation	0.787 in.

Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Elliptical Head, tr [Int. Press]
 $= P \cdot D \cdot K1 / (2 \cdot Sv \cdot E - 0.2 \cdot P)$ per Appendix 1-4(c)
 $= 12.29 \cdot 5703.2 \cdot 0.8995 / (2 \cdot 138.0 \cdot 1.0 - 0.2 \cdot 12.29)$
 $= 22.8607 \text{ mm.}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot Ro / (Sn \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 12.29 \cdot 177.8 / (138 \cdot 1.0 + 0.4 \cdot 12.29)$
 $= 1.5776 \text{ mm.}$

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	637.6000 mm.
Parallel to Vessel Wall, opening length	d	318.8000 mm.
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		63.5000 mm.

Weld Strength Reduction Factor [fr1]:

$= \min(1, Sn/Sv)$
 $= \min(1, 138.0/138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:

$= \min(1, Sn/Sv)$
 $= \min(1, 138.0/138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr4]:

$= \min(1, Sp/Sv)$
 $= \min(1, 138.0/138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:

$= \min(fr2, fr4)$
 $= \min(1.0, 1.0)$
 $= 1.000$

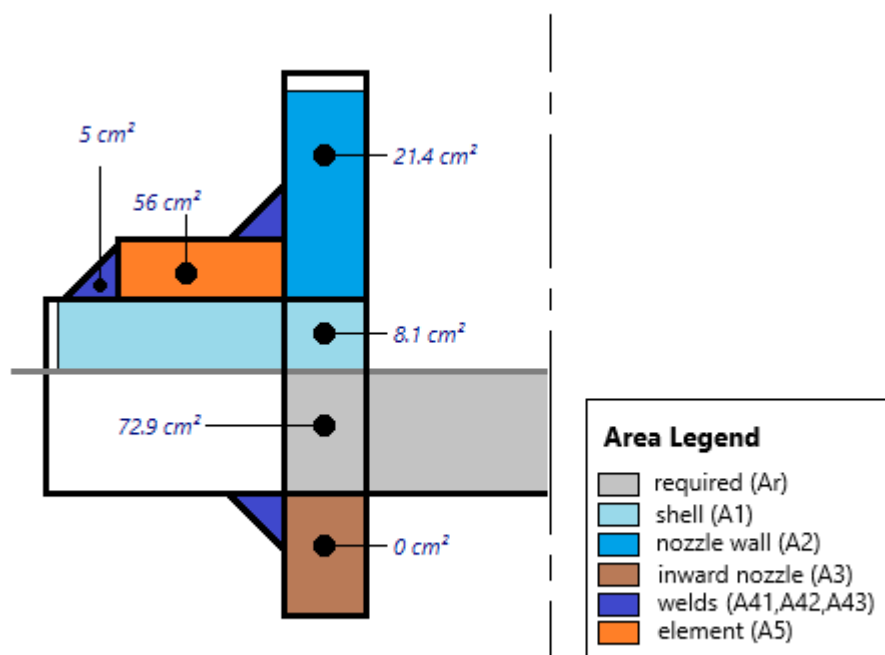


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	72.880	NA	NA
Area in Shell	A1	8.095	NA	NA
Area in Nozzle Wall	A2	21.364	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	5.000	NA	NA
Area in Element	A5	56.000	NA	NA
TOTAL AREA AVAILABLE	Atot	90.460	NA	NA

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Insufficient.
The area available with the given pad is Sufficient.

SELECTION OF POSSIBLE REINFORCING PADS: Diameter Thickness
Based on given Pad Thickness: 492.8156 28.0000 mm.
Based on given Pad Diameter: 555.6000 19.2102 mm.
Based on Shell or Nozzle Thickness: 547.7018 20.0000 mm.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1))$ UG-37(c)
 $= (318.8 \cdot 22.8607 \cdot 1.0 + 2 \cdot 18.4 \cdot 22.8607 \cdot 1.0 \cdot (1 - 1.0))$
 $= 72.880 \text{ cm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 318.8(1.0 \cdot 25.4 - 1.0 \cdot 22.861) - 2 \cdot 18.4$
 $(1.0 \cdot 25.4 - 1.0 \cdot 22.8607) \cdot (1 - 1.0)$
 $= 8.095 \text{ cm}^2$

Area Available in Nozzle Wall Projecting Outward [A2]:
 $= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2$



$$= (2 \cdot 63.5) \cdot (18.4 - 1.58) \cdot 1.0$$
$$= 21.364 \text{ cm}^2$$

Area Available in Welds [A41 + A42 + A43]:

$$= W_o^2 \cdot fr3 + (W_i - can/0.707)^2 \cdot fr2 + W_p^2 \cdot fr4$$
$$= 10.0^2 \cdot 1.0 + (0.0)^2 \cdot 1.0 + 20.0^2 \cdot 1.0$$
$$= 5.000 \text{ cm}^2$$

Area Available in Element [A5]:

$$= (\min(D_p, D_L) - (\text{Nozzle OD})) \cdot (\min(t_p, T_{lwp}, t_e)) \cdot fr4$$
$$= (555.6 - 355.6) \cdot 28.0 \cdot 1.0$$
$$= 56.000 \text{ cm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	$t_a = 3.1776 \text{ mm.}$
Wall Thickness per UG16(b),	$tr_{16b} = 4.1000 \text{ mm.}$
Wall Thickness, shell/head, internal pressure	$trb1 = 26.9954 \text{ mm.}$
Wall Thickness	$tb1 = \max(trb1, tr_{16b}) = 26.9954 \text{ mm.}$
Wall Thickness	$tb2 = \max(trb2, tr_{16b}) = 4.1000 \text{ mm.}$
Wall Thickness per table UG-45	$tb3 = 9.9312 \text{ mm.}$

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb3, \max(tb1, tb2)]$$
$$= \min[9.931, \max(26.9954, 4.1)]$$
$$= 9.9312 \text{ mm.}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(t_a, t_b)$$
$$= \max(3.1776, 9.9312)$$
$$= 9.9312 \text{ mm.}$$

Available Nozzle Neck Thickness = 20.0000 mm. -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 21.9,	Allowable	: 138.0 N./mm ²	Passed
Expansion	: 0.0,	Allowable	: 323.1 N./mm ²	Passed
Occasional	: 5.0,	Allowable	: 183.5 N./mm ²	Passed
Shear	: 11.0,	Allowable	: 96.6 N./mm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:D, Curve:A)

Govrn. thk, $t_g = 20.0$, $t_r = 1.578$, $c = 1.6 \text{ mm.}$, $E^* = 1.0$
Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.086$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	12 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Nozzle Neck to Pad Weld for the Nozzle, Curve: D

Govrn. thk, $t_g = 20.0$, $t_r = 1.578$, $c = 1.6 \text{ mm.}$, $E^* = 1.0$
Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.086$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve D	-40 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: B

Govrn. thk, $t_g = 20.0$, $t_r = 1.578$, $c = 1.6 \text{ mm.}$, $E^* = 1.0$
Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.086$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-7 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C



Min Metal Temp. w/o impact per UG-20(f) -29 °C

Shell to Pad Weld Junction at Pad OD, min(Curve:B, Curve:D)

Govrn. thk, tg = 27.0, tr = 22.861, c = 1.6 mm., E* = 1.0
 Thickness Ratio = $tr \cdot E^*/(tg - c) = 0.9$, Temp. Reduction = 6 °C

Min Metal Temp. w/o impact per UCS-66, Curve B 1 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -4 °C

Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: D

Govrn. thk, tg = 20.0, tr = 1.578, c = 1.6 mm., E* = 1.0
 Thickness Ratio = $tr \cdot E^*/(tg - c) = 0.086$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve D -40 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Gov. MDMT of the Nozzle : -104 °C
 Gov. MDMT of the Reinforcement Pad : -4 °C
 Gov. MDMT of the nozzle to shell joint welded assembly : -4 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
 Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -39 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = $12.29/19.60 = 0.627$

Weld Size Calculations, Description: N3-14"

Intermediate Calc. for nozzle/shell Welds Tmin 18.4000 mm.
 Intermediate Calc. for pad/shell Welds TminPad 19.0000 mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm.
Pad Weld	9.5000 = 0.5*TminPad	14.1400 = 0.7 * Wp mm.

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A-A1+2 \cdot tn \cdot fr1 \cdot (E1 \cdot t - tr)) \cdot Sv)$
 $= \max(0, (72.88 - 8.0952 + 2 \cdot 18.4 \cdot 1.0 \cdot (1.0 \cdot 25.4 - 22.8607)) \cdot 138)$
 $= 92479.02 \text{ Kg}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2+A5+A4-(Wi-Can/.707)^2 \cdot fr2) \cdot Sv$
 $= (21.3645 + 56.0 + 5.0 - 0.0 \cdot 1.0) \cdot 138$
 $= 115901.84 \text{ Kg}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot tn \cdot t \cdot fr1)) \cdot Sv$
 $= (21.3645 + 0.0 + 1.0 + (9.3472)) \cdot 138$
 $= 44624.09 \text{ Kg}$

Weld Load [W3]:
 $= (A2+A3+A4+A5+(2 \cdot tn \cdot t \cdot fr1)) \cdot S$
 $= (21.3645 + 0.0 + 5.0 + 56.0 + (9.3472)) \cdot 138$
 $= 129055.05 \text{ Kg}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot Dlo \cdot Wo \cdot 0.49 \cdot Snw$

$$= (3.1416/2.0) \cdot 355.6 \cdot 10.0 \cdot 0.49 \cdot 138$$

$$= 38515. \text{ Kg}$$

Shear, Pad Element Weld [Spew]:

$$= (\pi/2) \cdot DP \cdot WP \cdot 0.49 \cdot SEW$$

$$= (3.1416/2.0) \cdot 555.6 \cdot 20.0 \cdot 0.49 \cdot 138$$

$$= 120353. \text{ Kg}$$

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (Dlr + Dlo) / 4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn$$

$$= (3.1416 \cdot 168.6) \cdot (20.0 - 1.6) \cdot 0.7 \cdot 138$$

$$= 96001. \text{ Kg}$$

Tension, Pad Groove Weld [Tpgw]:

$$= (\pi/2) \cdot Dlo \cdot Wgpn \cdot 0.74 \cdot Seg$$

$$= (3.1416/2) \cdot 355.6 \cdot 28.0 \cdot 0.74 \cdot 138$$

$$= 162863. \text{ Kg}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot Dlo \cdot Wgnvi \cdot 0.74 \cdot Sng$$

$$= (3.1416/2.0) \cdot 355.6 \cdot (24.0 - 0.0) \cdot 0.74 \cdot 138$$

$$= 139597. \text{ Kg}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SPEW + SNW) = (120353 + 96001) = 216354 \text{ Kg}$$

Failure Path [Path 2-2]:

$$= (Sonw + Tpgw + Tngw + Sinw)$$

$$= (38515 + 162863 + 139597 + 0) = 340974 \text{ Kg}$$

Failure Path [Path 3-3]:

$$= (Spew + Tngw + Sinw)$$

$$= (120353 + 139597 + 0) = 259950 \text{ Kg}$$

Summary of Failure Path Calculations:

Path 1-1 = 216354 Kg, must exceed W = 92479 Kg	or W1 = 115901 Kg
Path 2-2 = 340974 Kg, must exceed W = 92479 Kg	or W2 = 44624 Kg
Path 3-3 = 259950 Kg, must exceed W = 92479 Kg	or W3 = 129055 Kg

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 12 bars

Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 3.0692 mm.
The Cut Length for this Nozzle is, Drop + Ho + H + T : 403.0691 mm.

Percent Elongation Calculations:

% Elongation per Table UG-79-1 ($50 \cdot t_{nom} / R_f \cdot (1 - R_f / R_o)$) 5.959 %
Note: Please Check Requirements of UCS-79 as Elongation is > 5%.

Input Echo, WRC107/537 Item 1, Description: N3-14" :

Diameter Basis for Vessel	Vbasis	ID
Cylindrical or Spherical Vessel	Cylsph	Spherical
Internal Corrosion Allowance	Cas	1.6000 mm.
Vessel Diameter	Dv	10260.000 mm.
Vessel Thickness	Tv	27.000 mm.
Design Temperature	T1	82.0 C
Vessel Material		SA-516 70
Vessel UNS Number		K02700
Vessel Cold S.I. Allowable	Smc	138.00 N./mm ²
Vessel Hot S.I. Allowable	Smh	138.00 N./mm ²



See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

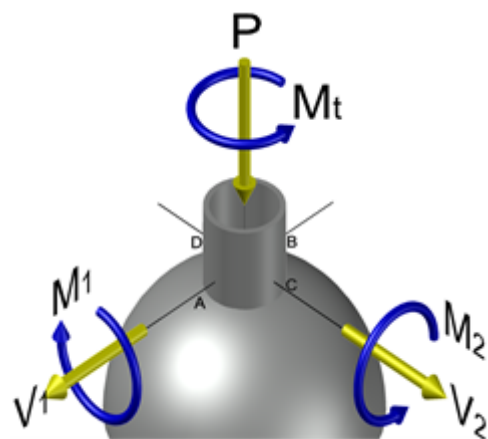
Attachment Type	Type	Round
WRC107 Attachment Classification	Holsol	Hollow
Diameter Basis for Nozzle	Nbasis	OD
Corrosion Allowance for Nozzle	Can	1.6000 mm.
Nozzle Diameter	Dn	355.600 mm.
Nozzle Thickness	Tn	20.000 mm.
Nozzle Material		SA-516 70
Nozzle UNS Number		K02700
Nozzle Cold S.I. Allowable	SNmc	138.00 N./mm ²
Nozzle Hot S.I. Allowable	SNmh	138.00 N./mm ²
Thickness of Reinforcing Pad	Tpad	28.000 mm.
Diameter of Reinforcing Pad	Dpad	555.600 mm.
Design Internal Pressure	Dp	12.287 bars
Include Pressure Thrust		No

External Forces and Moments in WRC 107/537 Convention:

Radial Load (SUS)	P	2676.9 Kgf
Longitudinal Shear (SUS)	(V1) V1	1784.6 Kgf
Circumferential Shear (SUS)	(Vc) V2	1784.6 Kgf
Circumferential Moment (SUS)	(Mc) M1	1748.3 Kg-m.
Longitudinal Moment (SUS)	(Ml) M2	1748.3 Kg-m.
Torsional Moment (SUS)	Mt	2497.5 Kg-m.

Use Interactive Control	No
WRC107 Version	Version March 1979
Include Pressure Stress Indices per Div. 2	No
Compute Pressure Stress per WRC-368	No
Local Loads applied at end of Nozzle/Attachment	No

WRC Bulletin 537 provides equations for the dimensionless curves found in bulletin 107. As noted in the foreword to bulletin 537, "537 is equivalent to WRC 107". Where 107 is printed in the results below, "537" can be interchanged with "107".



WRC 107/537 has dimensional limitations that can be found in the bulletin's Foreword and which can also be inferred by review of the non-dimensional curve limits.

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(\text{Rmean}(t - ca))}$$

$$= 355.6 + 2 \cdot 1.65 \cdot \sqrt{(5144.3 (27.0 - 1.6))}$$

$$= 1548.472 \text{ mm.}$$

WRC 107 Stress Calculation for SUStained loads:

Radial Load	P	2676.9 Kgf
-------------	---	------------

Circumferential Shear	(VC) V2	1784.6	Kgf
Longitudinal Shear	(VL) V1	1784.6	Kgf
Circumferential Moment	(MC) M1	1748.3	Kg-m.
Longitudinal Moment	(ML) M2	1748.3	Kg-m.
Torsional Moment	MT	2497.5	Kg-m.

Dimensionless Parameters: U = 0.34 TAU = 9.16 RHO = 2.90

Dimensionless Loads for Spherical Shells at Attachment Junction:

Curves read for 1979	Figure	Value	Location
N(x) * T / P	SP 3	0.07767	(A,B,C,D)
M(x) / P	SP 3	0.05334	(A,B,C,D)
N(x) * T * SQRT(Rm * T) / MC	SM 3	0.21314	(A,B,C,D)
M(x) * SQRT(Rm * T) / MC	SM 3	0.16228	(A,B,C,D)
N(x) * T * SQRT(Rm * T) / ML	SM 3	0.21314	(A,B,C,D)
M(x) * SQRT(Rm * T) / ML	SM 3	0.16228	(A,B,C,D)
N(y) * T / P	SP 3	0.27983	(A,B,C,D)
M(y) / P	SP 3	0.09551	(A,B,C,D)
N(y) * T * SQRT(Rm * T) / MC	SM 3	0.18016	(A,B,C,D)
M(y) * SQRT(Rm * T) / MC	SM 3	0.36578	(A,B,C,D)
N(y) * T * SQRT(Rm * T) / ML	SM 3	0.18016	(A,B,C,D)
M(y) * SQRT(Rm * T) / ML	SM 3	0.36578	(A,B,C,D)

Stress Concentration Factors: Kn = 1.00, Kb = 1.00

Stresses in the Vessel at the Attachment Junction (N./mm²)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad.	Memb. P	-0.7	-0.7	-0.7	-0.7	-0.7	-0.7	-0.7	-0.7
Rad.	Bend. P	-2.9	2.9	-2.9	2.9	-2.9	2.9	-2.9	2.9
Rad.	Memb. MC	0.0	0.0	0.0	0.0	-2.4	-2.4	2.4	2.4
Rad.	Bend. MC	0.0	0.0	0.0	0.0	-11.2	11.2	11.2	-11.2
Rad.	Memb. ML	-2.4	-2.4	2.4	2.4	0.0	0.0	0.0	0.0
Rad.	Bend. ML	-11.2	11.2	11.2	-11.2	0.0	0.0	0.0	0.0
Tot.	Rad. Str.	-17.3	10.9	9.9	-6.5	-17.3	10.9	9.9	-6.5
Tang.	Memb. P	-2.6	-2.6	-2.6	-2.6	-2.6	-2.6	-2.6	-2.6
Tang.	Bend. P	-5.3	5.3	-5.3	5.3	-5.3	5.3	-5.3	5.3
Tang.	Memb. MC	0.0	0.0	0.0	0.0	-2.1	-2.1	2.1	2.1
Tang.	Bend. MC	0.0	0.0	0.0	0.0	-25.1	25.1	25.1	-25.1
Tang.	Memb. ML	-2.1	-2.1	2.1	2.1	0.0	0.0	0.0	0.0
Tang.	Bend. ML	-25.1	25.1	25.1	-25.1	0.0	0.0	0.0	0.0
Tot.	Tang. Str.	-35.1	25.8	19.4	-20.4	-35.1	25.8	19.4	-20.4
	Shear VC	0.6	0.6	-0.6	-0.6	0.0	0.0	0.0	0.0
	Shear VL	0.0	0.0	0.0	0.0	-0.6	-0.6	0.6	0.6
	Shear MT	2.3	2.3	2.3	2.3	2.3	2.3	2.3	2.3
Tot.	Shear	2.9	2.9	1.7	1.7	1.7	1.7	2.9	2.9
Str.	Int.	35.5	26.3	19.7	20.6	35.2	26.0	20.2	21.0

Unitless Prm: U = 0.77 TAU = 0.00 (14.60) RHO = 0.00 (1.38)

Dimensionless Loads for Spherical Shells at Pad edge:

Curves read for 1979	Figure	Value	Location
N(x) * T / P	SR 2	0.08996	(A,B,C,D)
M(x) / P	SR 2	0.04924	(A,B,C,D)
N(x) * T * SQRT(Rm * T) / MC	SR 3	0.10459	(A,B,C,D)



$M(x) * T / P$ SR 3 0.12156 (A,B,C,D)
 $N(x) * T * \text{SQRT}(R_m * T) / MC$ SR 3 0.10459 (A,B,C,D)
 $M(x) * T * \text{SQRT}(R_m * T) / ML$ SR 3 0.12156 (A,B,C,D)

$N(y) * T / P$ SR 2 0.02706 (A,B,C,D)
 $M(y) / P$ SR 2 0.01525 (A,B,C,D)
 $N(y) * T * \text{SQRT}(R_m * T) / MC$ SR 3 0.03150 (A,B,C,D)
 $M(y) * \text{SQRT}(R_m * T) / MC$ SR 3 0.03638 (A,B,C,D)
 $N(y) * T * \text{SQRT}(R_m * T) / ML$ SR 3 0.03150 (A,B,C,D)
 $M(y) * \text{SQRT}(R_m * T) / ML$ SR 3 0.03638 (A,B,C,D)

Stress Concentration Factors: $K_n = 1.00$, $K_b = 1.00$

Stresses in the Vessel at the Edge of Reinforcing Pad (N./mm²)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad. Memb. P		-3.7	-3.7	-3.7	-3.7	-3.7	-3.7	-3.7	-3.7
Rad. Bend. P		-12.0	12.0	-12.0	12.0	-12.0	12.0	-12.0	12.0
Rad. Memb. MC		0.0	0.0	0.0	0.0	-7.7	-7.7	7.7	7.7
Rad. Bend. MC		0.0	0.0	0.0	0.0	-53.6	53.6	53.6	-53.6
Rad. Memb. ML		-7.7	-7.7	7.7	7.7	0.0	0.0	0.0	0.0
Rad. Bend. ML		-53.6	53.6	53.6	-53.6	0.0	0.0	0.0	0.0
Tot. Rad. Str.		-77.0	54.3	45.6	-37.6	-77.0	54.3	45.6	-37.6
Tang. Memb. P		-1.1	-1.1	-1.1	-1.1	-1.1	-1.1	-1.1	-1.1
Tang. Bend. P		-3.7	3.7	-3.7	3.7	-3.7	3.7	-3.7	3.7
Tang. Memb. MC		0.0	0.0	0.0	0.0	-2.3	-2.3	2.3	2.3
Tang. Bend. MC		0.0	0.0	0.0	0.0	-16.0	16.0	16.0	-16.0
Tang. Memb. ML		-2.3	-2.3	2.3	2.3	0.0	0.0	0.0	0.0
Tang. Bend. ML		-16.0	16.0	16.0	-16.0	0.0	0.0	0.0	0.0
Tot. Tang. Str.		-23.2	16.4	13.5	-11.1	-23.2	16.4	13.5	-11.1
Shear VC		0.8	0.8	-0.8	-0.8	0.0	0.0	0.0	0.0
Shear VL		0.0	0.0	0.0	0.0	-0.8	-0.8	0.8	0.8
Shear MT		2.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0
Tot. Shear		2.8	2.8	1.2	1.2	1.2	1.2	2.8	2.8
Str. Int.		77.1	54.5	45.7	37.6	77.0	54.3	45.9	37.9

WRC 107/537 Stress Summations:

Vessel Stress Summation at Attachment Junction (N./mm²)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad. Pm (SUS)		59.0	59.0	59.0	59.0	59.0	59.0	59.0	59.0
Rad. Pl (SUS)		-3.2	-3.2	1.7	1.7	-3.2	-3.2	1.7	1.7
Rad. Q (SUS)		-14.1	14.1	8.2	-8.2	-14.1	14.1	8.2	-8.2
Long. Pm (SUS)		59.0	59.0	59.0	59.0	59.0	59.0	59.0	59.0
Long. Pl (SUS)		-4.6	-4.6	-0.5	-0.5	-4.6	-4.6	-0.5	-0.5
Long. Q (SUS)		-30.4	30.4	19.9	-19.9	-30.4	30.4	19.9	-19.9
Shear Pm (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear Pl (SUS)		0.6	0.6	-0.6	-0.6	-0.6	-0.6	0.6	0.6
Shear Q (SUS)		2.3	2.3	2.3	2.3	2.3	2.3	2.3	2.3
Pm (SUS)		59.0	59.0	59.0	59.0	59.0	59.0	59.0	59.0
Pm+Pl (SUS)		56.1	56.1	60.9	60.9	56.1	56.1	60.9	60.9

Pm+Pl+Q (Total)	42.2	85.4	78.7	52.8	42.0	85.0	79.2	53.1
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Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	59.05	138.00	Passed
Pm+Pl (SUS)	60.92	207.00	Passed
Pm+Pl+Q (TOTAL)	85.37	414.00	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 107/537 Stress Summations:

Vessel Stress Summation at Reinforcing Pad Edge (N./mm²)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad.	Pm (SUS)	124.1	124.1	124.1	124.1	124.1	124.1	124.1	124.1
Rad.	Pl (SUS)	-11.3	-11.3	4.0	4.0	-11.3	-11.3	4.0	4.0
Rad.	Q (SUS)	-65.6	65.6	41.6	-41.6	-65.6	65.6	41.6	-41.6
Long.	Pm (SUS)	124.1	124.1	124.1	124.1	124.1	124.1	124.1	124.1
Long.	Pl (SUS)	-3.4	-3.4	1.2	1.2	-3.4	-3.4	1.2	1.2
Long.	Q (SUS)	-19.8	19.8	12.3	-12.3	-19.8	19.8	12.3	-12.3
Shear	Pm (SUS)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear	Pl (SUS)	0.8	0.8	-0.8	-0.8	-0.8	-0.8	0.8	0.8
Shear	Q (SUS)	2.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0
Pm (SUS)		124.1	124.1	124.1	124.1	124.1	124.1	124.1	124.1
Pm+Pl (SUS)		120.8	120.8	128.4	128.4	120.8	120.8	128.4	128.4
Pm+Pl+Q (Total)		101.1	178.6	169.8	113.1	101.0	178.5	170.0	113.3

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	124.13	138.00	Passed
Pm+Pl (SUS)	128.37	207.00	Passed
Pm+Pl+Q (TOTAL)	178.62	414.00	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

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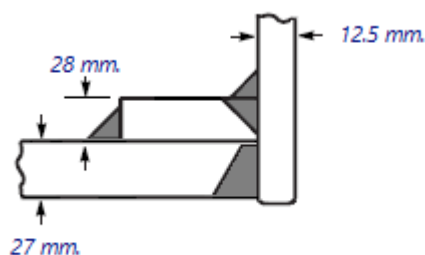
Input, Nozzle Desc: N4-6"
From: 90

Pressure for Reinforcement Calculations	P	12.287	bars
Temperature for Internal Pressure	Temp	82	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Elliptical Head	D	5700.00	mm.
Aspect Ratio of Elliptical Head	Ar	2.00	
Head Finished (Minimum) Thickness	t	27.0000	mm.
Head Internal Corrosion Allowance	c	1.6000	mm.
Head External Corrosion Allowance	co	0.0000	mm.
Distance from Head Centerline	L1	1500.0000	mm.
User Entered Minimum Design Metal Temperature		-1.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-106 B	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	118.00	N./mm ²
Allowable Stress At Ambient	Sna	118.00	N./mm ²
Diameter Basis (for tr calc only)		Outside	
Layout Angle		90.00	deg
Diameter		6.0000	in.
Size and Thickness Basis		Minimum	
Nominal Thickness		120	
Flange Material		SA-105	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	1.6000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	411.1000	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	24.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Pad Material		SA-516 70	
Pad Allowable Stress at Temperature	Sp	138.00	N./mm ²
Pad Allowable Stress At Ambient	Spa	138.00	N./mm ²
Diameter of Pad along vessel surface	Dp	268.2750	mm.
Thickness of Pad	te	28.0000	mm.
Weld leg size between Pad and Shell	Wp	18.0000	mm.
Groove weld depth between Pad and Nozzle	Wgpn	28.0000	mm.
Reinforcing Pad Width		50.0000	mm.
Flange Class		150	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).



Note : Checking Nozzle in the Meridional direction.

Reinforcement CALCULATION, Description: N4-6"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	6.625 in.
Actual Thickness Used in Calculation	0.492 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Elliptical Head, tr [Int. Press]
 $= P \cdot D \cdot K1 / (2 \cdot Sv \cdot E - 0.2 \cdot P)$ per Appendix 1-4(c)
 $= 12.29 \cdot 5703.2 \cdot 0.8995 / (2 \cdot 138.0 \cdot 1.0 - 0.2 \cdot 12.29)$
 $= 22.8608 \text{ mm.}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot Ro / (Sn \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 12.29 \cdot 84.1375 / (118 \cdot 1.0 + 0.4 \cdot 12.29)$
 $= 0.8725 \text{ mm.}$

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	306.7325 mm.
Parallel to Vessel Wall, opening length	d	153.3662 mm.
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		55.2261 mm.

Weld Strength Reduction Factor [fr1]:
 $= \min(1, Sn/Sv)$
 $= \min(1, 118.0/138.0)$
 $= 0.855$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, Sn/Sv)$
 $= \min(1, 118.0/138.0)$
 $= 0.855$

Weld Strength Reduction Factor [fr4]:
 $= \min(1, Sp/Sv)$
 $= \min(1, 138.0/138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(0.855, 1.0)$
 $= 0.855$

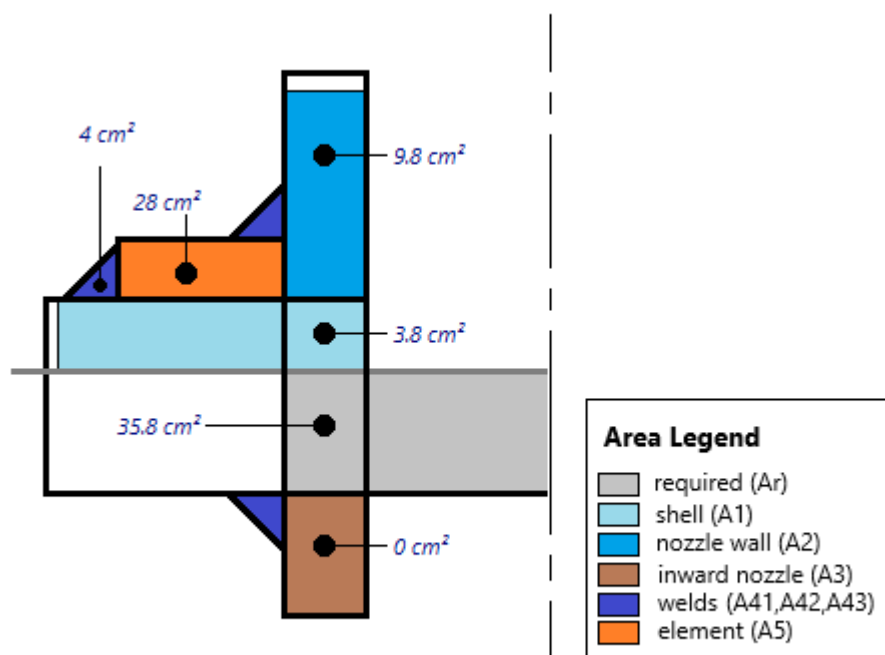


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	35.782	NA	NA
Area in Shell	A1	3.814	NA	NA
Area in Nozzle Wall	A2	9.840	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	4.021	NA	NA
Area in Element	A5	28.000	NA	NA
TOTAL AREA AVAILABLE	Atot	45.675	NA	NA

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 72.78 degs.

The area available without a pad is Insufficient.
The area available with the given pad is Sufficient.

SELECTION OF POSSIBLE REINFORCING PADS:	Diameter	Thickness
Based on given Pad Thickness:	232.9446	28.0000 mm.
Based on given Pad Diameter:	268.2750	18.1075 mm.
Based on the Estimated Diameter Limit:	305.1450	14.0394 mm.

Area Required [A]:

$$= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$$

$$= (153.3662 \cdot 22.8608 \cdot 1.0 + 2 \cdot 10.8904 \cdot 22.8608 \cdot 1.0 \cdot (1 - 0.86))$$

$$= 35.782 \text{ cm}^2$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d (E1 \cdot t - F \cdot tr) - 2 \cdot tn (E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 153.366 (1.0 \cdot 25.4 - 1.0 \cdot 22.861) - 2 \cdot 10.89$$

$$(1.0 \cdot 25.4 - 1.0 \cdot 22.8608) \cdot (1 - 0.855)$$

$$= 3.814 \text{ cm}^2$$

Area Available in Nozzle Wall Projecting Outward [A2]:

$$= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2 / \sin(\alpha3)$$

$$= (2 \cdot 55.23) \cdot (10.89 - 0.87) \cdot 0.8551 / \sin(74.1)$$

$$= 9.840 \text{ cm}^2$$

See ASME VIII-1 2011(a) Appendix L, L-7.7.7(b) for more information.

Area Available in Welds [A41 + A42 + A43]:

$$= (W_o^2 - A_r \text{ Lost}) \cdot Fr_3 + ((W_i - c_{an} / 0.707)^2 - A_r \text{ Lost}) \cdot fr_2 + \text{Trap} fr_4$$

$$= (1.) \cdot 0.86 + (0.0) \cdot 0.86 + 316.6113^2 \cdot 1.0$$

$$= 4.021 \text{ cm}^2$$

Area Available in Element [A5]:

$$= (\min(D_p, D_L) - (\text{Nozzle OD})) \cdot (\min(t_p, T_{lwp}, t_e)) \cdot fr_4$$

$$= (276.1689 - 176.1689) \cdot 28.0 \cdot 1.0$$

$$= 28.000 \text{ cm}^2$$

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

Govrn. thk, $t_g = 12.49$, $t_r = 0.873$, $c = 1.6 \text{ mm.}$, $E^* = 1.0$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.08$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	-1 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle Neck to Pad Weld for the Nozzle, Curve: B

Govrn. thk, $t_g = 12.49$, $t_r = 0.873$, $c = 1.6 \text{ mm.}$, $E^* = 1.0$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.08$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-22 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: B

Govrn. thk, $t_g = 12.49$, $t_r = 0.873$, $c = 1.6 \text{ mm.}$, $E^* = 1.0$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.08$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-22 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Shell to Pad Weld Junction at Pad OD, min(Curve:B, Curve:D)

Govrn. thk, $t_g = 27.0$, $t_r = 22.861$, $c = 1.6 \text{ mm.}$, $E^* = 1.0$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.9$, Temp. Reduction = 6 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	1 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-4 °C

Nozzle-Shell/Head Weld (UCS-66(a)(1)(b)), min(Curve:B, Curve:D)

Govrn. thk, $t_g = 12.49$, $t_r = 0.873$, $c = 1.6 \text{ mm.}$, $E^* = 1.0$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.08$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-22 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Gov. MDMT of the Nozzle	: -104 °C
Gov. MDMT of the Reinforcement Pad	: -4 °C
Gov. MDMT of the nozzle to shell joint welded assembly	: -4 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-39 °C



Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = 12.29/19.60 = 0.627

Weld Size Calculations, Description: N4-6"

Intermediate Calc. for nozzle/shell Welds T_{min} 10.8904 mm.
 Intermediate Calc. for pad/shell Welds T_{minPad} 19.0000 mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * W _o mm.
Pad Weld	9.5000 = 0.5*T _{minPad}	12.7260 = 0.7 * W _p mm.

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A-A1+2 \cdot t_n \cdot f_{r1} \cdot (E1 \cdot t - t_r)) \cdot S_v)$
 $= \max(0, (35.7824 - 3.8141 + 2 \cdot 10.8904 \cdot 0.8551 \cdot (1.0 \cdot 25.4 - 22.8608)) \cdot 138)$
 $= 45650.59 \text{ Kg}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2+A5+A4-(W_i - \text{Can}/.707)^2 \cdot f_{r2}) \cdot S_v$
 $= (9.8395 + 28.0 + 4.0212 - 0.0 \cdot 0.86) \cdot 138$
 $= 58905.68 \text{ Kg}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S_v$
 $= (9.8395 + 0.0 + 0.8551 + (4.7306)) \cdot 138$
 $= 21706.03 \text{ Kg}$

Weld Load [W3]:
 $= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S$
 $= (9.8395 + 0.0 + 4.0212 + 28.0 + (4.7306)) \cdot 138$
 $= 65562.44 \text{ Kg}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$
 $= (3.1416/2.0) \cdot 176.1689 \cdot 10.0 \cdot 0.49 \cdot 118$
 $= 16315. \text{ Kg}$

Shear, Pad Element Weld [Spew]:
 $= (\pi/2) \cdot D_P \cdot W_P \cdot 0.49 \cdot S_{EW}$
 $= (3.1416/2.0) \cdot 268.275 \cdot 18.0 \cdot 0.49 \cdot 138$
 $= 52302. \text{ Kg}$

Shear, Nozzle Wall [Snw]:
 $= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - \text{Can}) \cdot 0.7 \cdot S_n$
 $= (3.1416 \cdot 82.3838) \cdot (12.4904 - 1.6) \cdot 0.7 \cdot 118$
 $= 23740. \text{ Kg}$

Tension, Pad Groove Weld [Tpgw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_{gpn} \cdot 0.74 \cdot S_{eg}$
 $= (3.1416/2) \cdot 176.1689 \cdot 28.0 \cdot 0.74 \cdot 138$
 $= 80684. \text{ Kg}$

Tension, Shell Groove Weld [Tngw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_{gvi} \cdot 0.74 \cdot S_{ng}$
 $= (3.1416/2.0) \cdot 176.1689 \cdot (24.0 - 0.0) \cdot 0.74 \cdot 138$
 $= 69158. \text{ Kg}$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:



Failure Path [Path 1-1]:

$$= (\text{SPEW} + \text{SNW}) = (52302 + 23740) = 76043 \text{ Kgf}$$

Failure Path [Path 2-2]:

$$= (\text{Sonw} + \text{Tpgw} + \text{Tngw} + \text{Sinw}) \\ = (16315 + 80684 + 69158 + 0) = 166158 \text{ Kgf}$$

Failure Path [Path 3-3]:

$$= (\text{Spew} + \text{Tngw} + \text{Sinw}) \\ = (52302 + 69158 + 0) = 121460 \text{ Kgf}$$

Summary of Failure Path Calculations:

Path 1-1 = 76042 Kgf, must exceed W = 45650 Kgf	or W1 = 58905 Kgf
Path 2-2 = 166157 Kgf, must exceed W = 45650 Kgf	or W2 = 21706 Kgf
Path 3-3 = 121460 Kgf, must exceed W = 45650 Kgf	or W3 = 65562 Kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 12 bars

Note: The MAWP of this junction was limited by the parent Shell/Head.

Note : Checking Nozzle in the Latitudinal direction.

Reinforcement CALCULATION, Description: N4-6"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	6.625 in.
Actual Thickness Used in Calculation	0.492 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Elliptical Head, tr [Int. Press]
= $P \cdot D \cdot K1 / (2 \cdot Sv \cdot E - 0.2 \cdot P)$ per Appendix 1-4(c)
= $12.29 \cdot 5703.2 \cdot 0.8995 / (2 \cdot 138.0 \cdot 1.0 - 0.2 \cdot 12.29)$
= 22.8608 mm.

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
= $P \cdot Ro / (Sn \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
= $12.29 \cdot 84.1375 / (118 \cdot 1.0 + 0.4 \cdot 12.29)$
= 0.8725 mm.

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	292.9882 mm.
Parallel to Vessel Wall, opening length	d	146.4941 mm.
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		55.2261 mm.

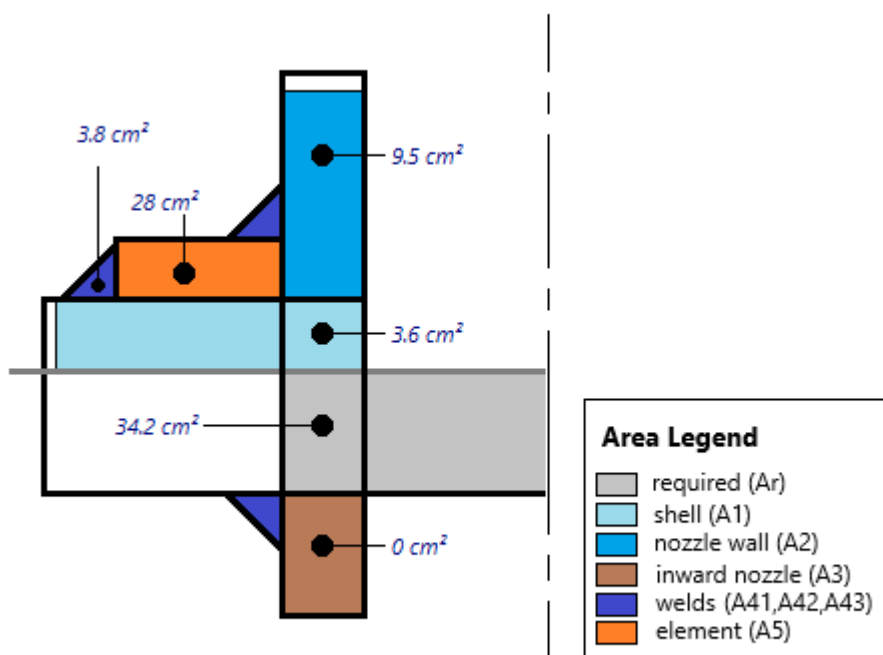


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	34.211	NA	NA
Area in Shell	A1	3.640	NA	NA
Area in Nozzle Wall	A2	9.461	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	3.777	NA	NA
Area in Element	A5	28.000	NA	NA
TOTAL AREA AVAILABLE	Atot	44.878	NA	NA

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Insufficient.
The area available with the given pad is Sufficient.

SELECTION OF POSSIBLE REINFORCING PADS: Diameter Thickness
Based on given Pad Thickness: 230.1812 28.0000 mm.
Based on given Pad Diameter: 268.2750 17.3337 mm.
Based on the Estimated Diameter Limit: 291.4007 14.0781 mm.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$
 $= (146.4941 \cdot 22.8608 \cdot 1.0 + 2 \cdot 10.8904 \cdot 22.8608 \cdot 1.0 \cdot (1 - 0.86))$
 $= 34.211 \text{ cm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d (E1 \cdot t - F \cdot tr) - 2 \cdot tn (E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 146.494 (1.0 \cdot 25.4 - 1.0 \cdot 22.861) - 2 \cdot 10.89$
 $(1.0 \cdot 25.4 - 1.0 \cdot 22.8608) \cdot (1 - 0.855)$
 $= 3.640 \text{ cm}^2$

Area Available in Nozzle Wall Projecting Outward [A2]:
 $= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2$



$$= (2 \cdot 55.23) \cdot (10.89 - 0.87) \cdot 0.8551$$

$$= 9.461 \text{ cm}^2$$

Area Available in Welds [A41 + A42 + A43]:

$$= (W_o^2 - \text{Ar Lost}) \cdot \text{Fr3} + ((W_i - \text{can}/0.707)^2 - \text{Ar Lost}) \cdot \text{fr2} + \text{Trapfr4}$$

$$= (1.) \cdot 0.86 + (0.0) \cdot 0.86 + 292.152^2 \cdot 1.0$$

$$= 3.777 \text{ cm}^2$$

Area Available in Element [A5]:

$$= (\min(D_p, D_L) - (\text{Nozzle OD})) \cdot (\min(t_p, T_{lp}, t_e)) \cdot \text{fr4}$$

$$= (268.275 - 168.275) \cdot 28.0 \cdot 1.0$$

$$= 28.000 \text{ cm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	$t_a = 2.4725 \text{ mm.}$
Wall Thickness per UG16(b),	$t_{r16b} = 4.1000 \text{ mm.}$
Wall Thickness, shell/head, internal pressure	$t_{rb1} = 26.9955 \text{ mm.}$
Wall Thickness	$t_{b1} = \max(t_{rb1}, t_{r16b}) = 26.9955 \text{ mm.}$
Wall Thickness	$t_{b2} = \max(t_{rb2}, t_{r16b}) = 4.1000 \text{ mm.}$
Wall Thickness per table UG-45	$t_{b3} = 7.8200 \text{ mm.}$

Determine Nozzle Thickness candidate [tb]:

$$= \min[t_{b3}, \max(t_{b1}, t_{b2})]$$

$$= \min[7.82, \max(26.9955, 4.1)]$$

$$= 7.8200 \text{ mm.}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(t_a, t_b)$$

$$= \max(2.4725, 7.82)$$

$$= 7.8200 \text{ mm.}$$

Available Nozzle Neck Thickness = 12.4904 mm. -> OK

Weld Size Calculations, Description: N4-6"

Intermediate Calc. for nozzle/shell Welds	$T_{min} = 10.8904 \text{ mm.}$
Intermediate Calc. for pad/shell Welds	$T_{minPad} = 19.0000 \text{ mm.}$

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$6.0000 = \text{Min per Code}$	$7.0700 = 0.7 \cdot W_o \text{ mm.}$
Pad Weld	$9.5000 = 0.5 \cdot T_{minPad}$	$12.7260 = 0.7 \cdot W_p \text{ mm.}$

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$$= \max(0, (A - A1 + 2 \cdot t_n \cdot \text{fr1} \cdot (E1 \cdot t - t_r)) \cdot S_v)$$

$$= \max(0, (34.2113 - 3.6396 + 2 \cdot 10.8904 \cdot 0.8551 \cdot (1.0 \cdot 25.4 - 22.8608)) \cdot 138)$$

$$= 43685.42 \text{ Kg}$$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$= (A2 + A5 + A4 - (W_i - \text{Can}/0.707)^2 \cdot \text{fr2}) \cdot S_v$$

$$= (9.4614 + 28.0 + 3.7766 - 0.0 \cdot 0.86) \cdot 138$$

$$= 58029.35 \text{ Kg}$$

Weld Load [W2]:

$$= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot \text{fr1})) \cdot S_v$$

$$= (9.4614 + 0.0 + 0.8551 + (4.7306)) \cdot 138$$

$$= 21173.89 \text{ Kg}$$

Weld Load [W3]:

$$= (A2 + A3 + A4 + A5 + (2 \cdot t_n \cdot t \cdot \text{fr1})) \cdot S$$

$$= (9.4614 + 0.0 + 3.7766 + 28.0 + (4.7306)) \cdot 138$$

$$= 64686.11 \text{ Kg}$$

Strength of Connection Elements for Failure Path Analysis



Shear, Outward Nozzle Weld [Sonw]:

$$\begin{aligned} &= (\pi/2) \cdot Dlo \cdot Wo \cdot 0.49 \cdot Snw \\ &= (3.1416/2.0) \cdot 168.275 \cdot 10.0 \cdot 0.49 \cdot 118 \\ &= 15584. \text{ Kgf} \end{aligned}$$

Shear, Pad Element Weld [Spew]:

$$\begin{aligned} &= (\pi/2) \cdot DP \cdot WP \cdot 0.49 \cdot SEW \\ &= (3.1416/2.0) \cdot 268.275 \cdot 18.0 \cdot 0.49 \cdot 138 \\ &= 52302. \text{ Kgf} \end{aligned}$$

Shear, Nozzle Wall [Snw]:

$$\begin{aligned} &= (\pi \cdot (Dlr + Dlo) / 4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn \\ &= (3.1416 \cdot 78.6923) \cdot (12.4904 - 1.6) \cdot 0.7 \cdot 118 \\ &= 22677. \text{ Kgf} \end{aligned}$$

Tension, Pad Groove Weld [Tpgw]:

$$\begin{aligned} &= (\pi/2) \cdot Dlo \cdot Wgpn \cdot 0.74 \cdot Seg \\ &= (3.1416/2) \cdot 168.275 \cdot 28.0 \cdot 0.74 \cdot 138 \\ &= 77069. \text{ Kgf} \end{aligned}$$

Tension, Shell Groove Weld [Tngw]:

$$\begin{aligned} &= (\pi/2) \cdot Dlo \cdot Wgnvi \cdot 0.74 \cdot Sng \\ &= (3.1416/2.0) \cdot 168.275 \cdot (24.0 - 0.0) \cdot 0.74 \cdot 138 \\ &= 66059. \text{ Kgf} \end{aligned}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SPEW + SNW) = (52302 + 22677) = 74979 \text{ Kgf}$$

Failure Path [Path 2-2]:

$$\begin{aligned} &= (Sonw + Tpgw + Tngw + Sinw) \\ &= (15584 + 77069 + 66059 + 0) = 158712 \text{ Kgf} \end{aligned}$$

Failure Path [Path 3-3]:

$$\begin{aligned} &= (Spew + Tngw + Sinw) \\ &= (52302 + 66059 + 0) = 118361 \text{ Kgf} \end{aligned}$$

Summary of Failure Path Calculations:

Path 1-1 = 74978	Kgf, must exceed W = 43685 Kg	or W1 = 58029 Kg
Path 2-2 = 158712	Kgf, must exceed W = 43685 Kg	or W2 = 21173 Kg
Path 3-3 = 118361	Kgf, must exceed W = 43685 Kg	or W3 = 64686 Kg

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 12 bars

Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 26.3968 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 465.7126 mm.

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Input, Nozzle Desc: N5-4"

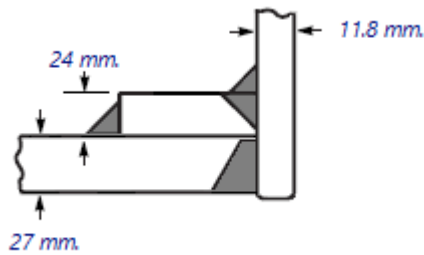
From: 90

Pressure for Reinforcement Calculations	P	12.287	bars
Temperature for Internal Pressure	Temp	82	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Elliptical Head	D	5700.00	mm.
Aspect Ratio of Elliptical Head	Ar	2.00	
Head Finished (Minimum) Thickness	t	27.0000	mm.
Head Internal Corrosion Allowance	c	1.6000	mm.
Head External Corrosion Allowance	co	0.0000	mm.
Distance from Head Centerline	L1	1600.0000	mm.
User Entered Minimum Design Metal Temperature		-1.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-106 B	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	118.00	N./mm ²
Allowable Stress At Ambient	Sna	118.00	N./mm ²
Diameter Basis (for tr calc only)		Outside	
Layout Angle		270.00	deg
Diameter		4.0000	in.
Size and Thickness Basis		Minimum	
Nominal Thickness		160	
Flange Material		SA-105	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	1.6000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	423.8000	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	24.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Pad Material		SA-516 70	
Pad Allowable Stress at Temperature	Sp	138.00	N./mm ²
Pad Allowable Stress At Ambient	Spa	138.00	N./mm ²
Diameter of Pad along vessel surface	Dp	216.0000	mm.
Thickness of Pad	te	24.0000	mm.
Weld leg size between Pad and Shell	Wp	18.0000	mm.
Groove weld depth between Pad and Nozzle	Wgpn	24.0000	mm.
Reinforcing Pad Width		50.8500	mm.
Flange Class		150	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).



Note : Checking Nozzle in the Meridional direction.

Reinforcement CALCULATION, Description: N5-4"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	4.500 in.
Actual Thickness Used in Calculation	0.465 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Elliptical Head, tr [Int. Press]
 $= P \cdot D \cdot K1 / (2 \cdot Sv \cdot E - 0.2 \cdot P)$ per Appendix 1-4(c)
 $= 12.29 \cdot 5703.2 \cdot 0.8995 / (2 \cdot 138.0 \cdot 1.0 - 0.2 \cdot 12.29)$
 $= 22.8608 \text{ mm.}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot Ro / (Sn \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 12.29 \cdot 57.15 / (118 \cdot 1.0 + 0.4 \cdot 12.29)$
 $= 0.5927 \text{ mm.}$

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	198.3050 mm.
Parallel to Vessel Wall, opening length	d	99.1525 mm.
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		49.5037 mm.

The Pad diameter is greater than the Diameter Limit. The excess will not be considered.

Weld Strength Reduction Factor [fr1]:
 $= \min(1, Sn/Sv)$
 $= \min(1, 118.0/138.0)$
 $= 0.855$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, Sn/Sv)$
 $= \min(1, 118.0/138.0)$
 $= 0.855$

Weld Strength Reduction Factor [fr4]:
 $= \min(1, Sp/Sv)$
 $= \min(1, 138.0/138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(0.855, 1.0)$
 $= 0.855$

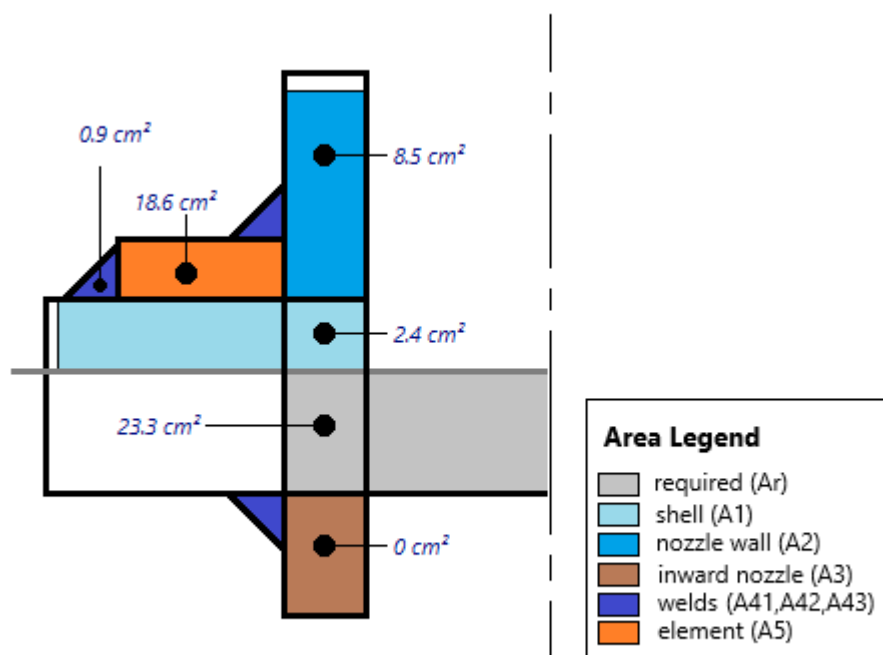


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	23.343	NA	NA
Area in Shell	A1	2.443	NA	NA
Area in Nozzle Wall	A2	8.543	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	0.855	NA	NA
Area in Element	A5	18.626	NA	NA
TOTAL AREA AVAILABLE	Atot	30.466	NA	NA

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 71.26 degs.

The area available without a pad is Insufficient.
The area available with the given pad is Sufficient.

SELECTION OF POSSIBLE REINFORCING PADS: Diameter Thickness
Based on given Pad Thickness: 162.2281 24.0000 mm.
Based on given Pad Diameter: 216.0000 14.8217 mm.
Based on the Estimated Diameter Limit: 196.7175 15.1312 mm.

Area Required [A]:
= (d • tr•F + 2 • tn • tr•F • (1-fr1)) UG-37(c)
= (99.1525•22.8608•1.0+2•10.2015•22.8608•1.0•(1-0.86))
= 23.343 cm²

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
= d(E1•t - F•tr) - 2 • tn(E1•t - F•tr) • (1 - fr1)
= 99.152(1.0 • 25.4 - 1.0 • 22.861) - 2 • 10.201
(1.0 • 25.4 - 1.0 • 22.8608) • (1 - 0.855)
= 2.443 cm²

Area Available in Nozzle Wall Projecting Outward [A2]:
= (2 • Tlwp) • (tn - trn) • fr2/sin(α3)



$$= (2 \cdot 49.5) \cdot (10.2 - 0.59) \cdot 0.8551 / \sin(72.2)$$

$$= 8.543 \text{ cm}^2$$

See ASME VIII-1 2011(a) Appendix L, L-7.7.7(b) for more information.

Area Available in Welds [A41 + A42 + A43]:

$$= W_o^2 \cdot fr_3 + (W_i - can / 0.707)^2 \cdot fr_2 + W_p^2 \cdot fr_4$$

$$= 10.0^2 \cdot 0.86 + (0.0)^2 \cdot 0.86 + 0.0^2 \cdot 1.0$$

$$= 0.855 \text{ cm}^2$$

Area Available in Element [A5]:

$$= (\min(D_p, DL) - (\text{Nozzle OD})) \cdot (\min(t_p, Tlwp, te)) \cdot fr_4$$

$$= (198.305 - 120.6974) \cdot 24.0 \cdot 1.0$$

$$= 18.626 \text{ cm}^2$$

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

Govrn. thk, tg = 11.801, tr = 0.593, c = 1.6 mm., E* = 1.0
 Thickness Ratio = tr * E* / (tg - c) = 0.058, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	-3 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle Neck to Pad Weld for the Nozzle, Curve: B

Govrn. thk, tg = 11.801, tr = 0.593, c = 1.6 mm., E* = 1.0
 Thickness Ratio = tr * E* / (tg - c) = 0.058, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-24 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: B

Govrn. thk, tg = 11.801, tr = 0.593, c = 1.6 mm., E* = 1.0
 Thickness Ratio = tr * E* / (tg - c) = 0.058, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-24 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Shell to Pad Weld Junction at Pad OD, min(Curve:B, Curve:D)

Govrn. thk, tg = 24.0, c = 1.6 mm., E* = 1.0
 Thickness Ratio = tr * E* / (tg - c) = 0.9, Temp. Reduction = 6 °C
 Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve B	-2 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-8 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), min(Curve:B, Curve:D)

Govrn. thk, tg = 11.801, tr = 0.593, c = 1.6 mm., E* = 1.0
 Thickness Ratio = tr * E* / (tg - c) = 0.058, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-24 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Gov. MDMT of the Nozzle	: -104 °C
Gov. MDMT of the Reinforcement Pad	: -29 °C
Gov. MDMT of the nozzle to shell joint welded assembly	: -29 °C

Flange MDMT including Temperature reduction per UCS-66.1:



Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
 Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -39 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = 12.29/19.60 = 0.627

Weld Size Calculations, Description: N5-4"

Intermediate Calc. for nozzle/shell Welds Tmin 10.2015 mm.
 Intermediate Calc. for pad/shell Welds TminPad 19.0000 mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm.
Pad Weld	9.5000 = 0.5*TminPad	12.7260 = 0.7 * Wp mm.

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A-A1+2 \cdot t_n \cdot f_{r1} \cdot (E1 \cdot t - t_r)) \cdot S_v)$
 $= \max(0, (23.343 - 2.4426 + 2 \cdot 10.2015 \cdot 0.8551 \cdot (1.0 \cdot 25.4 - 22.8608)) \cdot 138)$
 $= 30034.08 \text{ Kgf}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2+A5+A4-(W_i-\text{Can}/.707)^2 \cdot f_{r2}) \cdot S_v$
 $= (8.5426 + 18.6258 + 0.8551 - 0.0 \cdot 0.86) \cdot 138$
 $= 39434.16 \text{ Kgf}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S_v$
 $= (8.5426 + 0.0 + 0.8551 + (4.4313)) \cdot 138$
 $= 19459.86 \text{ Kgf}$

Weld Load [W3]:
 $= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S$
 $= (8.5426 + 0.0 + 0.8551 + 18.6258 + (4.4313)) \cdot 138$
 $= 45669.78 \text{ Kgf}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$
 $= (3.1416/2.0) \cdot 120.6974 \cdot 10.0 \cdot 0.49 \cdot 118$
 $= 11178. \text{ Kgf}$

Shear, Pad Element Weld [Spew]:
 $= (\pi/2) \cdot D_P \cdot W_P \cdot 0.49 \cdot S_{EW}$
 $= (3.1416/2.0) \cdot 216.0 \cdot 18.0 \cdot 0.49 \cdot 138$
 $= 42111. \text{ Kgf}$

Shear, Nozzle Wall [Snw]:
 $= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - \text{Can}) \cdot 0.7 \cdot S_n$
 $= (3.1416 \cdot 54.9625) \cdot (11.8015 - 1.6) \cdot 0.7 \cdot 118$
 $= 14836. \text{ Kgf}$

Tension, Pad Groove Weld [Tpgw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_{gpn} \cdot 0.74 \cdot S_{eg}$
 $= (3.1416/2) \cdot 120.6974 \cdot 24.0 \cdot 0.74 \cdot 138$
 $= 47382. \text{ Kgf}$

Tension, Shell Groove Weld [Tngw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_{gnvi} \cdot 0.74 \cdot S_{ng}$
 $= (3.1416/2.0) \cdot 120.6974 \cdot (24.0 - 0.0) \cdot 0.74 \cdot 138$
 $= 47382. \text{ Kgf}$

See UG-41 and UW-15 for clarification of allowable stresses.



Strength of Failure Paths:

Failure Path [Path 1-1]:
 $= (SPEW + SNW) = (42111 + 14836) = 56947 \text{ Kg}$

Failure Path [Path 2-2]:
 $= (Sonw + Tpgw + Tngw + Sinw)$
 $= (11178 + 47382 + 47382 + 0) = 105942 \text{ Kg}$

Failure Path [Path 3-3]:
 $= (Spew + Tngw + Sinw)$
 $= (42111 + 47382 + 0) = 89492 \text{ Kg}$

Summary of Failure Path Calculations:

Path 1-1 = 56947 Kg, must exceed W = 30034 Kg	or W1 = 39434 Kg
Path 2-2 = 105941 Kg, must exceed W = 30034 Kg	or W2 = 19459 Kg
Path 3-3 = 89492 Kg, must exceed W = 30034 Kg	or W3 = 45669 Kg

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 12 bars

Note: The MAWP of this junction was limited by the parent Shell/Head.

Note : Checking Nozzle in the Latitudinal direction.

Reinforcement CALCULATION, Description: N5-4"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	4.500 in.
Actual Thickness Used in Calculation	0.465 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Elliptical Head, tr [Int. Press]
 $= P \cdot D \cdot K1 / (2 \cdot Sv \cdot E - 0.2 \cdot P)$ per Appendix 1-4(c)
 $= 12.29 \cdot 5703.2 \cdot 0.8995 / (2 \cdot 138.0 \cdot 1.0 - 0.2 \cdot 12.29)$
 $= 22.8608 \text{ mm.}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot Ro / (Sn \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 12.29 \cdot 57.15 / (118 \cdot 1.0 + 0.4 \cdot 12.29)$
 $= 0.5927 \text{ mm.}$

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	187.7941 mm.
Parallel to Vessel Wall, opening length	d	93.8970 mm.
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		49.5037 mm.

The Pad diameter is greater than the Diameter Limit. The excess will not be considered.

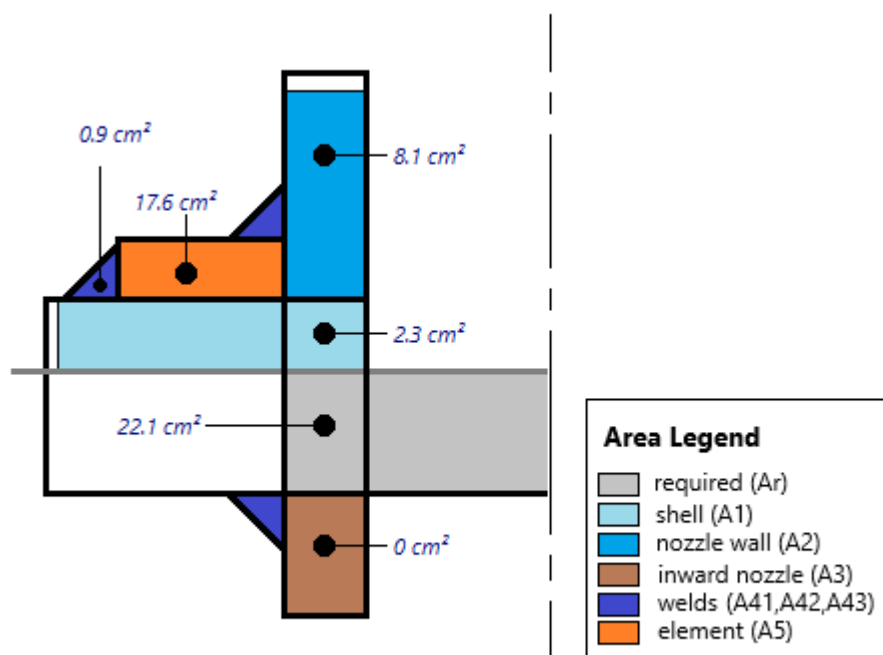


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	22.142	NA	NA
Area in Shell	A1	2.309	NA	NA
Area in Nozzle Wall	A2	8.135	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	0.855	NA	NA
Area in Element	A5	17.639	NA	NA
TOTAL AREA AVAILABLE	Atot	28.937	NA	NA

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Insufficient.
The area available with the given pad is Sufficient.

SELECTION OF POSSIBLE REINFORCING PADS: Diameter Thickness
Based on given Pad Thickness: 159.4779 24.0000 mm.
Based on given Pad Diameter: 216.0000 14.7532 mm.
Based on the Estimated Diameter Limit: 186.2066 15.0789 mm.

Area Required [A]:

$$= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$$

$$= (93.897 \cdot 1.0 \cdot 25.4 - 1.0 \cdot 2 \cdot 10.2015 \cdot 22.8608 \cdot 1.0 \cdot (1 - 0.86))$$

$$= 22.142 \text{ cm}^2$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d (E1 \cdot t - F \cdot tr) - 2 \cdot tn (E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 93.897 (1.0 \cdot 25.4 - 1.0 \cdot 22.861) - 2 \cdot 10.201$$

$$(1.0 \cdot 25.4 - 1.0 \cdot 22.8608) \cdot (1 - 0.855)$$

$$= 2.309 \text{ cm}^2$$

Area Available in Nozzle Wall Projecting Outward [A2]:

$$= (2 \cdot tlwp) \cdot (tn - trn) \cdot fr2$$



$$= (2 \cdot 49.5) \cdot (10.2 - 0.59) \cdot 0.8551$$

$$= 8.135 \text{ cm}^2$$

Area Available in Welds [A41 + A42 + A43]:

$$= W_o^2 \cdot fr_3 + (W_i\text{-can}/0.707)^2 \cdot fr_2 + W_p^2 \cdot fr_4$$

$$= 10.0^2 \cdot 0.86 + (0.0)^2 \cdot 0.86 + 0.0^2 \cdot 1.0$$

$$= 0.855 \text{ cm}^2$$

Area Available in Element [A5]:

$$= (\min(D_p, D_L) - (\text{Nozzle OD})) \cdot (\min(t_p, T_{lp}, t_e)) \cdot fr_4$$

$$= (187.7941 - 114.3) \cdot 24.0 \cdot 1.0$$

$$= 17.639 \text{ cm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	$t_a = 2.1927 \text{ mm.}$
Wall Thickness per UG16(b),	$tr_{16b} = 4.1000 \text{ mm.}$
Wall Thickness, shell/head, internal pressure	$trb_1 = 26.9955 \text{ mm.}$
Wall Thickness	$tb_1 = \max(trb_1, tr_{16b}) = 26.9955 \text{ mm.}$
Wall Thickness	$tb_2 = \max(trb_2, tr_{16b}) = 4.1000 \text{ mm.}$
Wall Thickness per table UG-45	$tb_3 = 6.8578 \text{ mm.}$

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb_3, \max(tb_1, tb_2)]$$

$$= \min[6.858, \max(26.9955, 4.1)]$$

$$= 6.8578 \text{ mm.}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(t_a, t_b)$$

$$= \max(2.1927, 6.8578)$$

$$= 6.8578 \text{ mm.}$$

Available Nozzle Neck Thickness = 11.8015 mm. -> OK

Weld Size Calculations, Description: N5-4"

Intermediate Calc. for nozzle/shell Welds	T_{min}	10.2015 mm.
Intermediate Calc. for pad/shell Welds	T_{minPad}	19.0000 mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$6.0000 = \text{Min per Code}$	$7.0700 = 0.7 \cdot W_o \text{ mm.}$
Pad Weld	$9.5000 = 0.5 \cdot T_{minPad}$	$12.7260 = 0.7 \cdot W_p \text{ mm.}$

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$$= \max(0, (A - A_1 + 2 \cdot t_n \cdot fr_1 \cdot (E_1 \cdot t - tr)) \cdot S_v)$$

$$= \max(0, (22.1416 - 2.3092 + 2 \cdot 10.2015 \cdot 0.8551 \cdot (1.0 \cdot 25.4 - 22.8608)) \cdot 138)$$

$$= 28531.22 \text{ Kg}$$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$= (A_2 + A_5 + A_4 - (W_i\text{-Can}/0.707)^2 \cdot fr_2) \cdot S_v$$

$$= (8.1347 + 17.6386 + 0.8551 - 0.0 \cdot 0.86) \cdot 138$$

$$= 37470.91 \text{ Kg}$$

Weld Load [W2]:

$$= (A_2 + A_3 + A_4 + (2 \cdot t_n \cdot t \cdot fr_1)) \cdot S_v$$

$$= (8.1347 + 0.0 + 0.8551 + (4.4313)) \cdot 138$$

$$= 18885.83 \text{ Kg}$$

Weld Load [W3]:

$$= (A_2 + A_3 + A_4 + A_5 + (2 \cdot t_n \cdot t \cdot fr_1)) \cdot S$$

$$= (8.1347 + 0.0 + 0.8551 + 17.6386 + (4.4313)) \cdot 138$$

$$= 43706.54 \text{ Kg}$$

Strength of Connection Elements for Failure Path Analysis



Shear, Outward Nozzle Weld [Sonw]:

$$\begin{aligned} &= (\pi/2) \cdot Dlo \cdot Wo \cdot 0.49 \cdot Snw \\ &= (3.1416/2.0) \cdot 114.3 \cdot 10.0 \cdot 0.49 \cdot 118 \\ &= 10586. \text{ Kg} \end{aligned}$$

Shear, Pad Element Weld [Spew]:

$$\begin{aligned} &= (\pi/2) \cdot DP \cdot WP \cdot 0.49 \cdot SEW \\ &= (3.1416/2.0) \cdot 216.0 \cdot 18.0 \cdot 0.49 \cdot 138 \\ &= 42111. \text{ Kg} \end{aligned}$$

Shear, Nozzle Wall [Snw]:

$$\begin{aligned} &= (\pi \cdot (Dlr + Dlo) / 4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn \\ &= (3.1416 \cdot 52.0493) \cdot (11.8015 - 1.6) \cdot 0.7 \cdot 118 \\ &= 14050. \text{ Kg} \end{aligned}$$

Tension, Pad Groove Weld [Tpgw]:

$$\begin{aligned} &= (\pi/2) \cdot Dlo \cdot Wgpn \cdot 0.74 \cdot Seg \\ &= (3.1416/2) \cdot 114.3 \cdot 24.0 \cdot 0.74 \cdot 138 \\ &= 44870. \text{ Kg} \end{aligned}$$

Tension, Shell Groove Weld [Tngw]:

$$\begin{aligned} &= (\pi/2) \cdot Dlo \cdot Wgnvi \cdot 0.74 \cdot Sng \\ &= (3.1416/2.0) \cdot 114.3 \cdot (24.0 - 0.0) \cdot 0.74 \cdot 138 \\ &= 44870. \text{ Kg} \end{aligned}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SPEW + SNW) = (42111 + 14050) = 56161 \text{ Kg}$$

Failure Path [Path 2-2]:

$$\begin{aligned} &= (Sonw + Tpgw + Tngw + Sinw) \\ &= (10586 + 44870 + 44870 + 0) = 100326 \text{ Kg} \end{aligned}$$

Failure Path [Path 3-3]:

$$\begin{aligned} &= (Spew + Tngw + Sinw) \\ &= (42111 + 44870 + 0) = 86981 \text{ Kg} \end{aligned}$$

Summary of Failure Path Calculations:

Path 1-1 = 56160	Kgf, must exceed W = 28531 Kg	or W1 = 37470 Kg
Path 2-2 = 100326	Kgf, must exceed W = 28531 Kg	or W2 = 18885 Kg
Path 3-3 = 86981	Kgf, must exceed W = 28531 Kg	or W3 = 43706 Kg

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 12 bars

Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 19.0438 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 471.2405 mm.

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Bill of Materials:

QTY	DESCRIPTION	MATERIAL
2	ELLIPTICAL HEAD: 2.0 X 1, 32.0mm. THK X 5700.0mm. ID X 50.0mm.	SA-516 70
4	PLATFORM: 1000mm X 1500mm WIDE	
1	LADDER: 3000.0mm	...
2	LADDER: 4000.0mm	...
1	LADDER: 5000.0mm	...
2	TRUNNIONS:	SA-516 70
2	TRUNNION PADS: 900.0mm OD X 550.0mm ID X 28.0mm THK	
1	PLATFORM: 1000mm X 6000mm WIDE	
1	LADDER: 2820.0mm	...
2	CLASS 150 GR 1.1, 24.0" BLIND FLANGE(S)	SA-105
1	BASERING: 6188.0mm. OD X 5588.0 ID X 38.0mm.	SA-36
24	BASERING BOLTS: 64.0mm. DIA	SA-307 A
24	BASERING BOLT NUTS: 64.0mm. DIA	...
1	BASERING TOP RING: 6188.0mm. OD X 5756.0 ID X 50.0mm.	...
48	BASERING GUSSET PLATES: 206.0mm. X 206.0mm. X 312.0mm. X 22.0mm.	...
5	CYLINDRICAL SKIRT SEGMENT 22.0mm. THK X 3054.4mm. X 1700.0mm.	SA-36
1	CYLINDRICAL SKIRT SEGMENT 22.0mm. THK X 2672.6mm. X 1700.0mm.	SA-36
5	CYLINDRICAL SKIRT SEGMENT 18.0mm. THK X 3056.5mm. X 1000.0mm.	SA-516 70
1	CYLINDRICAL SKIRT SEGMENT 18.0mm. THK X 2674.5mm. X 1000.0mm.	SA-516 70
17	CYLINDRICAL SEGMENT 32.0mm. THK X 3155.3mm. X 3000.0mm.	SA-516 70
1	CYLINDRICAL SEGMENT 32.0mm. THK X 2130.1mm. X 3000.0mm.	SA-516 70
3	CYLINDRICAL SEGMENT 32.0mm. THK X 14751.4mm. X 3000.0mm.	SA-516 70
3	CYLINDRICAL SEGMENT 32.0mm. THK X 9466.0mm. X 3000.0mm.	SA-516 70
4	CYLINDRICAL SEGMENT 32.0mm. THK X 3155.3mm. X 2400.0mm.	SA-516 70
1	CYLINDRICAL SEGMENT 32.0mm. THK X 14751.4mm. X 2400.0mm.	SA-516 70
1	CYLINDRICAL SEGMENT 32.0mm. THK X 9466.0mm. X 2400.0mm.	SA-516 70
1	NAMEPLATE	...



Nozzle Schedule:

Description	Nominal or Actual Size	Schd or FVC Type	Flg Type	Nozzle O/Dia in	Wall Thk mm	Reinforcing Diameter	Pad Thk mm	Cut Length mm	Flg Class
K1B-2"	2.000 in	Actual	LWN	3.323	16.800	...	...	238.69	300
N6-2"	2.000 in	Actual	LWN	3.323	16.800	...	...	238.69	300
K2B-2"	2.000 in	Actual	LWN	3.323	16.800	...	...	238.69	300
K1A-2"	2.000 in	Actual	LWN	3.323	16.800	...	...	238.69	300
K2A-2"	2.000 in	Actual	LWN	3.323	16.800	...	...	238.69	300
K5A-2"	2.000 in	Actual	LWN	3.323	16.800	...	...	238.69	300
K5B-2"	2.000 in	Actual	LWN	3.323	16.800	...	...	238.69	300
K3-2"	2.000 in	Actual	LWN	3.323	16.800	...	...	238.69	300
K4-2"	2.000 in	Actual	LWN	3.323	16.800	...	...	238.69	300
N2-3"	3.000 in	160	WNF	3.500	11.125	190.00	28.00	227.19	300
N5-4"	4.000 in	160	WNF	4.500	13.487	216.00	24.00	471.24	150
N4-6"	6.000 in	120	WNF	6.625	14.275	268.27	28.00	465.71	150
N1-14"	14.000 in	Actual	WNF	14.000	20.000	555.60	28.00	237.55	150
N3-14"	14.000 in	Actual	WNF	14.000	20.000	555.60	28.00	403.07	150
M1-24"	24.000 in	Actual	WNF	24.000	28.000	1009.60	28.00	298.35	150
M2-24"	24.000 in	Actual	WNF	24.000	28.000	1009.60	32.00	298.35	150

General Notes for the above table:

The cut length is the outside projection + inside projection + drop + in-plane shell thickness. This value does not include weld gaps, nor does it account for shrinkage.

In the case of oblique nozzles, the outside diameter must be increased. The re-pad width around the nozzle is calculated as follows:

Width of Pad = (Pad Outside Dia. (per above) - Nozzle Outside Dia.)/2

For hub nozzles, the thickness and diameter shown are those of the smaller and thinner section.

Nozzle Material and Weld Fillet Leg Size Details (mm.):

Description	Material	Shl Grve Weld	Noz Shl/Pad Weld	Pad OD Weld	Pad Grve Weld	Inside Weld
K1B-2"	SA-105	24.000	10.000	...	...	...
N6-2"	SA-105	24.000	10.000	...	...	...
K2B-2"	SA-105	24.000	10.000	...	...	...
K1A-2"	SA-105	24.000	10.000	...	...	...
K2A-2"	SA-105	24.000	10.000	...	...	...
K5A-2"	SA-105	24.000	10.000	...	...	...
K5B-2"	SA-105	24.000	10.000	...	...	...
K3-2"	SA-105	24.000	10.000	...	...	...
K4-2"	SA-105	24.000	10.000	...	...	...
N2-3"	SA-106 B	26.500	10.000	20.000	28.000	...
N5-4"	SA-106 B	24.000	10.000	18.000	24.000	...
N4-6"	SA-106 B	24.000	10.000	18.000	28.000	...
N1-14"	SA-516 70	26.000	10.000	20.000	28.000	...
N3-14"	SA-516 70	24.000	10.000	20.000	28.000	...
M1-24"	SA-516 70	26.000	10.000	20.000	28.000	...
M2-24"	SA-516 70	26.000	10.000	20.000	32.000	...

Note: The Outside projections below do not include the flange thickness.

Nozzle Miscellaneous Data:

Description	Elev/Distance From Datum mm.	Layout Angle deg	Proj Outside mm.	Proj Inside mm.	Installed in Component
K1B-2"	650.000	120.0	206.38	0.00	Shell-1
N6-2"	1500.000	270.0	206.38	0.00	Shell-1
K2B-2"	650.000	90.0	206.38	0.00	Shell-1
K1A-2"	1500.000	90.0	206.38	0.00	Shell-1



K2A-2"	1500.000	120.0	206.38	0.00	Shell-1
K5A-2"	10999.998	120.0	206.38	0.00	shell14
K5B-2"	10999.998	90.0	206.38	0.00	shell14
K3-2"	10699.999	90.0	206.38	0.00	shell14
K4-2"	13899.999	270.0	206.38	0.00	shell15
N2-3"	...	0.0	200.00	0.00	Bottom Dish
N5-4"	...	270.0	423.80	0.00	Top dish
N4-6"	...	90.0	411.10	0.00	Top dish
N1-14"	12549.999	270.0	200.00	0.00	shell15
N3-14"	...	0.0	373.00	0.00	Top dish
M1-24"	8549.999	90.0	250.00	0.00	Shell13
M2-24"	7550.000	220.0	250.00	0.00	Shell13

Nozzle Calculation Summary (bars & mm.):

Description	MAWP	Ext	MAPNC	UG-45	[tr]	Weld Path	Areas or Stresses
N2-3"	12	...	...	OK	6.40	OK	Passed
K1B-2"	12	...	...	OK	6.40	OK	No Calc[*]
N6-2"	12	...	...	OK	6.40	OK	No Calc[*]
K2B-2"	12	...	...	OK	6.40	OK	No Calc[*]
K1A-2"	12	...	...	OK	6.40	OK	No Calc[*]
K2A-2"	12	...	...	OK	6.40	OK	No Calc[*]
M1-24"	12	...	...	...		OK	Passed
M2-24"	12	...	...	...		OK	Passed
K5A-2"	12	...	...	OK	6.40	OK	No Calc[*]
K5B-2"	12	...	...	OK	6.40	OK	No Calc[*]
K3-2"	12	...	...	OK	6.40	OK	No Calc[*]
N1-14"	12	...	...	OK	9.93	OK	Passed
K4-2"	12	...	...	OK	6.40	OK	No Calc[*]
N3-14"	12	...	...	OK	9.93	OK	Passed
N4-6"	12	...	...	OK	7.82	OK	Passed
N5-4"	12	...	...	OK	6.86	OK	Passed

Nozzle MAWP Summary:

Minimum MAWP Nozzles : 12 Nozzle : N2-3"
Minimum MAWP Shells/Flanges : 12 Element : Bottom Dish
Minimum MAPnc Shells/Flanges : 13 Element : Top dish

Computed Vessel M.A.W.P. : 12 bars

[*] - This was a small opening and the areas were not computed or the MAWP of this connection could not be computed because the longitudinal bending stress was greater than the hoop stress.

Multiple output lines for the same nozzle indicates required Code calculations in both the longitudinal and circumferential planes of reinforcement where applicable.

Check the spatial relationship between the nozzles:

From Node	Nozzle Description	Y Coordinate mm.	Layout Angle deg	Dia. Limit mm.
30	N2-3"	0.000	0.000	145.262
40	K1B-2"	650.000	120.000	145.200
40	N6-2"	1500.000	270.000	145.200
40	K2B-2"	650.000	90.000	145.200
40	K1A-2"	1500.000	90.000	145.200
40	K2A-2"	1500.000	120.000	145.200
60	M1-24"	8549.999	90.000	1113.600
60	M2-24"	7550.000	220.000	1113.600
70	K5A-2"	10999.998	120.000	145.200
70	K5B-2"	10999.998	90.000	145.200
70	K3-2"	10699.999	90.000	145.200
80	N1-14"	12549.999	270.000	637.600
80	K4-2"	13899.999	270.000	145.200
90	N3-14"	0.000	0.000	637.600
90	N4-6"	0.000	90.000	292.988
90	N5-4"	0.000	270.000	187.794

The nozzle spacing is computed by the following:

$$= \sqrt{I_l^2 + I_c^2} \text{ where}$$

I_l - Arc length along the inside vessel surface in the long. direction.
 I_c - Arc length along the inside vessel surface in the circ. direction

If any interferences/violations are found, they will be noted below.



No interference violations have been detected!

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Minimum Design Metal Temperature Results Summary (°C):

Description	Notes	Curve	Basic MDMT	Reduced MDMT	UG-20(f) MDMT	Thickness ratio	Gov Thk mm.	E*	PWHT reqd
Bottom Dish	[10]	D	-33	-33		1.000	27.000	1.00	No
Bottom Dish	[7]	D	-29	-38		0.840	32.000	1.00	No
Shell1	[8]	D	-29	-38		0.841	32.000	0.85	No
Shell2	[8]	D	-29	-38		0.841	32.000	0.85	No
Shell3	[8]	D	-29	-38		0.841	32.000	0.85	No
shell4	[8]	D	-29	-38		0.841	32.000	0.85	No
shell5	[8]	D	-29	-38		0.841	32.000	0.85	No
Top dish	[10]	D	-33	-33		1.000	27.000	1.00	No
Top dish	[7]	D	-29	-38		0.840	32.000	1.00	No
N2-3"	[1]	B	1	-4		0.900	27.000	1.00	No
Nozzle Flg	[5]	A	-18	-104					
K1B-2"	[1]	A	7	-104		0.016	16.800	1.00	No
Nozzle Flg	[5]	A	-18	-104					
N6-2"	[1]	A	7	-104		0.016	16.800	1.00	No
Nozzle Flg	[5]	A	-18	-104					
K2B-2"	[1]	A	7	-104		0.016	16.800	1.00	No
Nozzle Flg	[5]	A	-18	-104					
K1A-2"	[1]	A	7	-104		0.016	16.800	1.00	No
Nozzle Flg	[5]	A	-18	-104					
K2A-2"	[1]	A	7	-104		0.016	16.800	1.00	No
Nozzle Flg	[5]	A	-18	-104					
M1-24"	[1]	D	-32	-41		0.840	28.000	1.00	No
Nozzle Flg	[5]	A	-18	-39					
M2-24"	[1]	D	-29	-38		0.840	32.000	1.00	No
Nozzle Flg	[5]	A	-18	-39					
K5A-2"	[1]	A	7	-104		0.016	16.800	1.00	No
Nozzle Flg	[5]	A	-18	-104					
K5B-2"	[1]	A	7	-104		0.016	16.800	1.00	No
Nozzle Flg	[5]	A	-18	-104					
K3-2"	[1]	A	7	-104		0.016	16.800	1.00	No
Nozzle Flg	[5]	A	-18	-104					
N1-14"	[1]	D	-32	-41		0.840	28.000	1.00	No
Nozzle Flg	[5]	A	-18	-39					
K4-2"	[1]	A	7	-104		0.016	16.800	1.00	No
Nozzle Flg	[5]	A	-18	-104					
N3-14"	[1]	B	1	-4		0.900	27.000	1.00	No
Nozzle Flg	[5]	A	-18	-39					
N4-6"	[1]	B	1	-4		0.900	27.000	1.00	No
Nozzle Flg	[5]	A	-18	-39					
N5-4"	[1]	B	-2	-29	-29	0.900	24.000	1.00	No
Nozzle Flg	[5]	A	-18	-39					
Nozl. Bolting[22]			-48			1.000	64.000	1.00	No

Warmest MDMT: 7 -4

Required Minimum Design Metal Temperature -1.0 °C
Warmest Computed Minimum Design Metal Temperature -4.0 °C

Notes:

- [!] - This was an impact tested material.
- [1] - Governing Nozzle Weld.
- [4] - ASME Flange MDMT Calcs; Thickness ratio per UCS-66(b)(1)(-c).
- [5] - ASME Flange MDMT Calcs; Thickness ratio per UCS-66(b)(1)(-b).
- [6] - MDMT Calculations at the Shell/Head Joint.
- [7] - MDMT Calculations for the Straight Flange.
- [8] - Cylinder/Cone/Flange Junction MDMT.
- [9] - Calculations in the Spherical Portion of the Head.
- [10] - Calculations in the Knuckle Portion of the Head.
- [11] - Calculated (Body Flange) Flange MDMT.
- [12] - Calculated Flat Head MDMT per UCS-66.3
- [13] - Tubesheet MDMT, shell side, if applicable
- [14] - Tubesheet MDMT, tube side, if applicable



- [15] - Nozzle Material
- [16] - Shell or Head Material
- [17] - Impact Testing required
- [18] - Impact Testing not required, see UCS-66(b)(3)
- [20] - Cylinder/Cone Junction MDMT based on Longitudinal Stress considerations
- [21] - Body Flange Bolting Material
- [22] - Nozzle Flange Bolting Material
- [23] - Stiffening Ring to Shell Weld
- [24] - Saddle to Shell Weld

UG-84(b)(2) was not considered.

UCS-66(g) was not considered.

UCS-66(i) was not considered.

Notes:

Impact test temps were not entered in and not considered in the analysis.

UCS-66(i) applies to impact tested materials not by specification and

UCS-66(g) applies to materials impact tested per UG-84.1 General Note (c).

The Basic MDMT includes the (30F) PWHT credit if applicable.

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Vessel Design Summary:

ASME Code, Section VIII Division 1, 2025

Diameter Spec : 5700.000 mm. ID
Vessel Design Length, Tangent to Tangent 14500.00 mm.
Distance of Bottom Tangent above Grade 3000.00 mm.
Distance of Base above Grade 300.00 mm.
Specified Datum Line Distance 2700.00 mm.
Internal Design Temperature 82 °C
Internal Design Pressure 12.000 bars
External Design Temperature 0 °C
Maximum Allowable Working Pressure 12 bars
Shop Test Pressure 12.750 bars
Field Test Pressure 15.600 bars
Required Minimum Design Metal Temperature -1.0 °C
Warmest Computed Minimum Design Metal Temperature -4.0 °C
Wind Design Code ASCE/SEI 7-16
Earthquake Design Code ASCE/SEI 7-16

Materials of Construction:

Component Type	Material	Class	Thickness	UNS #	Material Treatment Options
Shell	SA-516 70	...	...	K02700	
Head	SA-516 70	...	...	K02700	
Skirt	SA-36	...	...	K02600	
Skirt	SA-516 70	...	...	K02700	
Nozzle	SA-106 B	...	...	K03006	
Nozzle	SA-105	...	...	K03504	
Nozzle	SA-516 70	...	...	K02700	
Re-Pad	SA-516 70	...	...	K02700	
Nozzle Flg	SA-105	...	...	K03504	
Basering	SA-36	...	...	K02600	
Base Bolting	SA-307 A	...	6 <= t <= 100		

+FG - Fine Grained, +R - rolled, +T - Tempered, +N - Normalized
+Q - quenched, +CBV - Cb & V added, +IT - Impact tested

Element Pressures and MAWP (bars & mm.):

Element Description or Type	Design Pressure + Stat. head	Ext. Press.	Element M.A.W.P	Total Corrosion Allowance	Str. Flg. Gov.	In Creep Range
Skirt2	...	...	...	1.0000	N/A	No
Skirt-1	...	...	...	1.0000	N/A	No
Bottom Dish	12.002	0.00	12.000	1.6000	No	No
Shell-1	12.002	0.00	12.000	1.6000	N/A	No
Shell2	12.002	0.00	12.000	1.6000	N/A	No
Shell3	12.001	0.00	12.000	1.6000	N/A	No
shell4	12.001	0.00	12.000	1.6000	N/A	No
shell5	12.000	0.00	12.000	1.6000	N/A	No
Top dish	12.000	0.00	12.000	1.6000	No	No

Liquid Level: 17301.50 mm. Dens.: 0.000 kg./cm³ Sp. Gr.: 0.001

Element Types and Properties:



Element Type	"To" Elev mm.	Element Length mm.	Nominal Thickness mm.	Finished Thickness mm.	Reqd Thk Internal mm.	Reqd Thk External mm.	Long Eff	Circ Eff
Skirt	-1000.0	1700.0	22.0	22.0	...	...	0.70	0.70
Skirt	0.0	1000.0	18.0	18.0	...	...	0.70	0.50
Ellipse	50.0	50.0	32.0	27.0	26.4	...	1.00	1.00
Cylinder	3050.0	3000.0	32.0	32.0	31.0	...	0.85	0.85
Cylinder	6050.0	3000.0	32.0	32.0	31.0	...	0.85	0.85
Cylinder	9050.0	3000.0	32.0	32.0	31.0	...	0.85	0.85
Cylinder	12050.0	3000.0	32.0	32.0	31.0	...	0.85	0.85
Cylinder	14450.0	2400.0	32.0	32.0	31.0	...	0.85	0.85
Ellipse	14500.0	50.0	32.0	27.0	26.4	...	1.00	1.00

Local Stress Analysis Results:

Description	Analysis Type	Max Stress Ratio	High Stress Location	Pass Fail
Trunnion	WRC-107/537	0.758	n/a	Passed
Trunnion	WRC-107/537	0.759	n/a	Passed
Trunnion	WRC-107/537	0.761	n/a	Passed
Trunnion	WRC-107/537	0.762	n/a	Passed
Trunnion	WRC-107/537	0.763	n/a	Passed
Trunnion	WRC-107/537	0.763	n/a	Passed
Trunnion	WRC-107/537	0.764	n/a	Passed
Trunnion	WRC-107/537	0.764	n/a	Passed
Trunnion	WRC-107/537	0.764	n/a	Passed
Trunnion	WRC-107/537	0.763	n/a	Passed
Trunnion	WRC-107/537	0.763	n/a	Passed
Trunnion	WRC-107/537	0.762	n/a	Passed
Trunnion	WRC-107/537	0.761	n/a	Passed
Trunnion	WRC-107/537	0.759	n/a	Passed
Trunnion	WRC-107/537	0.758	n/a	Passed
Trunnion	WRC-107/537	0.756	n/a	Passed
Trunnion	WRC-107/537	0.754	n/a	Passed
Trunnion	WRC-107/537	0.752	n/a	Passed
Trunnion	WRC-107/537	0.749	n/a	Passed
Trunnion	WRC-107/537	0.746	n/a	Passed
Trunnion	WRC-107/537	0.743	n/a	Passed
Trunnion	WRC-107/537	0.740	n/a	Passed
Trunnion	WRC-107/537	0.737	n/a	Passed
Trunnion	WRC-107/537	0.733	n/a	Passed
Trunnion	WRC-107/537	0.729	n/a	Passed
Trunnion	WRC-107/537	0.725	n/a	Passed
Trunnion	WRC-107/537	0.721	n/a	Passed
Trunnion	WRC-107/537	0.716	n/a	Passed
Trunnion	WRC-107/537	0.711	n/a	Passed
Trunnion	WRC-107/537	0.706	n/a	Passed
Trunnion	WRC-107/537	0.701	n/a	Passed
Trunnion	WRC-107/537	0.695	n/a	Passed
Trunnion	WRC-107/537	0.689	n/a	Passed
Trunnion	WRC-107/537	0.683	n/a	Passed
Trunnion	WRC-107/537	0.677	n/a	Passed
Trunnion	WRC-107/537	0.671	n/a	Passed
Trunnion	WRC-107/537	0.664	n/a	Passed
Trunnion	WRC-107/537	0.657	n/a	Passed
Trunnion	WRC-107/537	0.650	n/a	Passed
Trunnion	WRC-107/537	0.643	n/a	Passed
Trunnion	WRC-107/537	0.636	n/a	Passed
Trunnion	WRC-107/537	0.628	n/a	Passed
Trunnion	WRC-107/537	0.620	n/a	Passed
Trunnion	WRC-107/537	0.612	n/a	Passed
Trunnion	WRC-107/537	0.604	n/a	Passed
Trunnion	WRC-107/537	0.595	n/a	Passed
Trunnion	WRC-107/537	0.587	n/a	Passed
Trunnion	WRC-107/537	0.578	n/a	Passed
Trunnion	WRC-107/537	0.569	n/a	Passed
Trunnion	WRC-107/537	0.560	n/a	Passed



Trunnion	WRC-107/537	0.550	n/a	Passed
Trunnion	WRC-107/537	0.541	n/a	Passed
Trunnion	WRC-107/537	0.531	n/a	Passed
Trunnion	WRC-107/537	0.521	n/a	Passed
Trunnion	WRC-107/537	0.511	n/a	Passed
Trunnion	WRC-107/537	0.500	n/a	Passed
Trunnion	WRC-107/537	0.490	n/a	Passed
Trunnion	WRC-107/537	0.479	n/a	Passed
Trunnion	WRC-107/537	0.469	n/a	Passed
Trunnion	WRC-107/537	0.458	n/a	Passed
Trunnion	WRC-107/537	0.446	n/a	Passed
Trunnion	WRC-107/537	0.435	n/a	Passed
Trunnion	WRC-107/537	0.424	n/a	Passed
Trunnion	WRC-107/537	0.412	n/a	Passed
Trunnion	WRC-107/537	0.417	n/a	Passed
Trunnion	WRC-107/537	0.422	n/a	Passed
Trunnion	WRC-107/537	0.428	n/a	Passed
Trunnion	WRC-107/537	0.433	n/a	Passed
Trunnion	WRC-107/537	0.439	n/a	Passed
Trunnion	WRC-107/537	0.445	n/a	Passed
Trunnion	WRC-107/537	0.450	n/a	Passed
Trunnion	WRC-107/537	0.456	n/a	Passed
Trunnion	WRC-107/537	0.462	n/a	Passed
Trunnion	WRC-107/537	0.468	n/a	Passed
Trunnion	WRC-107/537	0.474	n/a	Passed
Trunnion	WRC-107/537	0.480	n/a	Passed
Trunnion	WRC-107/537	0.487	n/a	Passed
Trunnion	WRC-107/537	0.494	n/a	Passed
Trunnion	WRC-107/537	0.501	n/a	Passed
Trunnion	WRC-107/537	0.508	n/a	Passed
Trunnion	WRC-107/537	0.516	n/a	Passed
Trunnion	WRC-107/537	0.524	n/a	Passed
Trunnion	WRC-107/537	0.533	n/a	Passed
Trunnion	WRC-107/537	0.543	n/a	Passed
Trunnion	WRC-107/537	0.553	n/a	Passed
Trunnion	WRC-107/537	0.565	n/a	Passed
Trunnion	WRC-107/537	0.577	n/a	Passed
Trunnion	WRC-107/537	0.592	n/a	Passed
Trunnion	WRC-107/537	0.608	n/a	Passed
Trunnion	WRC-107/537	0.626	n/a	Passed
Trunnion	WRC-107/537	0.648	n/a	Passed
Trunnion	WRC-107/537	0.764	n/a	Passed
N2-3"	WRC-107/537	0.900	n/a	Passed
N6-2"	WRC-297	0.833	Shell	Passed
N1-14"	WRC-297	0.833	Shell	Passed
N3-14"	WRC-107/537	0.900	n/a	Passed

Wind/Earthquake Shear, Bending:

From	To	Distance to Support mm.	Cumulative Wind Shear Kgf	Earthquake Shear Kgf	Wind Bending Kg-m.	Earthquake Bending Kg-m.
10	20	850	6531.82	23660.7	64373.9	321429
20	30	2200	6070.09	23528.4	53662.1	281318
30	40	2725	5798.67	23432.2	47727.6	257837
40	50	4250	5529.42	22630.3	47444.4	256685
50	60	7250	4713.47	21629.5	32079.7	190294
60	70	10250	3656.79	19543.4	19524.1	128534
70	80	13250	2570.82	16380.2	10182.5	74647.5
80	90	15950	1754.86	13264.8	3693.85	30179.1
90	100	17175	863.334	7753.29	551.953	4956.89

Abs Max of the all of the Stress Ratio's : 0.4822

Basing Data : Continuous Top Ring W/Gussets

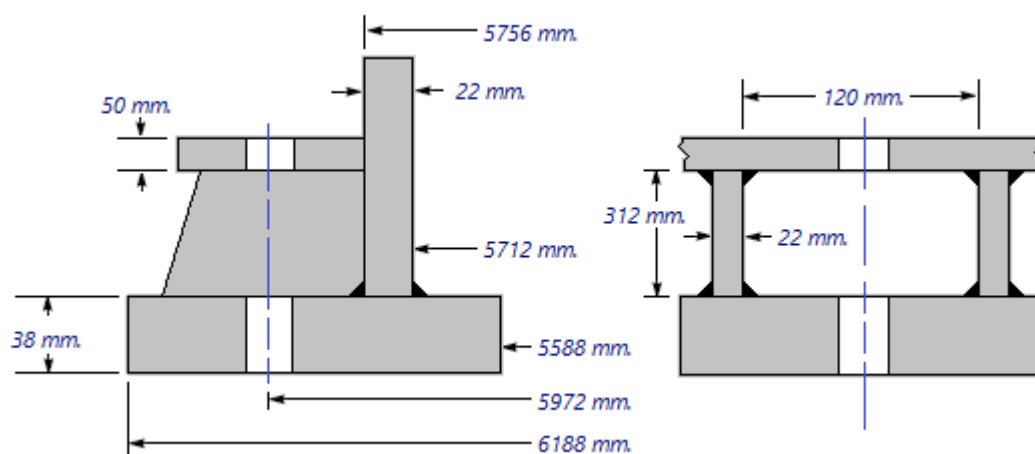
Thickness of Basing

38.0000 mm.



Inside Diameter of Basing	5588.0000 mm.
Outside Diameter of Basing	6188.0000 mm.
Nominal Diameter of Bolts	64.0000 mm.
Diameter of Bolt Circle	5972.0000 mm.
Number of Bolts	24
Thickness of Gusset Plates	22.0000 mm.
Average Width of Gusset Plates	206.0000 mm.
Height of Gussets	312.0000 mm.
Distance between Gussets	120.0000 mm.
Thickness of Top Plate or Ring	50.0000 mm.
Radial Width of the Top Plate	38 mm.

Basing Sketch for Cylindrical Skirts:



Basing Cross Section View

Loads for Foundation/Support Design:

Total Wind Shear on Support	6532. Kg
Total Earthquake Shear on Support	23661. Kg
Wind Moment on Support	73263. Kg-m.
Earthquake Moment on Support	330318. Kg-m.
Maximum Initial Bolt Load	3362. Kg

Wind and Earthquake moments include the effects of user defined forces and moments if any exist in the job and were specified to act (compute loads and stresses) during these cases. Also included are moment effects due to eccentric weights if any are present in the input.

Weights:

Fabricated - Bare w/o removable internals	107465.0 kg.
Shop Test - Fabricated + water (full)	538474.0 kg.
Shipping - Fab. + rem. intls.+ shipping app.	107465.0 kg.
Erected - Fab. + rem. intls.+ insul. (etc)	144549.6 kg.
Empty - Fab. + intls. + details + wghts.	144549.6 kg.
Operating - Empty + operating liquid (no ca)	145093.3 kg.
Field Test - Empty weight + test medium (full)	561482.5 kg.



PV Elite[®] Pressure Vessel Design Notes

Project Information	
Project Number	
Client	
Date	
Designer	
Design Code	
Additional Standards	

Design Conditions		
Internal Design Pressure:		
External Design Pressure:		
Design Int. Temperature:		
Design Ext. Temperature:		
Corrosion Allowance:		

Vessel Geometry		
Vessel Type		
Diameter		
Tangent-to-Tangent Length		
Head Type		
Number of Segments		

Material Specifications	
Shell:	
Head:	
Nozzle:	

Nozzle/Opening Details	ID/OD	Size	Rating	Matl.	Reinf.

Miscellaneous		
Support Type		
Number of Supports		
Support Material		
Seismic Code		
Wind Code		
Insulation		
Pressure Test		
Radiography		
Fatigue Analysis		

Analysis Notes: